



**Zebra<sup>®</sup> GX420t<sup>™</sup> / GX430t<sup>™</sup>**  
Desktop Thermal Printer

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**User Guide**





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- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced RF service technician for help.

The user is cautioned that any changes or modifications not expressly approved by Zebra Technologies could void the user's authority to operate the equipment. To ensure compliance, this printer must be used with fully shielded communication cables.



**Caution** • The optional RTC assembly has a three volt lithium battery. Battery replacement must be performed by a qualified service technician. Only use a Zebra approved replacement battery.



**Important** • Recycle batteries according to local your guidelines and regulations. Wrap the battery when disposing (or storing) to avoid a short circuit.

**DO NOT** short circuit the battery. Short circuiting the battery may result in heat generation, fire or bursting.

**DO NOT** heat, disassemble or dispose of battery in fire.

## Environmental Management



Do not dispose of this product in unsorted municipal waste. This product is recyclable, and should be recycled according to your local standards.

For more information, please see our website at:

**Web address:** [www.zebra.com/environment](http://www.zebra.com/environment)

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# About This Document



This section provides you with contact information, document structure and organization, and additional reference documents.

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## Who Should Use This Document

This User Guide is intended for use by any person who needs to operate or to troubleshoot problems with the printer.

## How This Document Is Organized

The User Guide is set up as follows:

| Section                                                        | Description                                                                                                                                                                                                      |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i><a href="#">Introduction on page 1</a></i>                  | This section describes what you get in your shipping box and provides an overview of printer features. This section also has procedures that describe how to open and close the printer and report any problems. |
| <i><a href="#">Getting Started on page 13</a></i>              | This section describes how to set up your printer for the first time and use the most common operating procedures for loading media.                                                                             |
| <i><a href="#">Print Operations on page 35</a></i>             | This section provides media and print handling, font and language support, and the set up of less common printer configurations.                                                                                 |
| <i><a href="#">Printer Options on page 51</a></i>              | This section covers the common printer options and accessories with brief descriptions and how to get you started using or configuring your printer option or accessory.                                         |
| <i><a href="#">Control Panel Operation on page 75</a></i>      | This section describes the operation, navigation and parameters displayed by printers equipped with the LCD control panel option.                                                                                |
| <i><a href="#">Maintenance on page 81</a></i>                  | This section provides routine cleaning and maintenance procedures.                                                                                                                                               |
| <i><a href="#">Troubleshooting on page 95</a></i>              | This section provides information about printer error reporting that you might need for printer troubleshooting. Assorted diagnostic tests are included.                                                         |
| <i><a href="#">Appendix: Interface Wiring on page 109</a></i>  | This section provides additional interface information and wiring diagrams to assist with printer to host system integration (typically a PC).                                                                   |
| <i><a href="#">Appendix: Dimensions on page 115</a></i>        | This section provides external printer dimensions for the standard printer and printer with options.                                                                                                             |
| <i><a href="#">Appendix: ZPL Configuration on page 119</a></i> | This section briefly covers printer configuration and includes a cross-reference to ZPL printer configuration commands.                                                                                          |

# Contacts

Technical Support via the Internet is available 24 hours per day, 365 days per year.

**Web Site:** [www.zebra.com](http://www.zebra.com)

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- **E-mail address:** [emb@zebra.com](mailto:emb@zebra.com)
- **Subject line:** Emailist

**Self Service Knowledge Base:** [www.zebra.com/knowledgebase](http://www.zebra.com/knowledgebase)

**Online Case Registration:** [www.zebra.com/techrequest](http://www.zebra.com/techrequest)

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**Key:** T: Telephone  
F: Facsimile  
E: E-mail

# Document Conventions

The following conventions are used throughout this document to convey certain information.

**Alternate Color** (online only) Cross-references contain hot links to other sections in this guide. If you are viewing this guide online in **.pdf** format, you can click the cross-reference (blue text) to jump directly to its location.

**Command Line Examples** Command line examples appear in **Courier New** font. For example, type **ZT00LS** to get to the Post-Install scripts in the **bin** directory.

**Files and Directories** File names and directories appear in **Courier New** font. For example, the **Zebra<version number>.tar** file and the **/root** directory.

## Icons Used

|                                                                                     |                                                                                                                                       |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Caution</b> • Warns you of the potential for electrostatic discharge.                                                              |
|    | <b>Caution</b> • Warns you of a potential electric shock situation.                                                                   |
|   | <b>Caution</b> • Warns you of a situation where excessive heat could cause a burn.                                                    |
|  | <b>Caution</b> • Advises you that failure to take or avoid a specific action could result in physical harm to you.                    |
|                                                                                     | <b>Caution</b> • (No icon) Advises you that failure to take or avoid a specific action could result in physical harm to the hardware. |
|  | <b>Important</b> • Advises you of information that is essential to complete a task.                                                   |
|  | <b>Note</b> • Indicates neutral or positive information that emphasizes or supplements important points of the main text.             |
|  | <b>Example</b> • Provides an example, often a scenario, to better clarify a section of text.                                          |



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# Introduction

This section describes what you get in your shipping box and provides an overview of printer features. This includes procedures that describe how to open and close the printer and report any problems.

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## GXGK Series Thermal Printers

The Zebra® GX models are the Best in Class™ desktop thermal printers with the widest range of features. The GX420t™ printer provides thermal transfer and direct thermal printing at speeds up to 6 ips (inches per second) at a 203 dpi (dots per inch) print density. The GX430t™ printers provide thermal transfer and direct thermal printing at speeds up to 4 ips at a 300 dpi print density. Both the GX420t and GX430t support both ZPL™ and EPL™ Zebra printer programming languages and a wide variety of interface and feature options.

The G-Series™ printer features:

- Automatic printer language detection and switching between ZPL and EPL programming and label formats.
- OpenAccess™ design for simplified media loading.
- The industry's easiest ribbon loading design.
- Color-coded operator controls and media guides.
- Zebra™ Global Printing Solution – supports Microsoft Windows keyboard encoding (and ANSI), Unicode UTF-8 and UTF 16 (Unicode Transformation Formats), XML, ASCII (7 and 8 bit used by legacy programs and systems), basic single and double byte font encoding, JIS and Shift-JIS (Japanese International Standards), Hexadecimal encoding, and custom character mapping (DAT table creation, font linking, and character remapping).
- On-The –Fly OpenType and TrueType font scaling and import, Unicode, preloaded Swiss 721 Latin 1 font, one resident scalable font, and a selection of resident bitmap fonts.
- Improved printer performance: faster print speeds and a 32-bit processor.
- Adaptive auto serial port cable sensing and configuration for plug-and-play integration.
- A complete suite of free software applications and drivers to configure printer settings, design and print labels and receipts, get printer status, import graphics and fonts, send programming commands, update firmware, and download files. Clone printer settings and send graphics, files, fonts, and firmware (updates) to one or more Zebra® Ethernet and locally connected printers with ZebraNet™ Bridge.
- Printhead test and maintenance reporting enabled and customizable by the user.

The GX printers offer a wide range of printer options:

- Label Dispense (Peel-Off).
- Internal 10/100 Print Server and Ethernet Interface.
- Wi-Fi (802.11b/g) with user display.
- Bluetooth with user display.
- Moveable Blackline Sensor: A full width moveable black mark sensor and multi position transmissive (label gap) sensor.”
- General purpose media cutter.
- On-board RTC (Real Time Clock) and 68MB Full Flash memory (12MB Full Flash Memory prior to July 24, 2009).
- Asian Language support with printer configuration options for the large Simplified and Traditional Chinese, Japanese, or Korean character sets.

- Zebra® ZBI 2.0™ (Zebra BASIC Interpreter) programming language. ZBI allows you to create custom printer operations that can automate processes, use peripherals (i.e. scanners, scales, keyboards, Zebra® KDU or KDU Plus™, etc.) all without being attached to a PC or network.

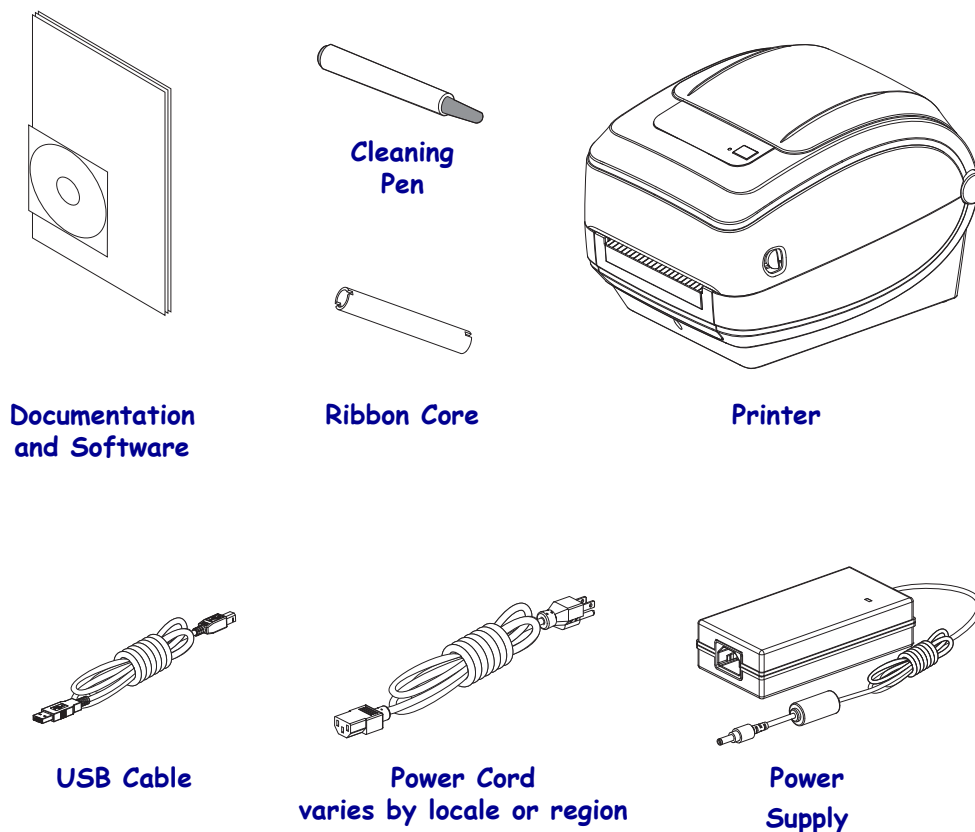
This users guide provides all the information you need to operate your printer on a daily basis. To create label formats, refer to your programming guides or label design applications such as ZebraDesigner™.

Your printer, when connected to a host computer, functions as a complete system for printing labels and tags.

**Note •** Many printer settings may also be controlled by your printer's driver or label design software. Refer to the driver or software documentation for more information.

## What's in the Box?

Save the carton and all packing materials in case you need to ship or store the printer later. After unpacking, make sure you have all parts. Follow the procedures for inspecting the printer to familiarize yourself with printer parts so you can follow the instructions in this book.



## Unpack and Inspect the Printer

When you receive the printer, immediately unpack it and inspect for shipping damage.

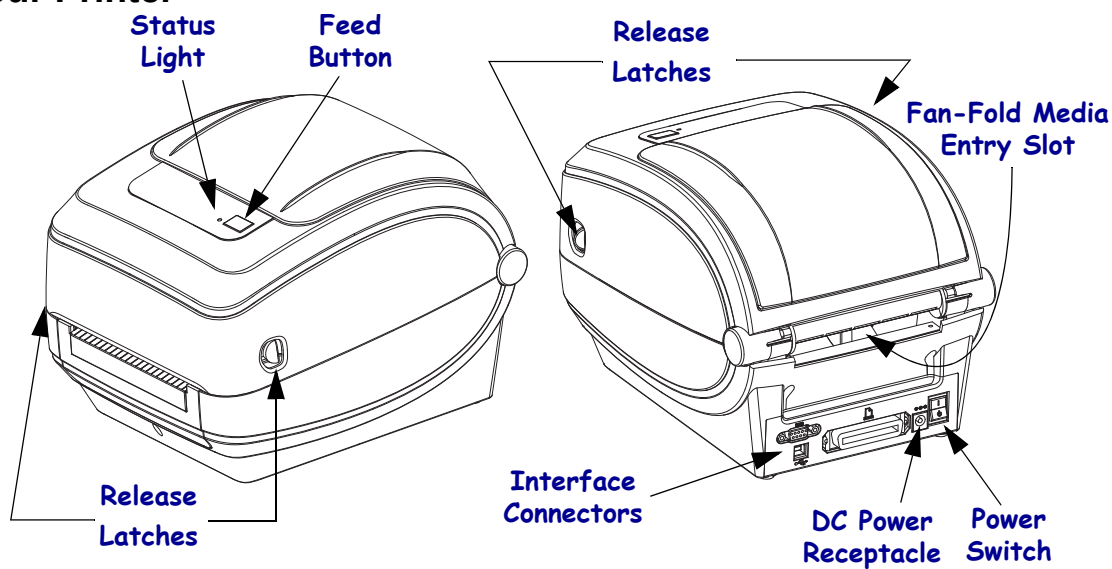
- Save all packing materials.
- Check all exterior surfaces for damage.
- Open the printer and inspect the media compartment for damage to components.

If you discover shipping damage upon inspection:

- Immediately notify the shipping company and file a damage report. Zebra Technologies Corporation is not responsible for any damage to the printer incurred during shipment, and will not cover the repair of this damage under its warranty policy.
- Keep all packaging material for shipping company inspection.
- Notify your authorized Zebra<sup>®</sup> reseller.

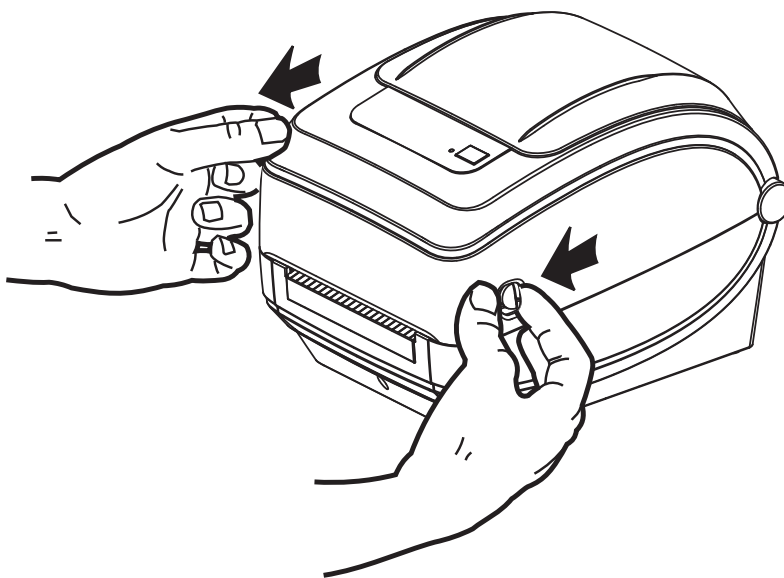


## Your Printer



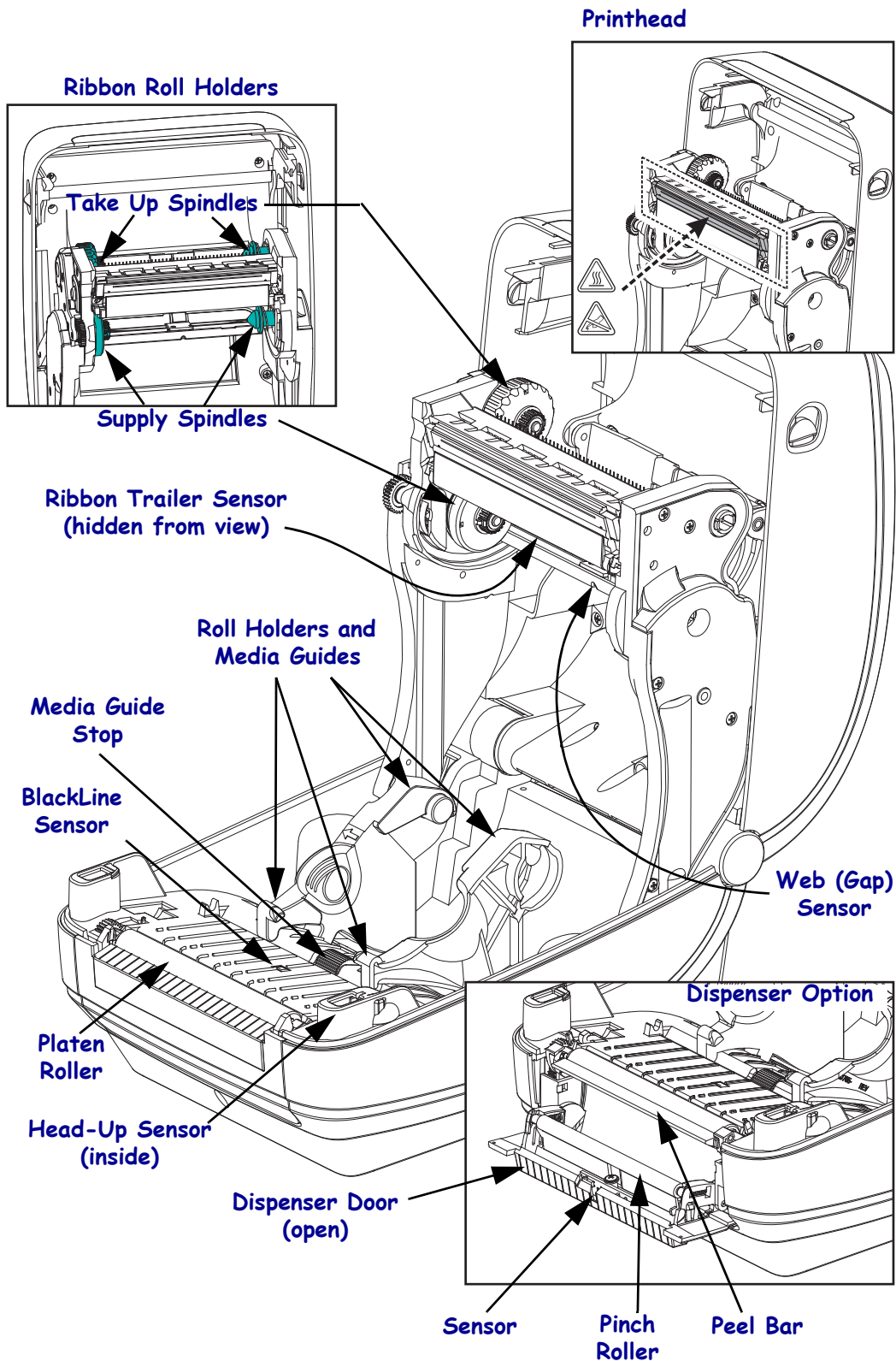
## Opening the printer

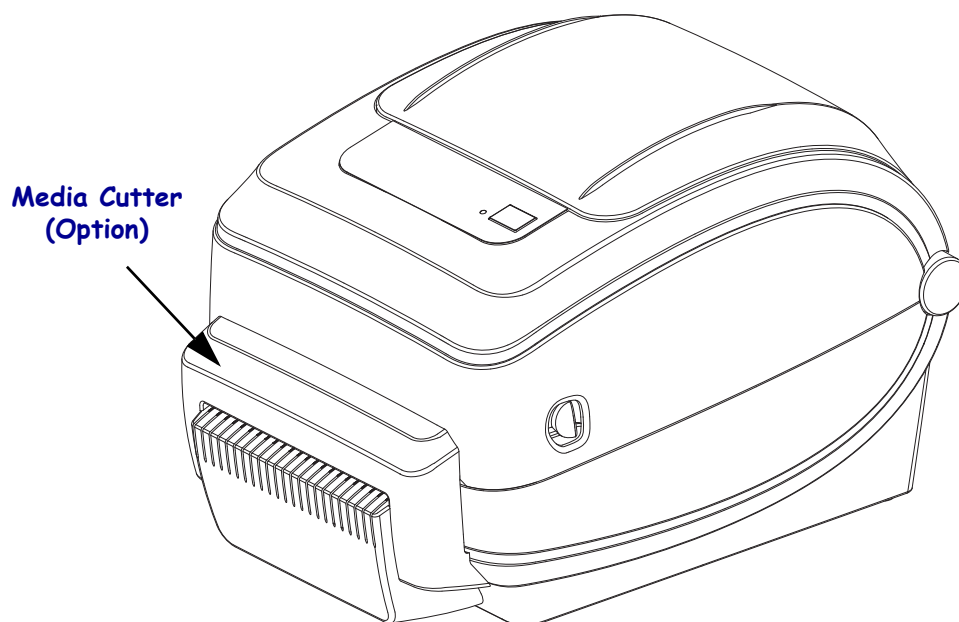
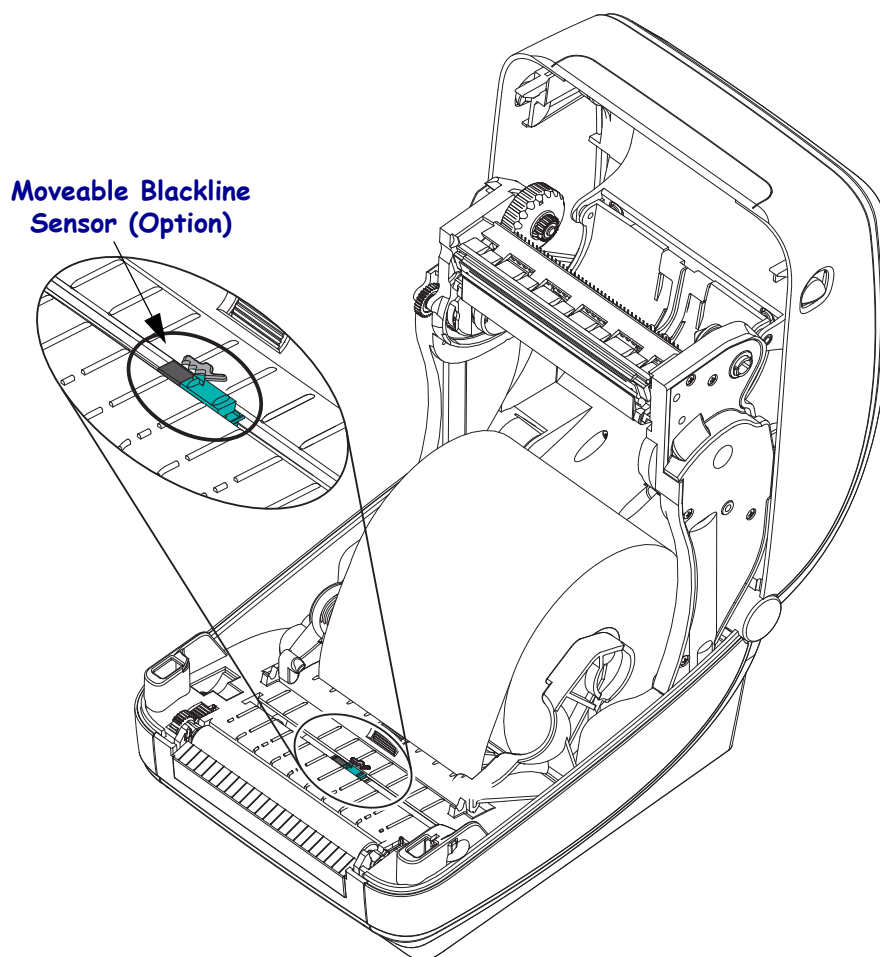
To access the media compartment, you must open the printer. Pull the release latches towards you and lift the cover. Check the media compartment for loose or damaged components.



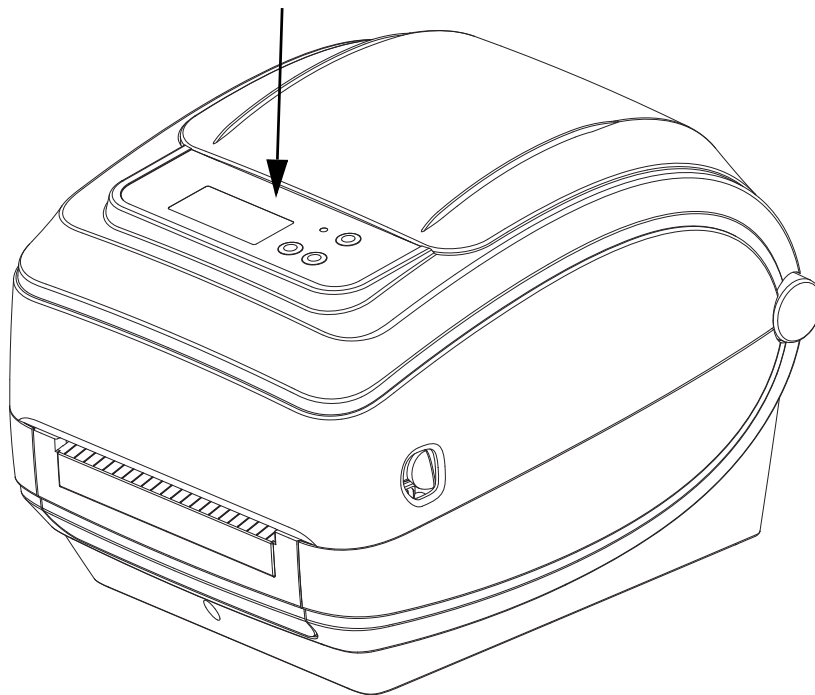
**Caution** • The discharge of electrostatic energy that accumulates on the surface of the human body or other surfaces can damage or destroy the printhead or electronic components used in this device. You must observe static-safe procedures when working with the printhead or the electronic components under the top cover.

## Printer Features





**Wireless Printer Display and Controls  
(Wi-Fi and Bluetooth Options)**



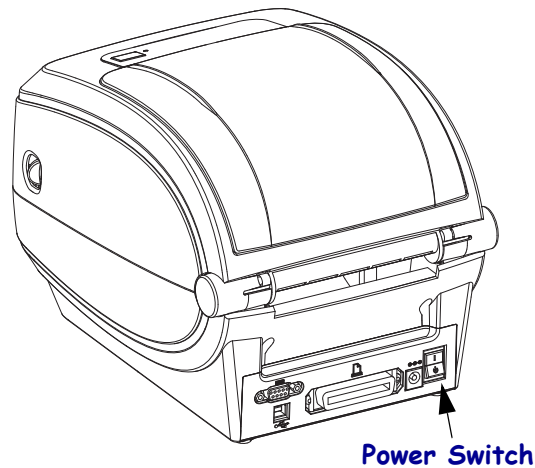
## Operator Controls

### Power Switch

Press up to turn **ON** or *down* to turn **OFF** the printer.



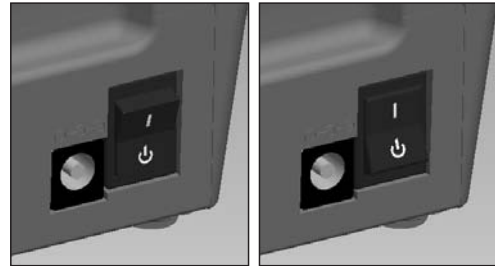
**Caution** • The printer power should be turned off before connecting or disconnecting the communications and power cables.



Power Switch

Power OFF

Power ON



### Feed Button

- Tap the Feed button once to force the printer to feed one blank label.
- Press the Feed button to take the printer out of a “pause” condition. The printer is put into “pause” by either a programming command or an error condition. See [What the Status Light is Telling You on page 96](#) in the Troubleshooting chapter.
- Use the Feed button for printer setup and status (see [Feed Button Modes on page 106](#) in the Troubleshooting chapter).

### Scroll Button (Wireless Printers Only)

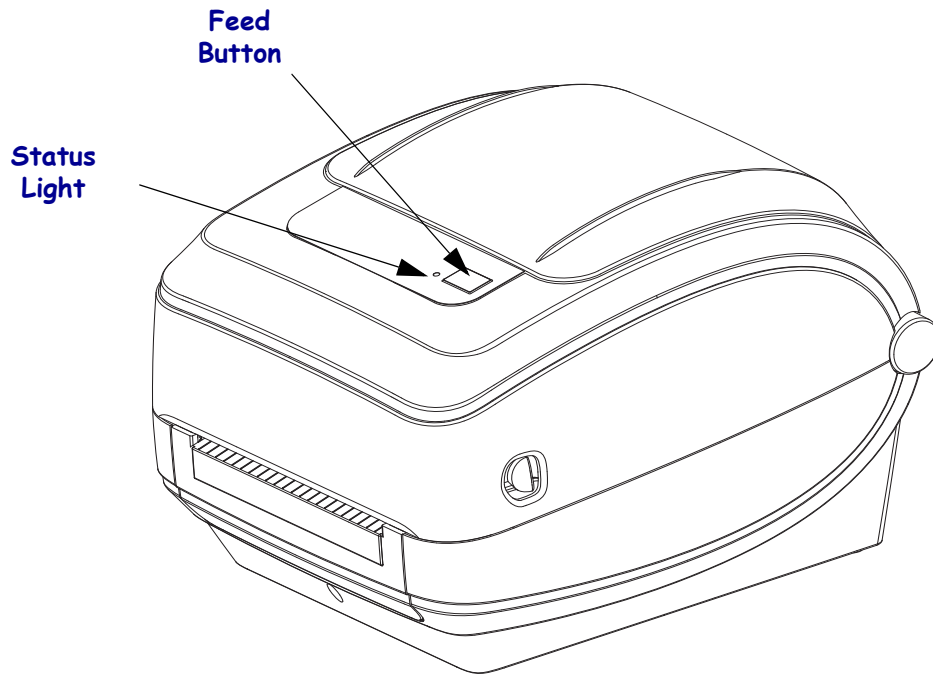
Press the Scroll button to step through the menu, displayed printer parameters or settings.

### Select Button (Wireless Printers Only)

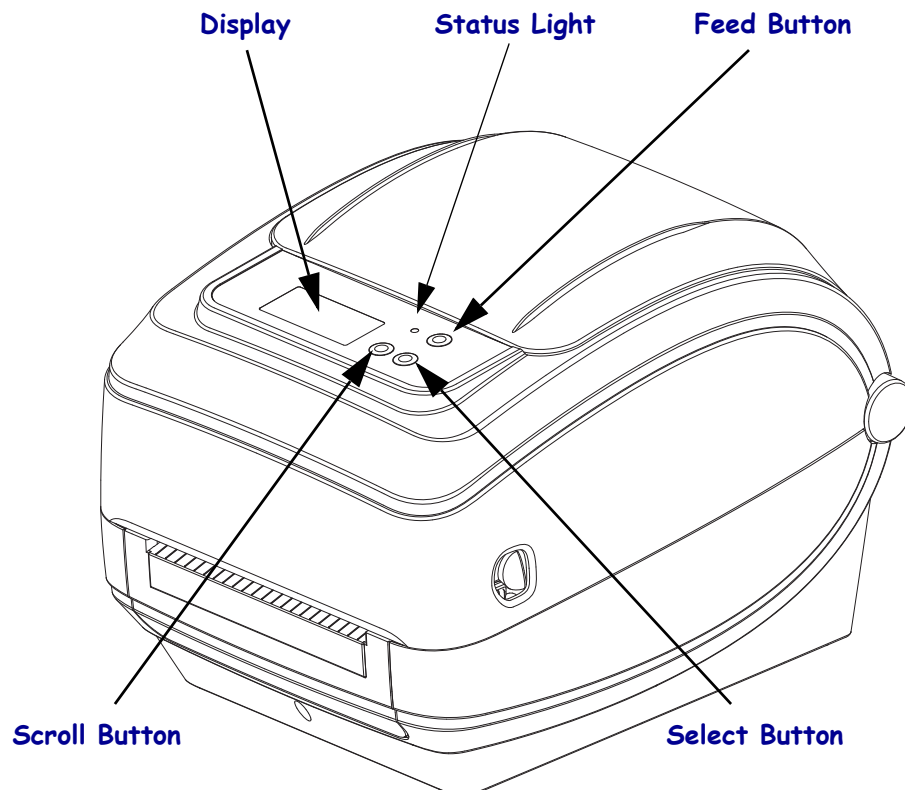
Press the Select button to choose a menu branch or select a printer setting.

## Status Light

Located on the top case next to the feed button, the status light functions as a printer operational indicator (see [Status Light Descriptions on page 96](#)).

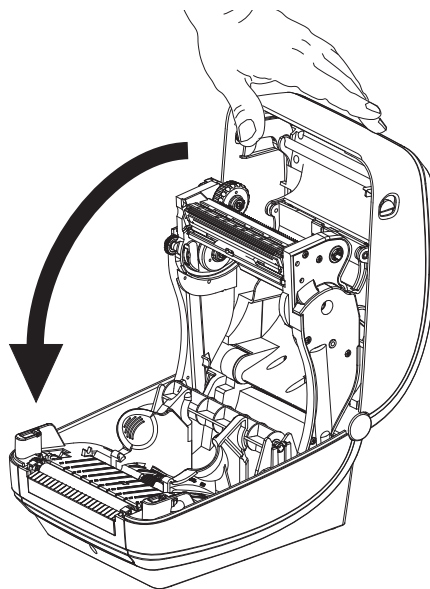


## Wireless Printer Display and Controls (Wi-Fi and Bluetooth Options)

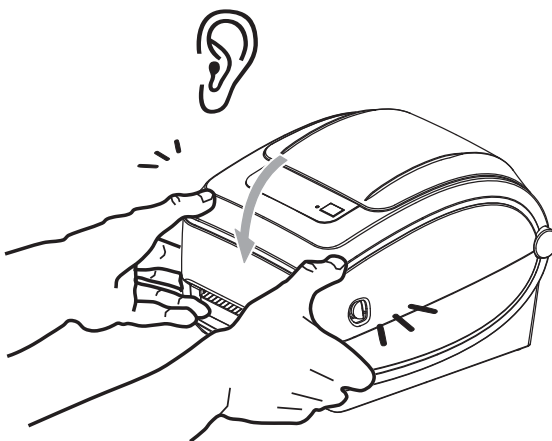


## Closing the printer

1. Lower the top cover.



2. Press down until the cover snaps closed.





Notes • \_\_\_\_\_

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# Getting Started

This section describes how to set up your printer for the first time and use the most common operating procedures for loading media.

## Contents

|                                                         |    |
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| Attaching Power . . . . .                               | 14 |
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## Attaching Power

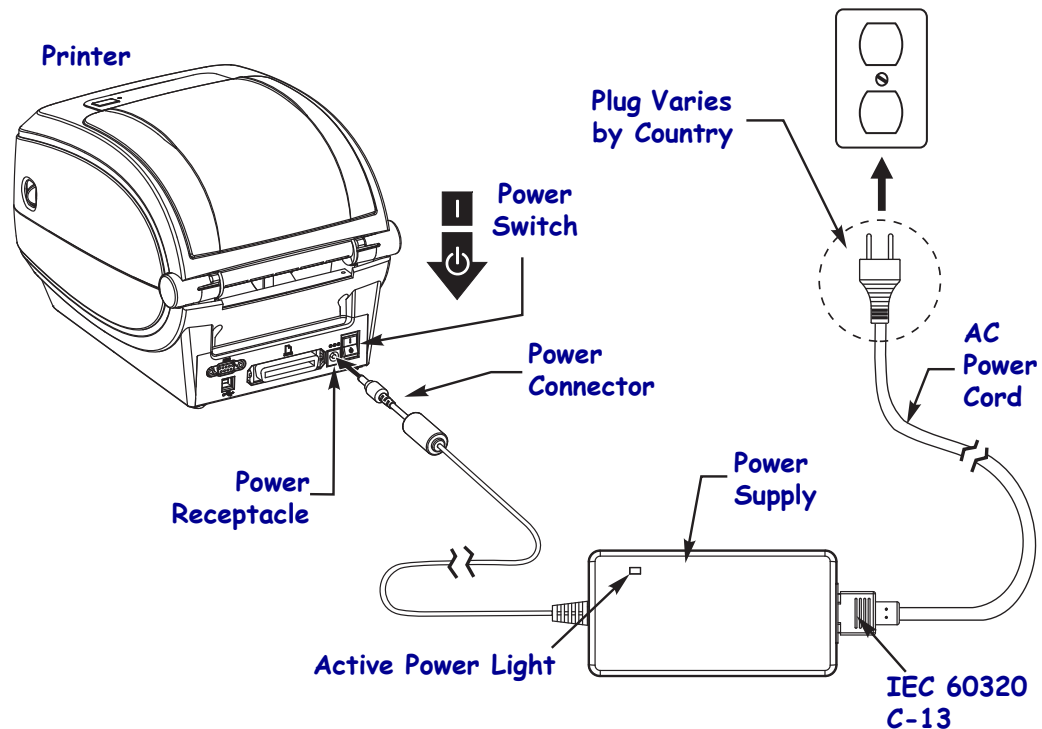


**Important** • Set up your printer so that you can handle the power cord easily if needed. To make certain the printer cannot carry electrical current, you must separate the power cord from the power supply receptacle or AC electrical outlet.



**Caution** • Never operate the printer and power supply in an area where they can get wet. Serious personal injury could result!

1. Make sure the printer's power switch is in the off position (down).
2. Insert the AC power cord into the power supply.
3. Plug the other end of the cord into an appropriate AC electrical outlet. Note: The active power light will go on if power is on at the AC outlet.
4. Insert the power supply's power connector into the printer's power receptacle.



**Note** • Ensure the appropriate power cord with a three (3) prong plug and an IEC 60320-C13 connector are used at all times. These power cords must bear the relevant certification mark of the country in which the product is being used.

## Loading Roll Media

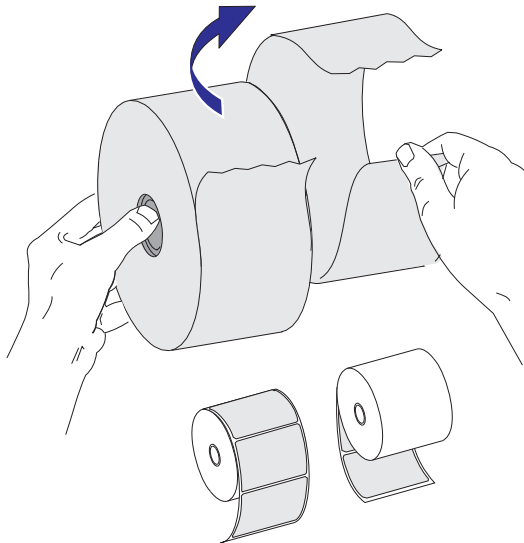
When you load media, you must place the roll onto the media hangers.

You must use the correct media for the type of printing you require.

### Preparing Media

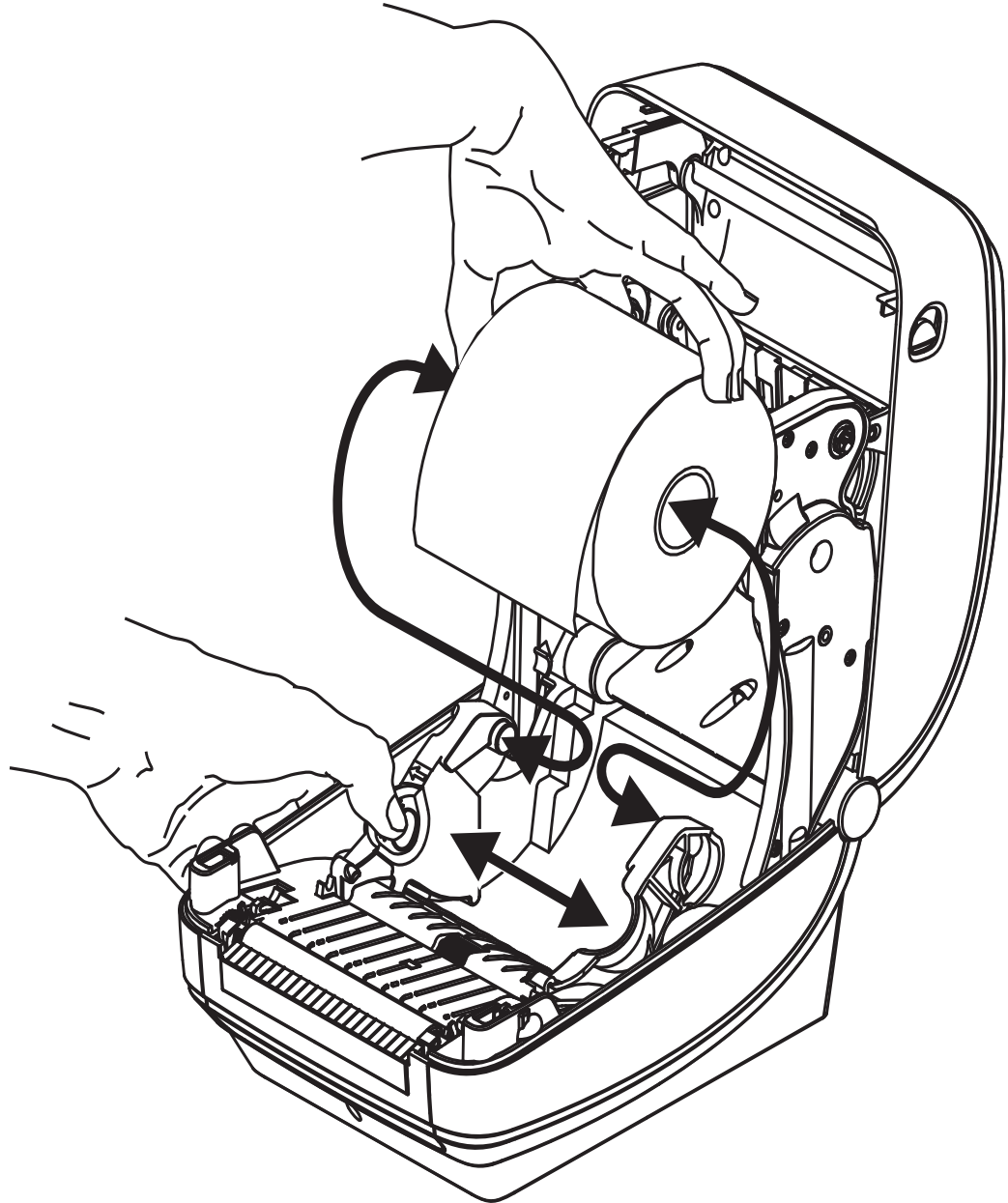
Whether your roll media is inside or outside wound, you load it into the printer the same way.

- Remove the outside length of media. During shipment, handling, or storage, the roll may become dirty or dusty. Removing the outside length of media avoids dragging adhesive or dirty media between the printhead and platen.

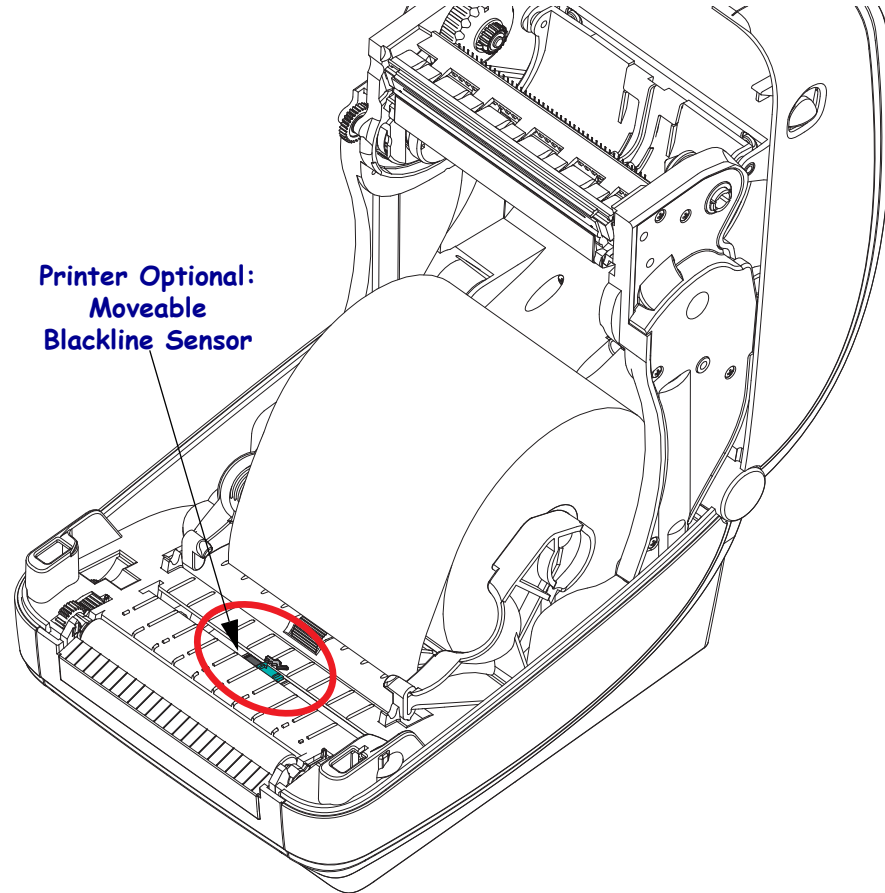


## Placing the Roll in the Media Compartment

1. Open the printer. Remember that you need to pull the release latch levers toward the front of the printer.
2. Open the media roll holders. Pull the media guides open with your free hand and place the media roll on the roll holders and release the guides. Orient the media roll so that its printing surface will face up as it passes over the platen (drive) roller.

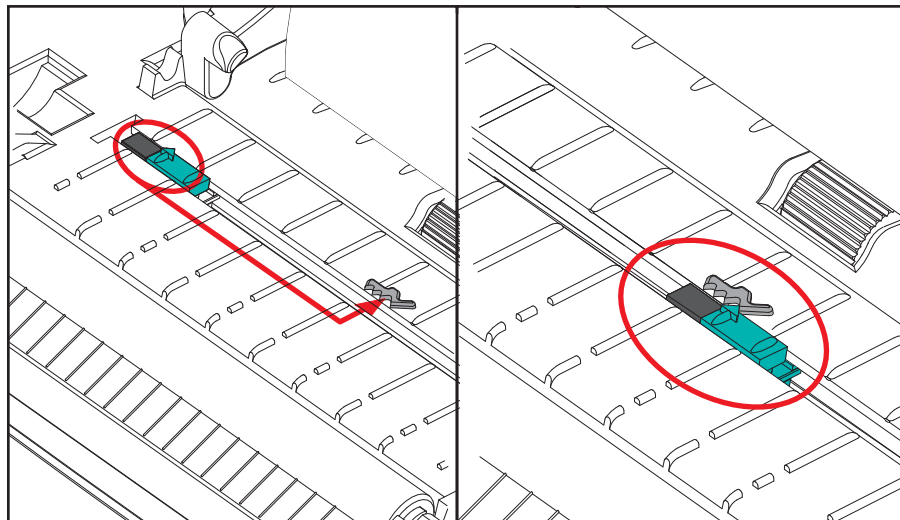


3. If your printer includes the moveable blackline sensor option, verify that the sensor is aligned in the center default position. This is the standard operating position for media sensing. When the sensor is moved from the default label web (gap) sensing area, the printer will only detect blackline, black mark and die-cut notched media. See [Moveable Blackline Sensor Option](#) on page 55 for more details on adjusting this sensor option.

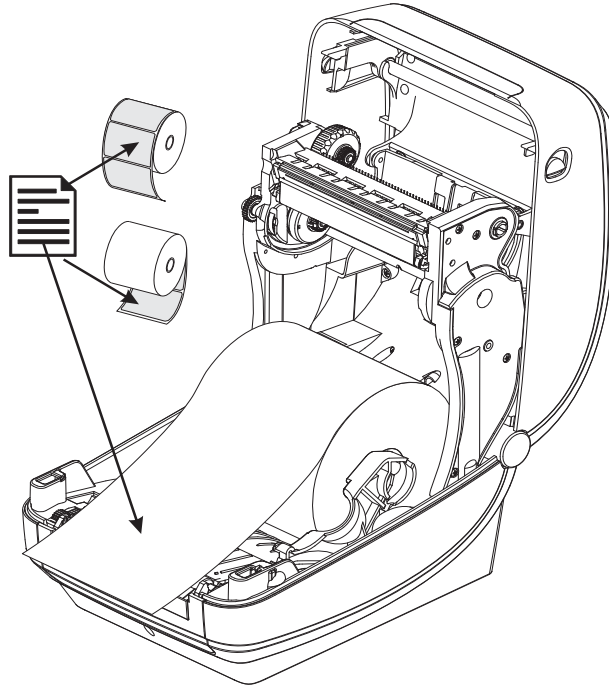


Off Center  
Blackline Sensing Only

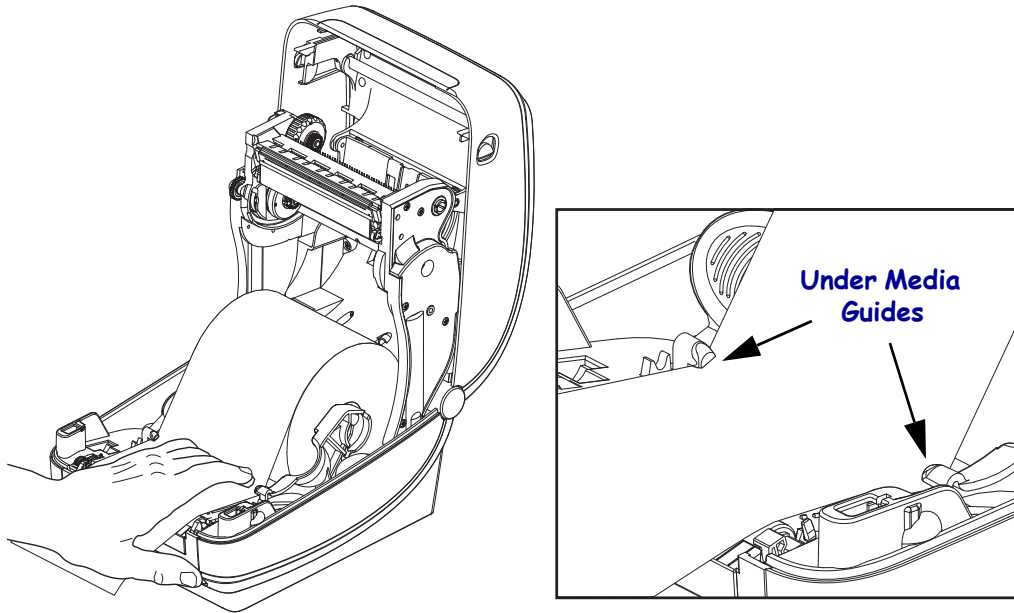
Default - Web (Gap) Sensing  
Standard Operating Position



4. Pull the media so that it extends out of the front of the printer. Verify the roll turns freely. The roll must not sit in the bottom of the media compartment. Verify that the media's printing surface is facing up.



5. Push the media under both of the media guides.



6. Close the printer. Press down until the cover snaps closed.

## Loading Transfer Ribbon

Transfer ribbons come in several varieties and in some cases colors to match your application needs. Genuine Zebra® transfer ribbons are specifically designed for your printer and Zebra brand media. Using non-Zebra media or ribbons not approved for use in your Zebra® printer may damage your printer or printhead.

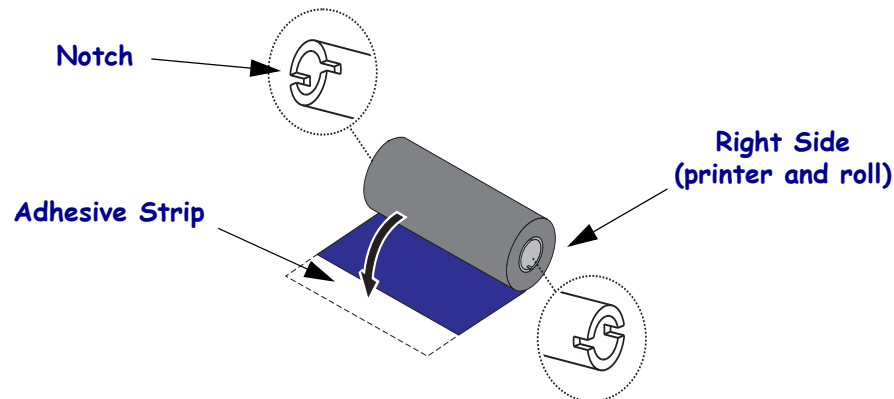
- The media and ribbon types should be matched to provide you with optimal print results.
- Always use ribbon that is wider than the media to protect the printhead from wear.
- For direct thermal printing, do not load ribbon in the printer.

Your printer needs to use Zebra® Uni-Ribbon™ (universal ribbon). Uni-Ribbons have a ribbon out trailer (reflector) to stop printing when empty. Zebra® Uni-Ribbon™ will work on older model Zebra® desktop printers too.

Uni-Ribbons are identified with a color coded leader.

- **Blue** - Performance Wax (5319)
- **Silver** - Premium Wax/Resin (3200)
- **Gold** - Performance Resin (5095) for synthetics (6 ips max. speed) and coated paper (4 ips max. speed)
- **Red** - Premium Resin (5100) for synthetics (4 ips max. speed)

Before following these steps, prepare the ribbon by removing its wrapping and pulling its adhesive strip free.



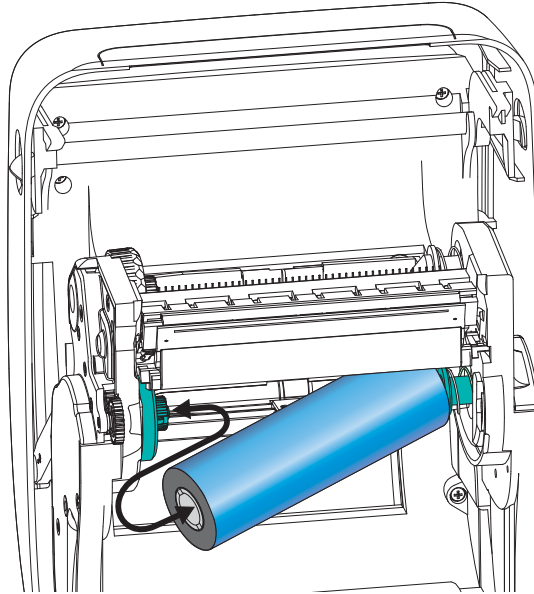
See [Ribbon Overview](#) on page 42 for more information on ribbon use.



**Important** • DO NOT USE early model desktop printer ribbon cores! The older ribbon cores can be identified by notches on only one side of the ribbon core. These older cores are too large and will cause take-up spool to bind.

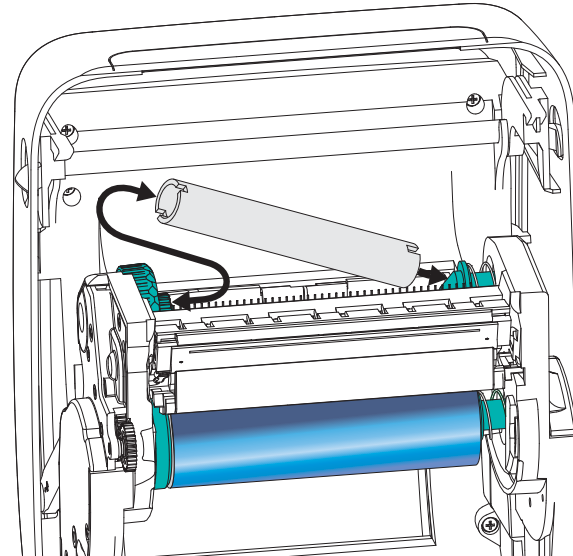
**Note** • DO NOT USE ribbon cores with damaged notches - rounded, frayed, smashed, etc. The core notches should be square to lock the core on the spindle or the core may slip causing ribbon wrinkle, poor end of ribbon sensing or other intermittent failures.

1. With the printer open, place a new ribbon roll on printer's lower ribbon supply spindles.



Rotate the roll until the notches align and lock into the left side of the supply hub.

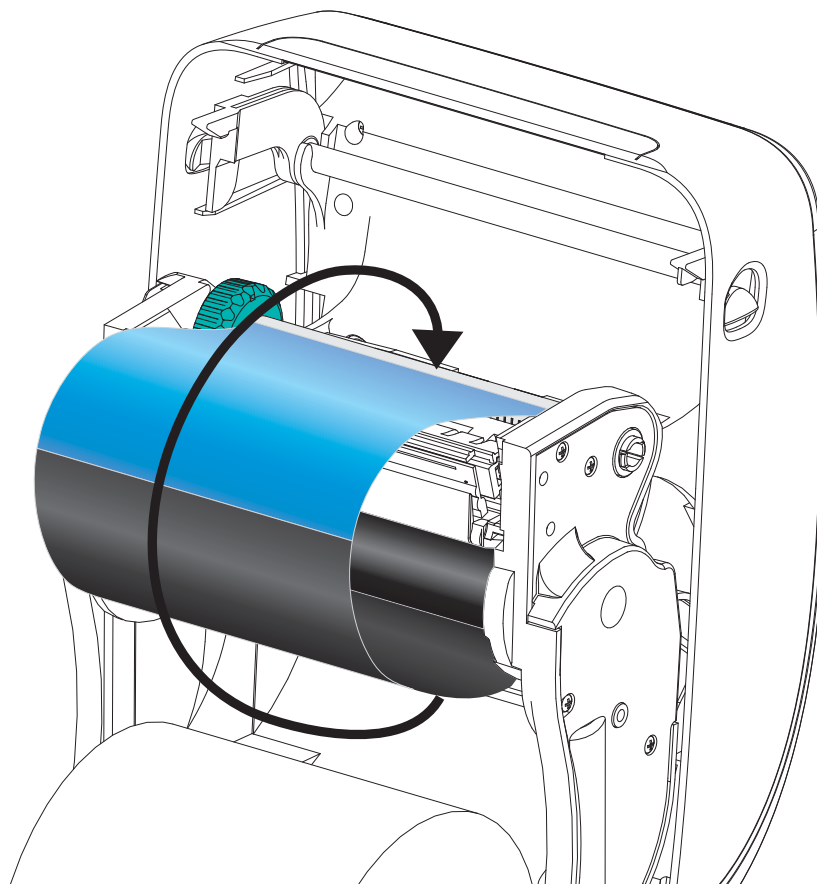
2. Place an empty ribbon core on printer's take-up spindles. Rotate the ribbon core until the notches align and lock into the left side of the take-up hub.



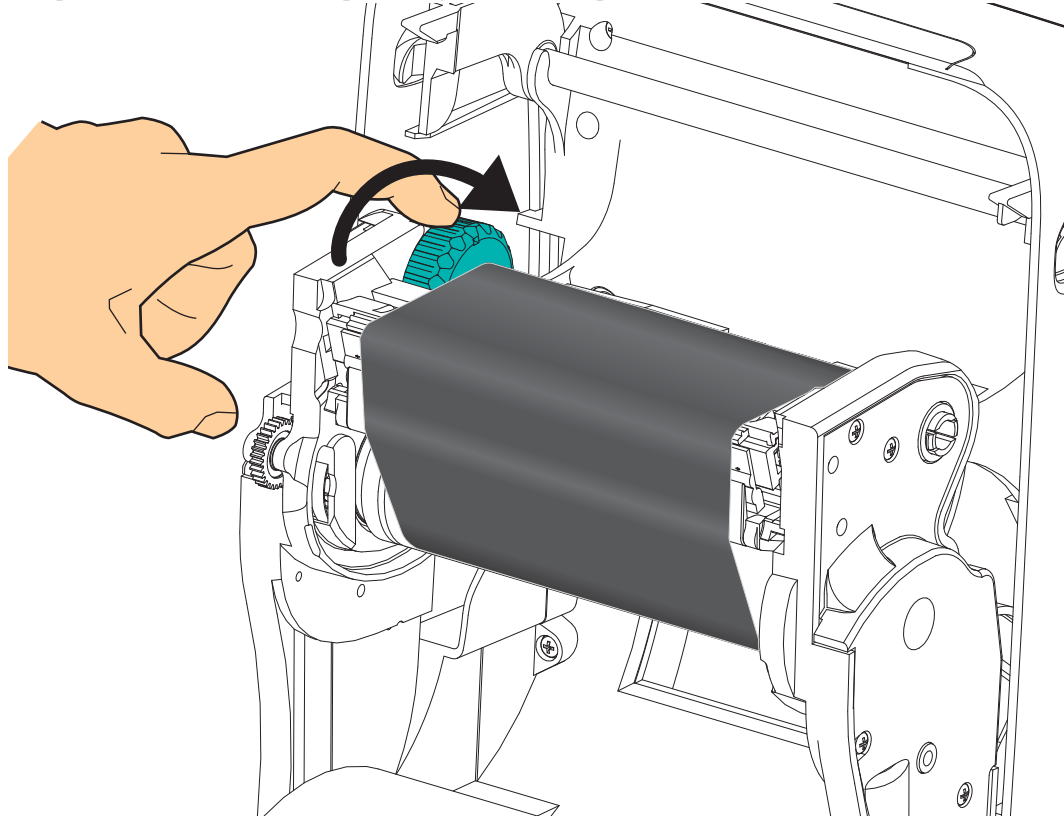
You can find your first ribbon take-up core in the packing box. Subsequently, use the empty supply core to take up the next roll of ribbon.



3. Pull the transfer ribbon's leader off the roll and attach it with the adhesive strip on the leader to the empty ribbon core on the supply spindle. Center the ribbon on the ribbon core.



4. Rotate the thumb wheel on the left side of the supply spindle towards the rear of the printer until the ribbon is pulled tight across the printhead.



5. Verify that the media is loaded and ready to print and then close the printer cover.
6. Press the Feed button to have the printer feed a minimum of 10 cm (4 inches) of media to remove slack and ribbon wrinkle, and to align the ribbon on the spindles.
7. Change the print mode setting from direct thermal printing to thermal transfer to set the printer's temperature profiles for thermal transfer media. This can be done with the printer driver, application software, or printer programming commands.
  - When controlling printer operations with ZPL programming, refer to the Media Type (**^MT**) ZPL II command (follow the instructions in the ZPL Programming Guide).
  - When controlling printer operations with EPL Page Mode, refer to the Options (**O**) EPL command (follow the instructions in the *EPL Page Mode Programmer's Guide*).
8. To verify the mode change from direct thermal printing to thermal transfer printing, use the [Printing a Test \(Printer Configuration\) Label on page 23](#) to print a configuration label. The 'PRINT METHOD' should read 'THERMAL-TRANS' on the printer configuration status label.

Your printer is now ready to print.

## Printing a Test (Printer Configuration) Label

**Before you connect the printer to your computer,** make sure that the printer is in proper working order.

You can do this by printing a configuration status label.

1. Make sure the media is properly loaded and the top cover of the printer is closed. Then, turn the printer power on if you have not already done so. If the printer initializes with the status light blinking green (pause mode), press the Feed button once to set the printer in Ready (to print) mode. See the [Troubleshooting on page 95](#) if the printer's status light does not turn solid green (Ready).
2. Press the Feed button two to three times to allow the printer to calibrate the printer for the installed media. The printer may feed several labels during this process.
3. When the status light is solid green, press and hold the Feed button until the status light flashes once.
4. Release the Feed button. A configuration label will print.

If you cannot get this label to print, refer to [Getting Started on page 13](#).

| PRINTER CONFIGURATION            |                 |
|----------------------------------|-----------------|
| Zebra Technologies<br>ZTC GX420t |                 |
| 15.0.....                        | DARKNESS        |
| 6 IPS.....                       | PRINT SPEED     |
| +000.....                        | TEAR OFF        |
| TEAR OFF.....                    | PRINT MODE      |
| GAP/NOTCH.....                   | MEDIA TYPE      |
| WEB.....                         | SENSOR TYPE     |
| MANUAL.....                      | SENSOR SELECT   |
| THERMAL-TRANS.....               | PRINT METHOD    |
| 800.....                         | PRINT WIDTH     |
| 1234.....                        | LABEL LENGTH    |
| 39.0IN 989MM.....                | MAXIMUM LENGTH  |
| CONNECTED.....                   | USB COMM.       |
| BIDIRECTIONAL.....               | PARALLEL COMM.  |
| 9600.....                        | BAUD            |
| 8 BITS.....                      | DATA BITS       |
| NONE.....                        | PARITY          |
| XON/XOFF.....                    | HOST HANDSHAKE  |
| NONE.....                        | PROTOCOL        |
| <~> 7EH.....                     | CONTROL CHAR    |
| <^> 5EH.....                     | COMMAND CHAR    |
| <,> 2CH.....                     | DELIM. CHAR     |
| ZPL II.....                      | ZPL MODE        |
| CALIBRATION.....                 | MEDIA POWER UP  |
| CALIBRATION.....                 | HEAD CLOSE      |
| DEFAULT.....                     | BACKFEED        |
| +000.....                        | LABEL TOP       |
| +0000.....                       | LEFT POSITION   |
| NO.....                          | HEXDUMP         |
| 034.....                         | WEB S.          |
| 096.....                         | MEDIA S.        |
| 011.....                         | WEB GAIN        |
| 050.....                         | MARK S.         |
| 013.....                         | MARK GAIN       |
| 095.....                         | MARK MED S.     |
| 015.....                         | MARK MEDIA GAIN |
| 095.....                         | CONT MEDIA S.   |
| 007.....                         | CONT MEDIA GAIN |
| 075.....                         | RIBBON OUT      |
| 040.....                         | RIBBON GAIN     |
| 000.....                         | TAKE LABEL      |
| CWF.....                         | MODES ENABLED   |
| .....                            | MODES DISABLED  |
| 832 8/MM FULL.....               | RESOLUTION      |
| V56.15.1ZP32 <-.....             | FIRMWARE        |
| V06.00.0207.....                 | HARDWARE ID     |
| CUSTOMIZED.....                  | CONFIGURATION   |
| 2944k.....R:                     | RAM             |
| 1536k.....E:                     | ONBOARD FLASH   |
| NONE.....                        | FORMAT CONVERT  |
| 77,291 IN.....                   | LAST CLEANED    |
| 77,291 IN.....                   | HEAD USAGE      |
| 77,291 IN.....                   | TOTAL USAGE     |
| 77,291 IN.....                   | RESET CNTR1     |
| 77,291 IN.....                   | RESET CNTR2     |
| 31A07330008.....                 | SERIAL NUMBER   |
| 2007-09-20 22:48:18              | TIME STAMP      |

FIRMWARE IN THIS PRINTER IS COPYRIGHTED

## Pre-install Windows® Printer Drivers

Zebra is changing the way you install and use your printer with your Windows based PC systems. We recommend that at minimum, the ZebraDesigner™ Windows® driver be pre-installed to take advantage of the advancements in ease of use and simplicity of the Windows operating systems since the release of Windows XP® SP2.

Zebra provides the Zebra Setup Utilities (ZSU); a suite of Zebra® printer drivers, utilities, and communications and installation tools for use with most Windows PC operating systems. Zebra Setup Utilities and Zebra Windows printer drivers are available on the user's CD and the Zebra web site ([www.zebra.com](http://www.zebra.com)) for latest versions.

**ZebraDesigner™ Driver and Zebra Setup Utilities (includes the driver):** Supports Windows 7®, Windows Vista®, Windows XP®, Windows® 2000®, Windows Server® 2008, and Windows Server® 2003 operating systems. The driver supports 32 bit and 64 bit Windows operating systems and is Microsoft Certified. Zebra Setup Utilities and the ZebraDesigner driver support the following G-Series printer communication interfaces: USB, parallel, serial, Wired and Wireless Ethernet and Bluetooth (using a Bluetooth virtual printer port).

**Install the Zebra Setup Utilities before applying power to the printer** connected to the PC (running a Zebra driver supported Windows operating system). The Zebra Setup Utility will prompt you to apply the printer power. Continue to follow the instructions to complete your printer installation.

### Plug'N'Play (PnP) Printer Detection and Windows® operating systems

More recent Windows operating systems automatically detect the printer when it is connected via the USB interface. Depending on the hardware configuration and the Windows version, your printer may be Plug-and-Play (PnP) detected when connecting to the USB, parallel, or serial port interfaces. The printer drivers do not support serial port PnP installation at this time. The printer's PC interface configuration for the parallel port must support and have bidirectional communications for PnP operations.

The operating system automatically starts a "Add new hardware" wizard when connecting the printer for the first time to the PC. If you have pre-loaded the driver suite with the Zebra Setup Utility, then the printer driver will automatically install. Go to your Windows printer directory and right-click and select 'Properties'. Click on the 'Print test page' button to verify a successful installation.

The Windows operating system will detect and re-link a previously installed printer if it is reconnected to the USB interface or if printer power is turned on after the PC has finished its restart of the operating system. Ignore the new device detected warnings and close the Task bar prompts. Wait several seconds for the operating system to match the printer to the driver software. The warnings will quit and the printer now should be ready to begin printing.

### Universal Serial Bus (USB) Device Communications

The printer is a terminal device (not a host or hub) when using a USB interface. You can refer to the USB Specification for details regarding this interface.

**Note** • Scanners, scales or other data input (terminal) devices must use the serial port (not the USB port) to send data to the printer.

## Serial Port and Windows® operating systems

The Windows operating system default settings for the serial port communication closely match the printer's default settings with one exception; the data *Flow Control* settings. The Windows default data *Flow Control* setting is **NONE**. The G-Series printer requires data *Flow Control* set to **Hardware**.



**Note** • The G-Series™ printer does not support Windows® Serial Port Plug and Play (PnP) device detection at this time.

## Ethernet

This printer option has a variety of methods and utilities to assist with printer connection to a network (WAN or LAN) and to configure Wired and Wireless (WiFi) printers. The Zebra Setup Utility configuration wizards support creating a connection to the printer on a shared network with Windows based systems by using the printer's IP address. The printer includes internal Web pages to provide easy access to printer and network configuration. The web pages are accessible via the printers IP address using any web browser. The free version of ZebraNet™ Bridge software allows you to centrally deploy, manage, and monitor your Zebra® printers with automatic Zebra® printer discovery up to 3 printers from a single PC screen anywhere on your global network. ZebraNet™ Bridge Enterprise is available for purchase to manage larger numbers of Zebra® printers.

## Connecting your Printer to a Computer

The Zebra® G-Series™ printers support a variety of interface options and configurations. These include: Universal Serial Bus (USB) interface, RS232 Serial, Parallel (IEEE 1284.4) and 10/100 Ethernet.

- USB, Serial and Parallel
- Optional: USB, Serial, and Ethernet (Wired)
- Optional: USB, Serial, and Ethernet (Wireless-WiFi) with LCD display
- Optional: USB, Serial, and Bluetooth with LCD display

**The Zebra Setup Utility is designed to assist you with installing these interfaces.** The cabling and unique parameters for each of these physical printer communication interfaces is discussed in the following pages to assist you with making configuration setup choices prior to and immediately following applying power. The Zebra Setup Utilities configuration wizards will instruct you to turn the printer's power on at the appropriate time to complete the installation of your printer.

### For Connecting to Wireless Communication Interfaces:

- Wi-Fi (Ethernet), see the ZebraNet® 10/100 Internal Print Server manual.
- Bluetooth, see the [Bluetooth Option on page 67](#).



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**Caution** • Keep the power switch in the OFF position when attaching the interface cable. The power cord must be inserted into the power supply and the power receptacle on the back of the printer before connecting or disconnecting the communications cables.

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## Interface Cable Requirements

Data cables must be of fully shielded construction and fitted with metal or metallized connector shells. Shielded cables and connectors are required to prevent radiation and reception of electrical noise.

To minimize electrical noise pickup in the cable:

- Keep data cables as short as possible (6 foot [1.83 m] recommended).
- Do not tightly bundle the data cables with power cords.
- Do not tie the data cables to power wire conduits.

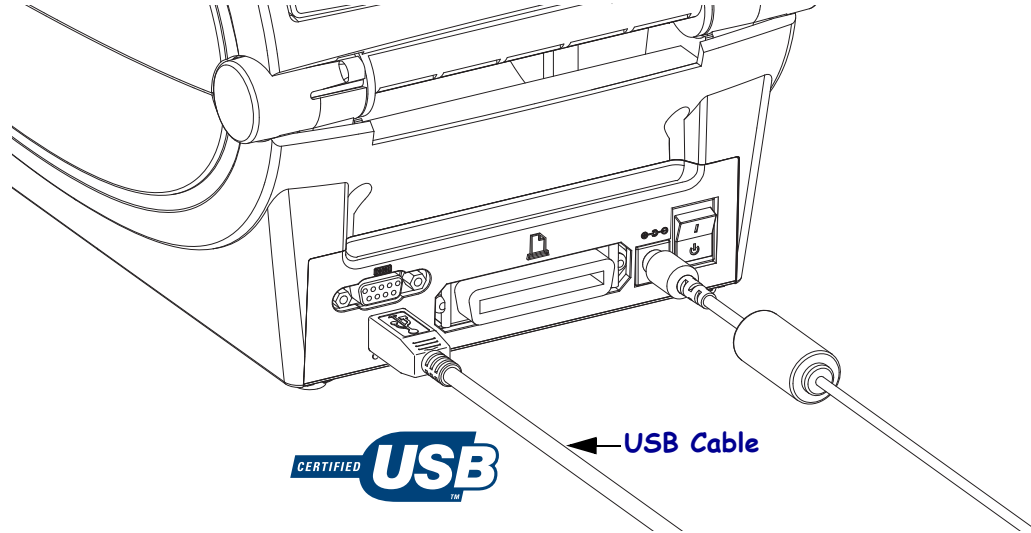


**Important** • This printer complies with FCC “Rules and Regulations,” Part 15, for Class B Equipment, using fully shielded data cables. Use of un-shielded cables may increase radiated emissions above the Class B limits.

## USB Interface Requirements

Universal Serial Bus (version 2.0 compliant) provides a fast interface that is compatible with your existing PC hardware. USB's "plug and play" design makes installation easy. Multiple printers can share a single USB port/hub.

When using a USB cable (not supplied with your printer), verify that the cable or cable packaging bears the "Certified USB™" mark (see below) to guarantee USB 2.0 compliance.



## Serial Communications

The printer automatically detects and switches the serial port to match common serial port cabling and signal connection configurations for DTE and DCE communications. The required cable must have a nine-pin “D” type (DB-9P) male connector on one end which plugs into the mating (DB-9S) serial port located on the back of the printer. The other end of this signal interface cable connects to a serial port on the host computer. This allows for the use of two common cable types and drop-in replacement for Zebra and other printer models. Zebra® printers use a Null Modem (cross-over) cable. Early models of Zebra® printers (a DCE device) running the EPL programming utilized a straight through (no cross-over) signal connections cable. For pinout information, refer to Appendix A.

The serial port communication settings between the printer and host (typically a PC) must match for reliable communication. The Bits per second (or Baud rate) and Flow control are the most common settings that get changed. The host (typically a Windows PC) needs to have the data Flow control changed to match the printer’s default communication method: Hardware and is noted by the Host Handshake setting **DTR/Xon/Xoff** for legacy printer. This combined hardware (DTR) and software (Xon/Xoff) mode may need to change depending upon the non-Zebra application software and the serial cable variation in use.

Serial communications between the printer and the host computer can be set by:

- Autobaud synchronization
- ZPL programming **^SC** command
- EPL programming **Y** command
- Resetting the printer to its default printer configuration.

### Autobaud

Autobaud synchronization allows the printer to automatically match the communication parameters of the host computer. To autobaud:

1. Press and hold the feed button until the green status LED flashes once, twice, and then three times.
2. While the status LED flashes, send the **^XA^XZ** command sequence to the printer.
3. When the printer and host are synchronized, the LED changes to solid green. (No labels will print during autobaud synchronization.)

### ZPL ^SC Command

Use the Set Communications (**^SC**) command to change the communications settings on the printer.

1. With the host computer set at the same communications settings as the printer, send the **^SC** command to change the printer to the desired settings.
2. Change the host computer settings to match the new printer settings.

Refer to the *ZPL Programming Guide* for more information about this command.



## EPL Y Command

Use the serial port setup (**Y**) command to change the communications settings on the printer.

1. With the host computer set at the same communications settings as the printer, send the **Y** command to change the printer to the desired settings. Note: the Y command does not support setting the data flow control, use the **Xon/Xoff** setting.
2. Change the host computer settings to match the new printer settings.

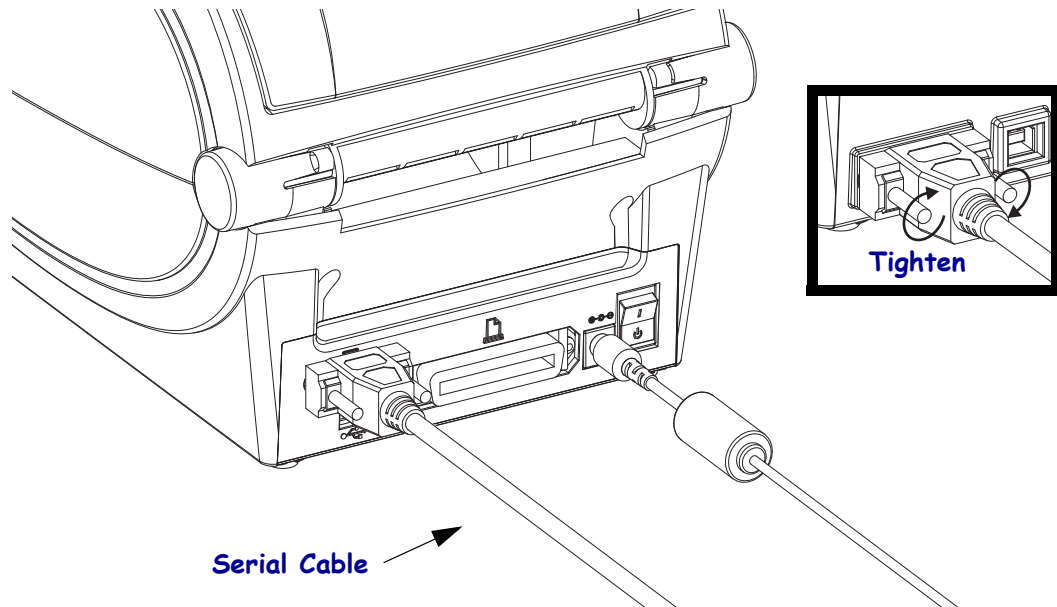
Refer to the *EPL Page Mode Programming Guide* for more information about this command.

## Resetting the Default Serial Port Parameters

Do the following to reset the communications parameters on the printer to the factory defaults (serial communication settings are: **9600** baud, **8** bit word length, **NO** parity, **1** stop bit, and **DTR/XON/XOFF** data flow control).

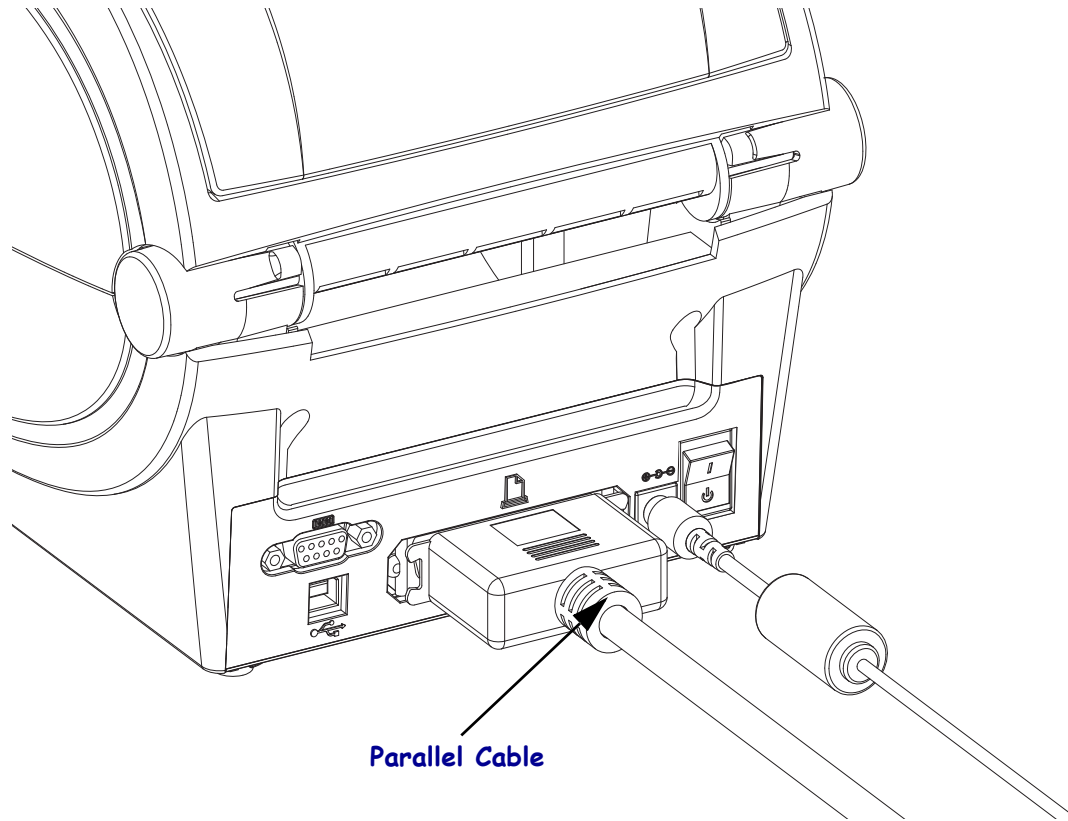
1. Press and hold the feed button until the green status LED flashes once, waits a moment and flashes twice, and then waits again before flashing three times (release immediately).
2. While the status LED rapidly flashes amber and green, press the feed button. Serial communications between the printer and the host computer can be set by the ZPL **^SC** command or the EPL **Y** command.

**Note** • Early models of Zebra® printers running the EPL programming language had **9600** baud, **NO** parity, **8** data bits, **1** stop bit and a **HARDWARE** and **SOFTWARE** (merged) data control (essentially DTR/Xon/Xoff) as the default serial port settings. The Windows operating system flow control setting was Hardware for most applications.



## Parallel Port

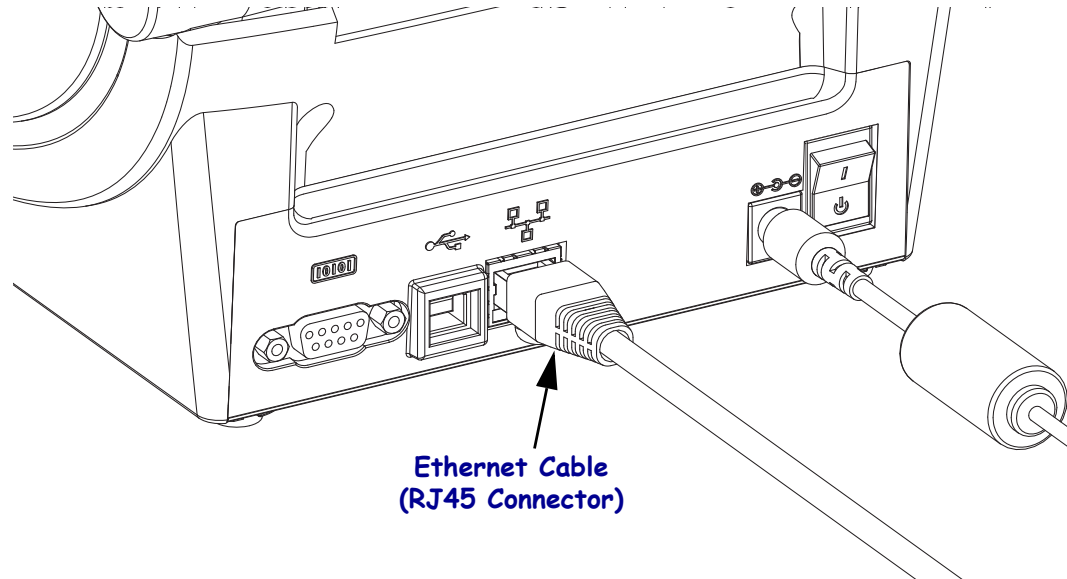
The required cable must have a 25 pin “D” type (DB-25P) male connector on one end and a Centronics on the end (IEEE 1284 A-B parallel interface specification). Early models of G-Series printers originally supported parallel cables with two 25 pin “D” type (DB-25P) male connectors on both ends (IEEE 1284 A-A parallel interface specification).



## Ethernet

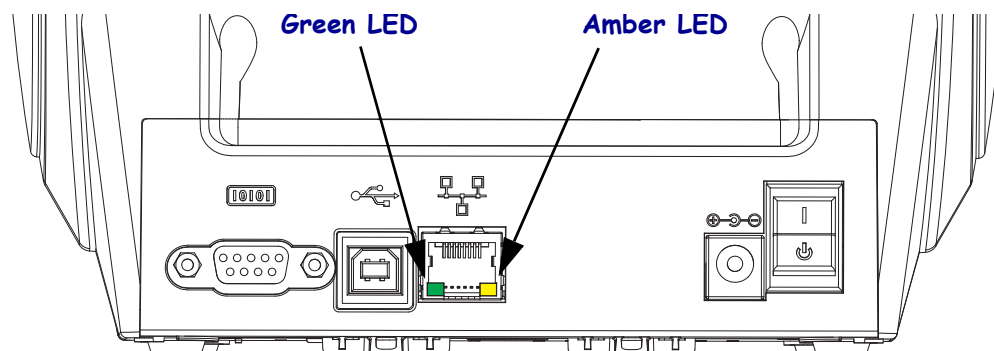
The printer requires UTP RJ45 Ethernet cable rated CAT-5 or better.

For more information on configuring your printer to run on a compatible Ethernet based network, see the ZebraNet® 10/100 Internal Print Server manual. The printer must be configured to run on your LAN (local area network) or WAN (wide area network). The print server on board your printer can be accessed through the printer's Web pages.



### Ethernet Status/Activity Indicators

| LED Status                                        | Description                                  |
|---------------------------------------------------|----------------------------------------------|
| <b>Both OFF</b>                                   | No Ethernet link detected                    |
| <b>Green</b>                                      | 100 Mbps link detected                       |
| <b>Green with the Amber flickering on and off</b> | 100 Mbps link and Ethernet activity detected |
| <b>Amber</b>                                      | 10 Mbps link detected                        |
| <b>Amber with the Green flickering on and off</b> | 10 Mbps link and Ethernet activity detected  |



## After Your Printer is Connected

Now that you have basic communication to your printer, you may want to test printer communications and then install other printer related applications, drivers or utilities.

### Testing Communications by Printing

Verifying the print system's operation is a relatively simple process. For Windows operating systems, the Zebra Setup Utility or the Windows 'Printers and Faxes' Control Panel to access and print a test label. For non-Windows operation systems, coping a basic ASCII text file with a single command (**~WC**) to print a configuration status label.

#### Test Printing with Zebra Setup Utility:

1. Open the Zebra Setup Utility.
2. Click on the newly install printer's icon to select the printer and active the printer configuration buttons below it in the window.
3. Click the 'Open Printer Tools' button.
4. In the 'Print' tab window, click on the 'Print configuration label' line and click on the 'Send' button. The printer should print a configuration status label.

#### Test Printing with Windows 'Printer and Faxes' menu:

1. Click on the Windows 'Start' menu button, to access the 'Printers and Faxes' menu or 'Control Panel' to access the 'Printers and Faxes' menu. Open the menu.
2. Select the newly install printer's icon to select the printer and right-click the mouse to access the printer 'Properties' menu.
3. From the printer's 'General' tab window, click on the 'Print Test Page' button. The printer should print a Windows test print page.

#### Test Print on a Ethernet printer connected to a network (LAN or WAN) with a (MS-DOS) 'Command Prompt' (or 'Run' from the Windows XP Start menu):

1. Create a text file with the following ASCII three characters: **~WC**
2. Save the file as: TEST.ZPL (an arbitrary file name and extension name).
3. Read the IP address off the printer's Configuration Status Label's Network Status printout. On a system that is connected to the same LAN or WAN as the printer, type the following in the Web browser window's address bar and enter:  
**fpt (IP address)**  
*(for IP address 123.45.67.01 it would be: ftp 123.45.67.01)*
4. Type the word 'put' followed by the file name and enter. For this 'test print' file, it would be: **put TEST.ZPL**  
The printer should print a new Print Configuration Status label.

**Test Print with a Copied ZPL Command File for Non-Windows Operating Systems:**

1. Create a text file with the following ASCII three characters: **~WC**
2. Save the file as: TEST.ZPL (an arbitrary file name and extension name).
3. Copy the file to the printer. For DOS, a file sent to a printer connected to the system's parallel port would be as simple as:

**COPY TEST.ZPL LPT1**

Other interface connection types and operating systems will have different command strings. See your operating system documentation for detailed instructions to copy to the appropriate printer interface for this test.



Notes • \_\_\_\_\_

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# Print Operations

This section provides media and print handling, font and language support, and the setup of less common printer configurations.

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## Determining Printer Configuration

The G-Series printer uses a ZPL printer configuration status label to report the printer's configuration status for both EPL and ZPL operations. The ZPL style label provides a more intuitive and functionally descriptive naming conventions than the EPL style printer status label. Operational status (darkness, speed, media type, etc.), installed printer options (network, interface settings, cutter, etc.) and printer description information (serial number, model name, firmware version, etc.) are all included on the status label. See [Printing a Test \(Printer Configuration\) Label on page 23](#) to print this label. See [Appendix: ZPL Configuration on page 119](#) for detail on printer configuration and the ZPL commands that control the printer settings listed on the Printer Configuration Status Label.

To get an EPL style printer configuration status label, send the printer the EPL **U** command. See the EPL programmer's guide for more information on the various EPL **U** commands and interpreting the settings displayed on these labels.

## Localizing the Printer Status Configuration Label

The printer configuration status label can be localized for up to 16 languages. Use the ZPL programming command **^KD** to modify the displayed language for most status items on this label.

See [Printing a Test \(Printer Configuration\) Label on page 23](#) or [Feed Button Modes on page 106](#) for information on accessing the printer configuration status label.

## Long Term Printer In-activity or Storage

Overtime the printhead may stick to the platen (drive) roller. To prevent this, always store the printer with a piece of media (a label or paper) between the printhead and platen roller. Do not ship the printer with a roll of media installed or damage to the printer or media may result.

## Thermal Printing



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**Caution** • The printhead becomes hot while printing. To protect from damaging the printhead and risk of personal injury, avoid touching the printhead. Only use the cleaning pen to perform printhead maintenance.

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**Caution** • The discharge of electrostatic energy that accumulates on the surface of the human body or other surfaces can damage or destroy the printhead or electronic components used in this device. You must observe static-safe procedures when working with the printhead or the electronic components under the top cover.

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## Modes of Printing

You can operate this printer in many different modes and media configurations:

- Direct thermal printing (which uses heat sensitive media to print).
- Thermal transfer printing (which uses ribbon to heat transfer print to the media).
- Standard Tear-Off mode allows you to tear off each label (or batch print a strip of labels) after it is printed.
- Label Dispense Mode: If an optional dispenser is installed, the backing material can be peeled away from the label as it is printed. After this label is removed, the next one is printed.
- Media Cutting: If an optional media cutter is installed, the printer can cut the label liner between labels, receipt paper or tag stock depending upon the cutter option purchased.
- Stand-alone: The printer can print without being connected to a computer using the printer's auto running label form feature (programming based) or by using a data input device connected to the printer's serial port. This mode accommodates data input devices, such as scanners, weigh scales, Zebra® KDU Plus™ or KDU (Keyboard Display Unit) with a KDU Adapter, etc.
- Shared network printing: Printers configured with the Ethernet interface option include an internal print server with ZebraLink™ printer configuration web page and ZebraNet™ Bridge software for managing and monitoring status of Zebra® printers on a network.

## Print Media Types



**Important** • Zebra strongly recommends the use of Zebra-brand supplies for continuous high-quality printing. A wide range of paper, polypropylene, polyester, and vinyl stock has been specifically engineered to enhance the printing capabilities of the printer and to prevent premature printhead wear. To purchase supplies, go to <http://www.zebra.com/howtobuy>.

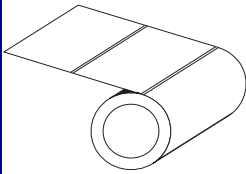
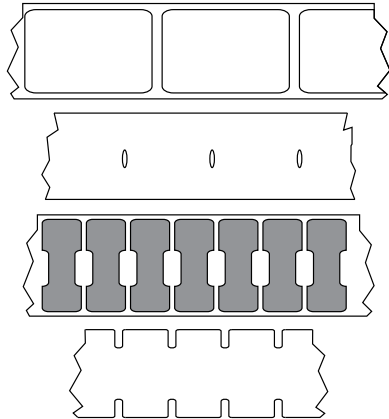
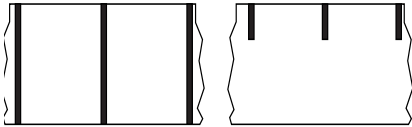
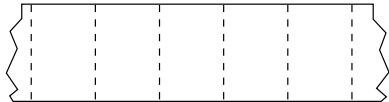
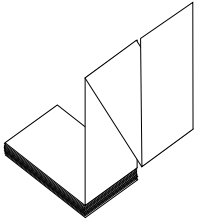
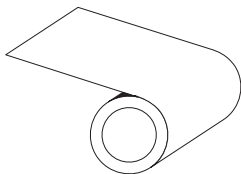
Your printer can use various types of media:

- *Standard media*—Most standard (non-continuous) media uses an adhesive backing that sticks individual labels or a continuous length of labels to a liner.
- *Continuous roll media*—Most continuous roll media is direct thermal media (similar to FAX paper) and is used for receipt or ticket style printing.
- *Tag stock*—Tags are usually made from a heavy paper (up to 0.0075in./0.19mm thick). Tag stock does not have adhesive or a liner, and it is typically perforated between tags.

For more information on basic media types, see [Table 1](#).

The printer typically uses roll media, but you can use fan-fold or other continuous media too. Use the correct media for the type of printing you require. You must use direct thermal media. When printing without a ribbon, you must use direct thermal media. When using ribbon, you must use thermal transfer media.

**Table 1 • Types of Media Roll and Fanfold Media**

| Media Type                          | How It Looks                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Non-Continuous Roll Media</b>    |    | <p>Roll media is wound on a core that can be 0.5 to 1.5 in. (12.7 to 38.1 mm) in diameter. Labels have adhesive backing that sticks them to a liner, and they are separated by gaps, holes, notches, or black marks. Tags are separated by perforations. Individual labels are separated by one or more of the following methods:</p> <ul style="list-style-type: none"> <li>• <i>Web media</i> separates labels by gaps, holes, or notches.</li> </ul>  <ul style="list-style-type: none"> <li>• <i>Black mark media</i> uses pre-printed black marks on the back side of the media to indicate label separations.</li> </ul>  <ul style="list-style-type: none"> <li>• <i>Perforated media</i> has perforations that allow the labels or tags to be separated from each other easily. The media may also have black marks or other separations between labels or tags.</li> </ul>  |
| <b>Non-Continuous Fanfold Media</b> |  | <p>Fan-Fold media is folded in a zigzag pattern. Fanfold media can have the same label separations as non-continuous roll media. The separations would fall on or near the folds.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Continuous Roll Media</b>        |  | <p>Roll media is wound on a core that can be 0.5 to 1.5 in. (12.7 to 38.1 mm) in diameter. Continuous roll media does not have gaps, holes, notches, or black marks to indicate label separations. This allows the image to be printed anywhere on the label. A cutter can be used to cut apart individual labels. With continuous media, use the transmissive (gap) sensor so the printer can detect when the media runs out.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Determining Thermal Media Types

Thermal transfer media requires ribbon for printing while direct thermal media does not. To determine if ribbon must be used with a particular media, perform a media scratch test.

### To perform a media scratch test, complete these steps:

1. Scratch the print surface of the media with a finger nail or pen cap. Press firmly and quickly while dragging it across the media surface. Direct thermal media is chemically treated to print (expose) when heat is applied. This test method uses friction heat to expose the media.
2. Did a black mark appear on the media?

| If a black mark...           | Then the media is...                           |
|------------------------------|------------------------------------------------|
| Does not appear on the media | <b>Thermal transfer.</b> A ribbon is required. |
| Appears on the media         | <b>Direct thermal.</b> No ribbon is required.  |

## Replacing Supplies

If labels or ribbon run out while printing, leave the printer power on while reloading (data loss results if you turn off the printer). After you load a new label or ribbon roll, press the Feed button to restart.

Always use high quality, approved labels, tags and ribbons. If adhesive backed labels are used that don't lay flat on the backing liner, the exposed edges may stick to the label guides and rollers inside the printer, causing the label to peel off from the liner and jam the printer. Permanent damage to the printhead may result if a non-approved ribbon is used as it may be wound incorrectly for the printer or contain chemicals corrosive to the printhead.

## Adding a New Transfer Ribbon

If ribbon runs out in the middle of a print job, the indicator lights red and the printer waits for you to add a fresh roll.

1. Keep the power on as you change ribbon.
2. Open the top cover, then cut the used ribbon so you can remove the cores.
3. Load a new ribbon roll and empty ribbon core. If necessary, review the Ribbon Loading steps. *Note: Do Not Use ribbon cores with damaged notch. The notches should have square corners.*
4. Close the top cover.
5. Press the Feed button to restart printing.

## Replacing a Partially Used Transfer Ribbon

To remove used transfer ribbon, perform the following steps.

1. Cut the ribbon from the take-up roll.
2. Remove the take-up roll and discard used ribbon.
3. Remove the supply roll and tape the end of any fresh ribbon to prevent it from unwrapping. When reinstalling a partially used supply roll, tape the cut end onto the empty take-up roll.

## Adjusting the Print Width

Print width must be set when:

- You are using the printer for the first time.
- There is a change in media width.

Print width may be set by:

- The Windows printer driver or application software such as ZebraDesigner™.
- The five-flash sequence in [Feed Button Modes on page 106](#).
- Controlling printer operations with ZPL programming; refer to the Print Width (^PW) command (consult your *ZPL Programming Guide*).
- Controlling printer operations with EPL Page Mode programming, refer to the Set Label Width (q) command (consult your *EPL Programmer's Guide*).

## Adjusting the Print Quality

Print quality is influenced by the heat (density) setting of the printhead, the print speed, and the media in use. Experiment with these settings to find the optimal mix for your application. Print quality can be set with the Zebra Setup Utility's 'Configure Print Quality' routine.



**Note** • Media manufactures may have specific recommendations for speed settings for your printer and the media. Some media types have lower maximum speeds than your printer's maximum speed.

The relative darkness (or density) setting can be controlled by:

- The six-flash sequence in [Feed Button Modes on page 106](#). This will overwrite any ZPL and EPL programmed darkness/density settings.
- The Set Darkness (~SD) ZPL command (consult your *ZPL Programming Guide*).
- The Density (D) EPL command (consult your *EPL Programmer's Guide*).

If you find that the print speed needs to be adjusted, use:

- The Windows printer driver or application software such as ZebraDesigner™.
- The Print Rate (^PR) command (consult your *ZPL Programming Guide*).
- The Speed Select (S) command (consult your *EPL Programmer's Guide*).

## Media Sensing

The G-Series printer has automatic media sensing capability. The printer is designed to continuously check and adjust media length sensing for minor variations. Once the printer is printing or feeding media, the printer continually checks and adjusts the media sensing to accommodate for minor changes in media parameters from label to label on a roll and from roll to roll of media. The printer will automatically initiate a media length calibration if the expected media length or the label to label gap distance has exceeded the acceptable variation range when starting a print job or feeding media. The automatic media sensing in the G-Series printers works the same for printer operations that use EPL and ZPL label formats and programming.

If the printer does not detect labels or black marks (or notches with blackline sensing) after feeding the media the default maximum label length distance of 39 inches (1 meter), then the printer will switch to continuous (receipt) media mode. The printer will keep these settings until changed by software, programming or a manual calibration with different media.

Optionally, the printer can be set to do a short media calibration after printer power up or when closing the printer with power on. The printer will then feed several labels while calibrating.

The printer's media settings can be verified by printing a Printer Configuration label. See the [Printing a Test \(Printer Configuration\) Label on page 23](#) for more details.

The maximum distance that the automatic media type detection and sensing will check can be reduced by using the ZPL Maximum Label Length command (**^ML**). It is recommended that this distance be set to no less than two times the longest label being printed. If the largest label being printed was a 4 by 6 inch label, then the maximum label (media) length detection distance can be reduced from the default distance of 39 inches down to 12 inches.

If the printer has difficulty automatically detecting the media type and auto-calibrating, see [Manual Calibration on page 102](#) to perform an extensive calibration. It includes a printed graph of sensor operation for your media. This method disables the printer's automatic media sensing capability until the printer's default parameters are reset to the factory defaults with the four flash Feed button mode. See the [Feed Button Modes on page 106](#) for more details.

The automatic media calibration can be modified, turned on or turned off to meet your needs. Sometimes print job conditions require that the printer use all the media on a roll. The two automatic media conditions, power up with media loaded and closing the printer with power on, can be controlled individually with the ZPL Media Feed command, **^MF**. The feed action discussed in the ZPL programmers guide for the **^MF** command is primarily for automatic media sensing and calibration. The automatic media calibration that controls the dynamic media (label to label) calibration is the **^XS** command. If multiple media types of different lengths, material or detection methods (web/gap, black mark, notches or continuous) are used, you should not change these settings.

The media calibration and detection process can also be refined to match the media type loaded into the printer. Use the ZPL Media Tracking command (**^MN**) to set the media type. Sometimes the printer can detect preprinted media as the gap between labels or the liner backing with print as a black marks. If the **^MN** parameter for continuous media is set, then the print does not preform the automatic calibration. The **^MN** command also includes an automatic calibration parameter (**^MNA**) to return the printer to its default setting to automatically detect all media types.

## Ribbon Overview

Ribbon is a thin film that is coated on one side with wax, resin, or wax resin, which is transferred to the media during the thermal transfer process. The media determines whether you need to use ribbon and how wide the ribbon must be. When ribbon is used, it must be as wide as or wider than the media being used. If the ribbon is narrower than the media, areas of the printhead are unprotected and subject to premature wear.

### When to Use Ribbon

Thermal transfer media requires ribbon for printing while direct thermal media does not. Direct thermal media should never be used with ribbon. Bar codes and graphics can get distorted. To determine if ribbon must be used with a particular media, perform a media scratch test.

### Coated Side of Ribbon

Ribbon can be wound with the coated side on the inside or outside. This printer can only use ribbon that is coated on the outside. If you are unsure which side of a particular roll of ribbon is coated, perform an adhesive test or a ribbon scratch test to determine which side is coated.



**To determine which side of a ribbon is coated, complete these steps:**

### Ribbon Test with Adhesive

If you have labels available, perform the adhesive test to determine which side of a ribbon is coated. This method works well for ribbon that is already installed.

**To perform an adhesive test, complete these steps:**

1. Peel a label from its liner.
2. Press a corner of the sticky side of the label to the outer surface of the roll of ribbon.
3. Peel the label off of the ribbon.

4. Observe the results. Did flakes or particles of ink from the ribbon adhere to the label?

| If ink from the ribbon...   | Then...                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------|
| Adhered to the label        | The ribbon is coated on the <b>outside</b> and can be used with the G-Series printer.  |
| Did not adhere to the label | The ribbon is coated on the <b>inside</b> and can not be used in the G-Series printer. |

## Ribbon Scratch Test

Perform the ribbon scratch test when labels are unavailable.

### To perform a ribbon scratch test, complete these steps:

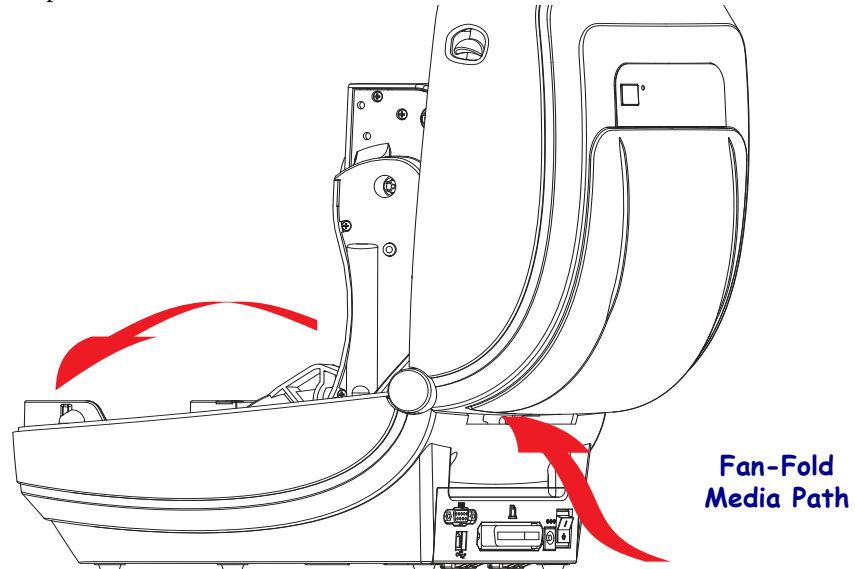
1. Unroll a short length of ribbon.
2. Place the unrolled section of ribbon on a piece of paper with the outer surface of the ribbon in contact with the paper.
3. Scratch the inner surface of the unrolled ribbon with your fingernail.
4. Lift the ribbon from the paper.
5. Observe the results. Did the ribbon leave a mark on the paper?

| If the ribbon...                  | Then...                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------|
| Left a mark on the paper          | The ribbon is coated on the <b>outside</b> and can be used with the G-Series printer.  |
| Did not leave a mark on the paper | The ribbon is coated on the <b>inside</b> and can not be used in the G-Series printer. |

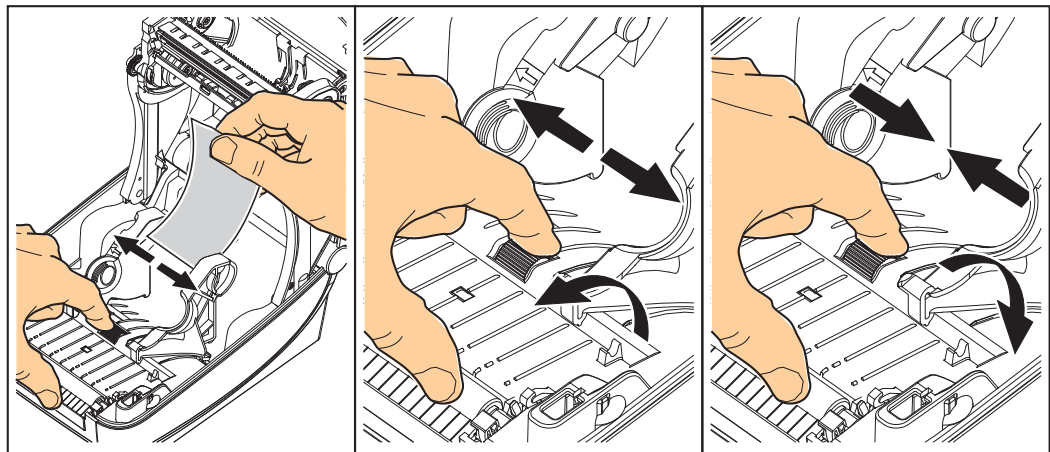
## Printing on Fan-Fold Media

Printing on fan-fold media requires you to adjust the media guide's stop position.

1. Open the top cover.

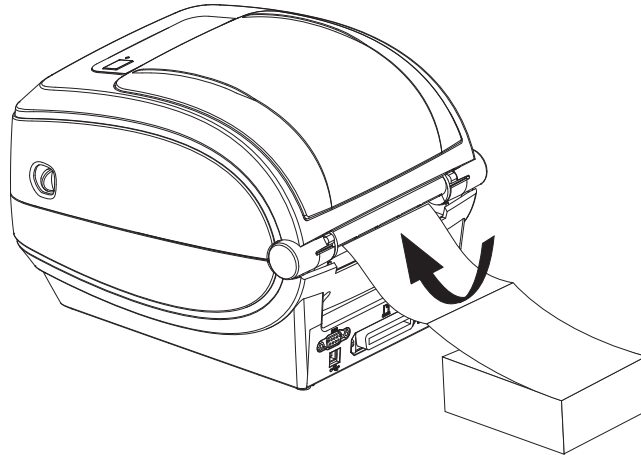


2. Adjust the media guide stop position with the green thumb wheel. Use a piece of the fan-fold media to set the stop position. Rotate the wheel toward you to open the guides wider. Rotate the wheel away from you to make the media guide narrower.

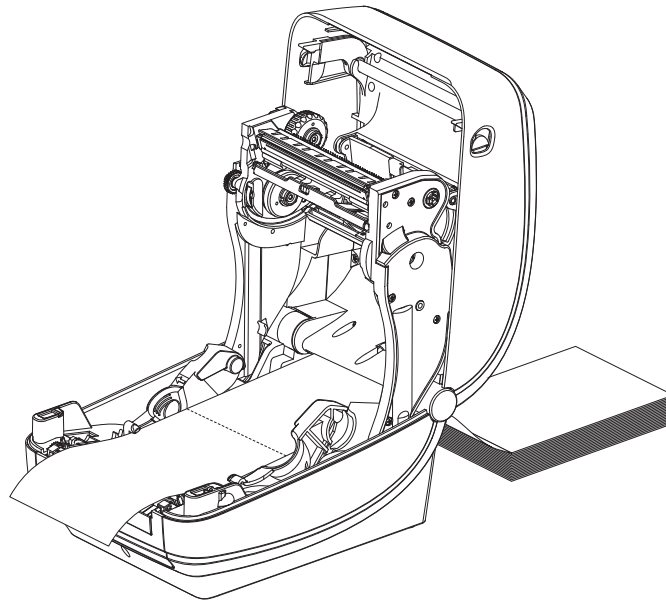




3. Insert the media through the slot at the rear of the printer.



4. Run the media between the media guide and roll holders.



5. Close the top cover.

**After printing or feeding several labels:** If the media does not track down the center (moves from side to side) or the sides of the media (liner, tag, paper, etc.) are frayed or damaged when exiting the printer, then the media guides may need further adjustment. If this does not correct the problem, then the media can be routed over the two roll holding pins on the media guide. An empty roll core of the same width as the stack of the fan-fold media can be placed between the roll holders to provide additional support for thin media.

## Printing with Externally Mounted Roll Media

The G-Series printer accommodates externally mounted roll media similar to the printer's support of fan-fold media. The printer requires the media roll and stand combination to have a low initial inertia to pull the media off the roll. The G-Series™ motor has 30% more torque than the Zebra® 2800™ Series desktop printers to provide improved media handling and speed.

Zebra at this time does not offer an external media option for the G-Series™ printer.

### Externally Mounted Roll Media Considerations:

- The media ideally should enter the printer directly behind the printer through the fan-fold media slot in the rear of the printer. See [Printing on Fan-Fold Media on page 44](#) for media loading.
- Lower the print speed to decrease the chance of motor stalls. The roll typically has the highest inertia when trying to start the roll moving. Larger media roll diameters require the printer to have more torque applied to get the roll moving.
- The media should move smoothly and freely. The media should not slip, skip, jerk, bind and then move, etc. when mounted on your media stand.
- The printer should not touch the media roll.
- The printer should not slip or lift up off the operating surface.

## Fonts and Your Printer

The G-Series™ printer supports your language and font requirements with a variety of internal fonts, on-board font scaling, international font sets and character code page support, Unicode support, and font download.

The G-Series™ printer's font capabilities are programming language dependent. The EPL™ programming language provides for basic bitmapped fonts and international code pages. The ZPL™ programming language provides advanced font mapping and scaling technology to support outline fonts (TrueType™ or OpenType™) and Unicode character mapping as well as basic bitmapped fonts and character code pages. ZPL and EPL programming guides describe and document the fonts, code pages, character access, listing fonts, and limitations for their respective printer programming languages. See the printer programming guides for information on Text, Fonts and Character support.

The G-Series printer includes utilities and application software that support font download into the printer for both printer programming languages.

## Identifying Fonts in Your Printer

Fonts and memory are shared by the programming languages in the printer. Fonts can be loaded in many memory areas in the G-Series printer. ZPL programming can recognize EPL and ZPL fonts. EPL programming can only recognize EPL fonts. See the respective programmer guides for more information on fonts and printer memory.

### ZPL Fonts:

- To manage and download fonts for ZPL print operation, use the Zebra Setup utility or ZebraNet™ Bridge.
- To display all of the fonts loaded in your G-Series printer, send the printer the ZPL command **^WD**. See the *ZPL Programmers Guide* for details.
  - Bitmap fonts in the various printer memory areas are identified by the **.FNT** file extension in ZPL.
  - Scalable fonts are identified with the **.TTF**, **.TTE** or **.OTF** file extensions in ZPL. EPL does not support these fonts.

### EPL Fonts:

- To download fonts for EPL print operations, use the Zebra Setup Utility or ZebraNet™ Bridge to send files to the printer.
- To display Soft Fonts (ext.) available to EPL, send the printer the EPL command **EI**.
  - Optional Asian Fonts in G-Series printers are displayed as Soft Fonts, but are still accessible as documented in the **A** command in the EPL programmer's guide.
  - All EPL fonts displayed are bitmap fonts. They do not include the **.FNT** file extension or the horizontal (**H**) or vertical (**V**) designators that are displayed with the ZPL command **^WD** as described above in ZPL Fonts.
- To remove non Asian EPL fonts with EPL programming, use the **EK** command.
- To remove EPL Asian fonts from the printer, use the ZPL **^ID** command.

## Localizing the Printer with Code Pages

The G-Series printer supports two sets of language, region and character sets for permanent fonts loaded into the printer for each printer programming language, ZPL and EPL. The printer supports localization with common international character map code pages.

- For ZPL code page support, including Unicode, see the **^CI** command in the ZPL programmer's guide.
- For EPL code page support, see the **I** command in the EPL programmer's guide.

## Asian fonts and Other Large Font Sets

Asian language pictographic fonts have large character sets with thousands of characters that support single language code page. In order to support the large Asian character sets, the industry adopted a double-byte (67840 maximum) character system instead of the single-byte characters (256 maximum) used by Latin based language characters to address large font sets. In order to address multiple languages with a single font set, Unicode was invented. A Unicode font supports one or more code points (relate these to code page character maps) and is accessed in a standard method that resolves character mapping conflicts. The ZPL programming language supports Unicode; EPL does not. Both of the printer's programming languages support the large pictographic double-byte character Asian font sets.

Asian language support requires additional memory support by the G-Series printer configured with the RTC and Full Flash Memory factory option. The number of fonts that can be downloaded is dependent upon the amount of available flash memory not already in use and the size of the font to be downloaded.

Some large Unicode fonts are too large to be installed in the early model G-Series printers (12MB Full Flash Memory prior to July 24, 2009) with the RTC and Full Flash Memory factory option such as MS (Microsoft) Arial Unicode font (23MB) available from Microsoft or the Andale font (22MB) offered by Zebra. These large font sets typically support a large number of languages as well.

## Getting Asian Fonts

Asian bitmap font sets are downloaded into the printer by the user or integrator. The ZPL fonts are purchased separately from the printer. EPL Asian Fonts are available for free to download from the Zebra Web site.

- Simplified and Traditional Chinese (ZPL and EPL)
- Japanese — JIS and Shift-JIS mappings (ZPL and EPL)
- Korean (ZPL including Johab and EPL)
- Thai (ZPL only)

## Stand Alone Printing

Your printer can be configured to work without being attached to a computer. The printer has the ability to automatically run a single label form. One or more downloaded label forms can be accessed and run with a terminal or wedge device or a Zebra® KDU (keyboard display unit) to call a label form. These methods allow the developer to incorporate data input devices, such as scanners or weight scales, to the printer via the serial port.

Label formats can be developed and stored in the printer to support labels that have:

- No data entry required and print when the Feed button is pressed.
- No data entry required and print when a label is removed from the printer's optional label dispenser.
- One or more data variables to be entered via the terminal or wedge device. The label will print after the last variable data field has been input.
- One or more label formats that are called by scanning bar codes containing programming to run a label form.
- Label forms designed to work as a process chain with each label including a bar code containing the programming to run the next label in the process sequence.

Both printer programming languages support special label forms that will automatically run after a power cycle or reset. The ZPL looks for a file named **AUTOEXEC.ZPL** and EPL looks for a label form named **AUTOFR**. If both files are loaded in the G-Series printer, then only the AUTOEXEC.ZPL will run. The EPL AUTOFR form runs until it is disabled. Both files must be deleted from the printer followed by a reset or a power cycle to completely remove the files.



Note • The G-Series EPL **AUTOFR** command can only be disabled with **NULL** character (00 hex or ASCII 0). The G-Series™ printer will ignore the other character that normally disables the AUTOFR form operation in most other EPL printers, the XOFF character (13 hex or ASCII 19).

The G-Series™ printer can source up to 750mA through the serial port's 5 volt line; see Appendix A for more information on the printer's serial port interface.

## Sending Files to the Printer

Graphics, fonts and programming files can be sent to the printer from the Microsoft Windows operating systems by using the Zebra Setup Utilities (and driver), ZebraNet™ Bridge or Zebra® ZDownloader found on the user's CD or at [www.zebra.com](http://www.zebra.com). These methods are common for both programming languages and the G-Series printers.

## Print Meter

The G-Series printer has the capability to report printhead maintenance alerts. The printer can provide cleaning and early warning of end of calculated printhead life alerts. If the RTC (Real Time Clock) is installed in the printer, the printhead life and history reporting will also include the date. By default, print metering alerts are disabled.

Many of the print meter messages and reports are customize able. See the *ZPL or EPL Programming Guides* for more information on the Print Meter.

To active the Print Meter Alerts, send one of the following commands to the printer:

- EPL command **oLY**
- ZPL command **^JH,,,,,E**
- by printing out a printer configuration.



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# Printer Options

This section covers the common printer options and accessories with brief descriptions and how to get you started using or configuring your printer option or accessory.

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## Label Dispenser Option

The factory installed label dispenser option allows you to print a label with backing (liner/web) being removed from the label as it prints, ready for application. When printing multiple labels, removing the dispensed (peeled) label notifies the printer to print and dispense the next label.

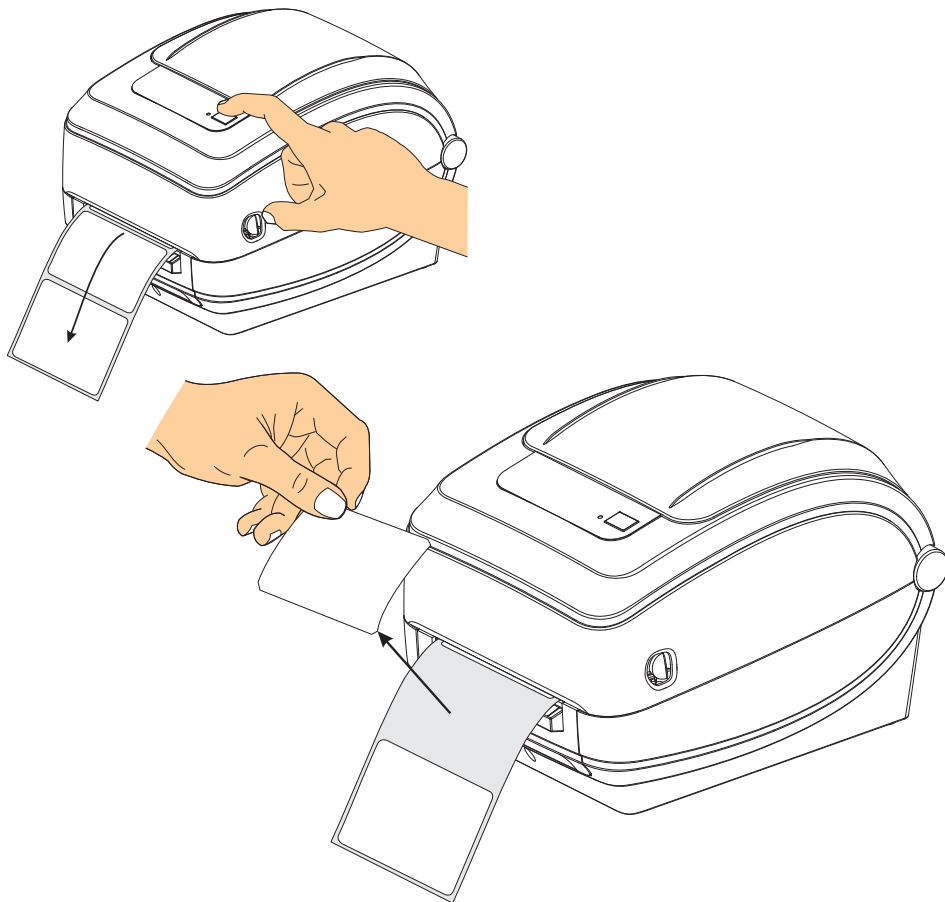
To use dispenser-mode correctly, use your printer driver to activate the label (taken) sensor along with these typical label settings which include but are not limited to length, non-continuous (gap), and web (liner). Otherwise, you must send the ZPL or EPL programming commands to the printer.

**When programming in ZPL**, you can use the following command sequences shown below and refer to your *ZPL Programming Guide* for more information on ZPL programming.

```
^XA ^MMP ^XZ  
^XA ^JUS ^XZ
```

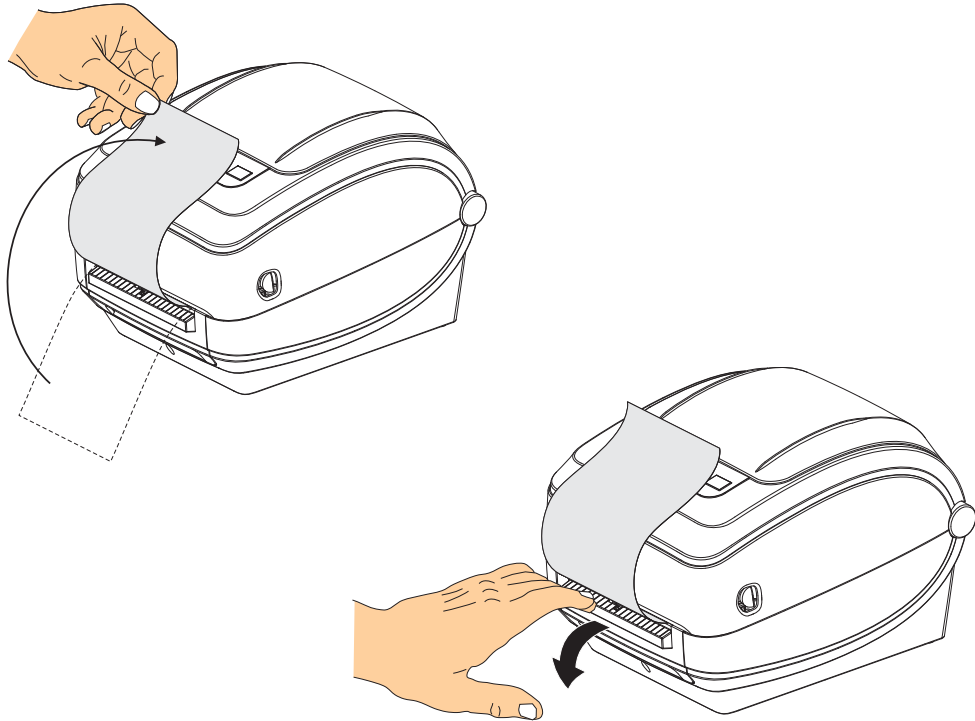
**When programming in EPL**, send the Options (**O**) command with the 'P' command parameter (**OP**) to enable the Label Taken sensor. Other printer option parameters may also be included with the Options command string. Refer to your *EPL Programmer's Guide* for more information on programming with EPL and the Options (**O**) command behaviors.

1. Load your labels into the printer. Close the printer and press the Feed button until a minimum of 4 inches or 100 millimeters of exposed labels exit the printer. Remove the exposed labels from the liner.

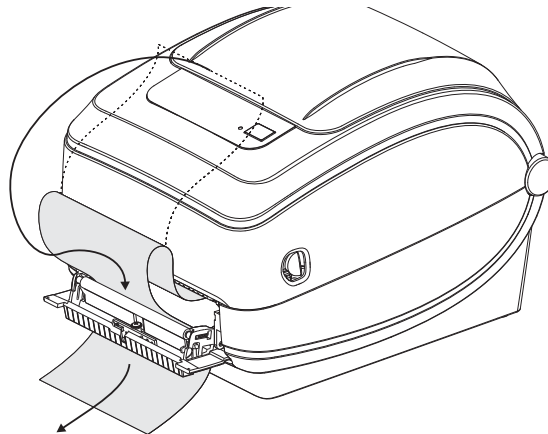




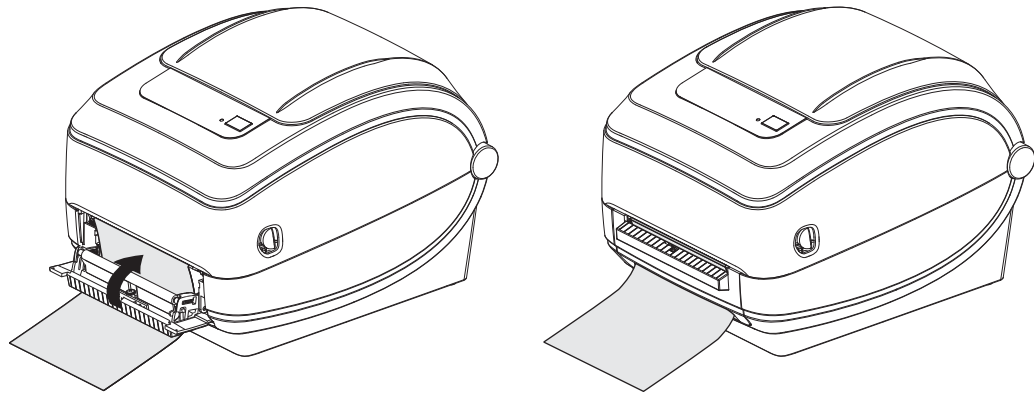
2. Lift the liner over the top of the printer and open the dispenser door.



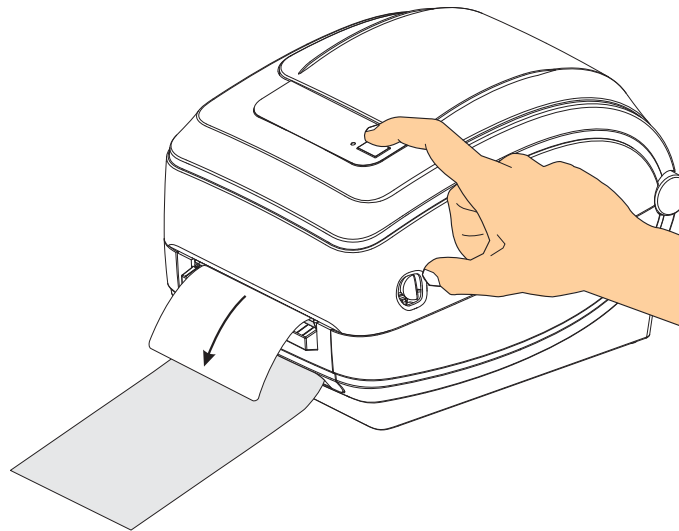
3. Insert the label liner between the dispenser door and the printer body.



4. Close the dispenser door.



5. Press the Feed button to advance the media.



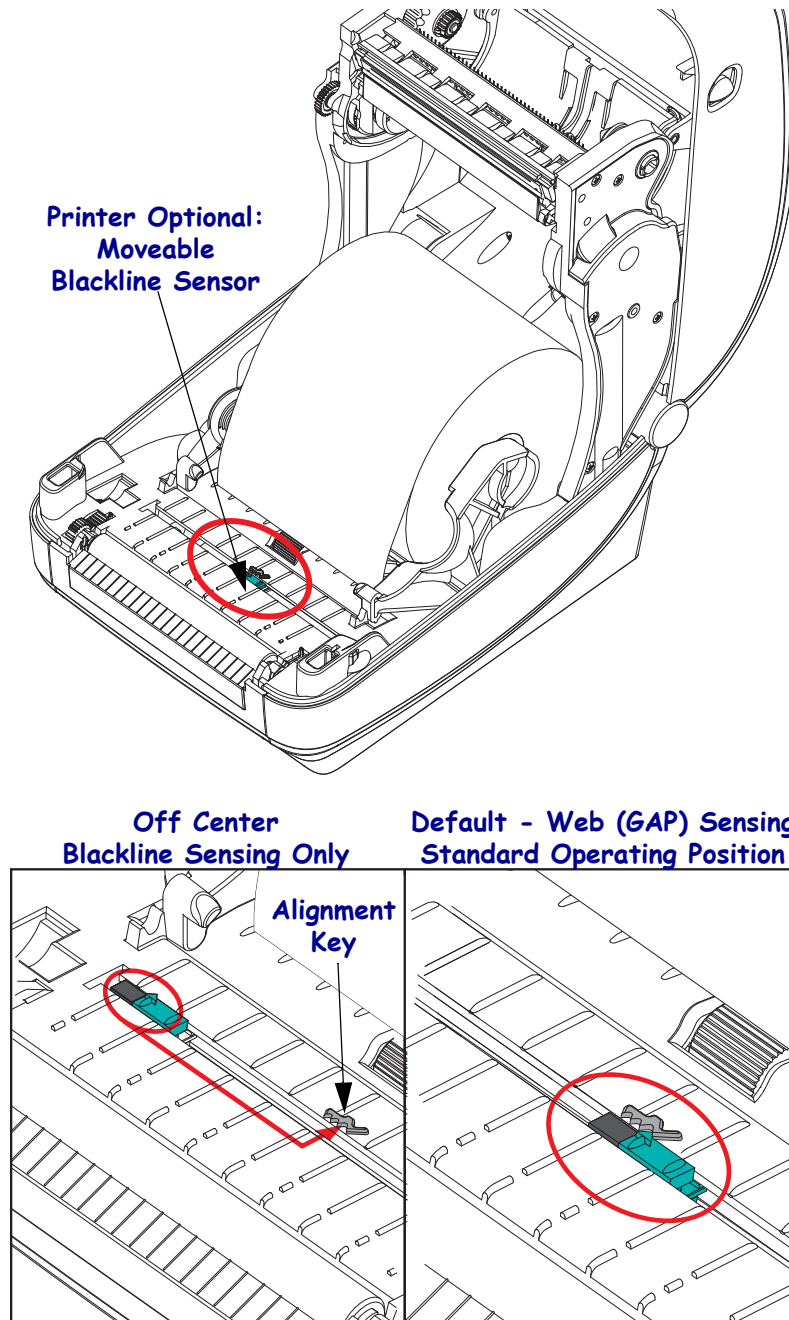
6. During the print job, the printer will peel off the backing and present a single label. Take the label from the printer to allow the printer to print the next label. Note: If you did not activate the label taken sensor to detect the removal of the dispensed (peeled) label with software commands, the printer will stack and eject peeled labels.

## Moveable Blackline Sensor Option

The factory installed moveable blackline sensor option allows the printer to use media with black marks or notches (holes) on the back side of the media (or media liner) that are not located in the center of the media.

The moveable sensor's second functional feature is adjustable transmissive web (gap) sensing to positions that match legacy Zebra® desktop printer sensor locations and in between. This provides for use of some non standard media variations or irregularly shaped media.

The moveable blackline sensor can not properly sense continuous media or labels (for inter-label gap sensing) if the sensor is not in the default sensing area for web (gap) sensing. See [Adjusting the Moveable Sensor for Web \(Gap\) Sensing on page 58](#)



## Adjusting the Moveable Sensor for Black Marks or Notches

The blackline sensor looks for non-reflective surfaces such as black marks, black lines, notches or holes in the media which do not reflect the sensor's beam back to the sensor's detector. The sensor and its black mark detector are next to each other under the sensor cover (dark plastic cover translucent to the sensor's non-visible light).

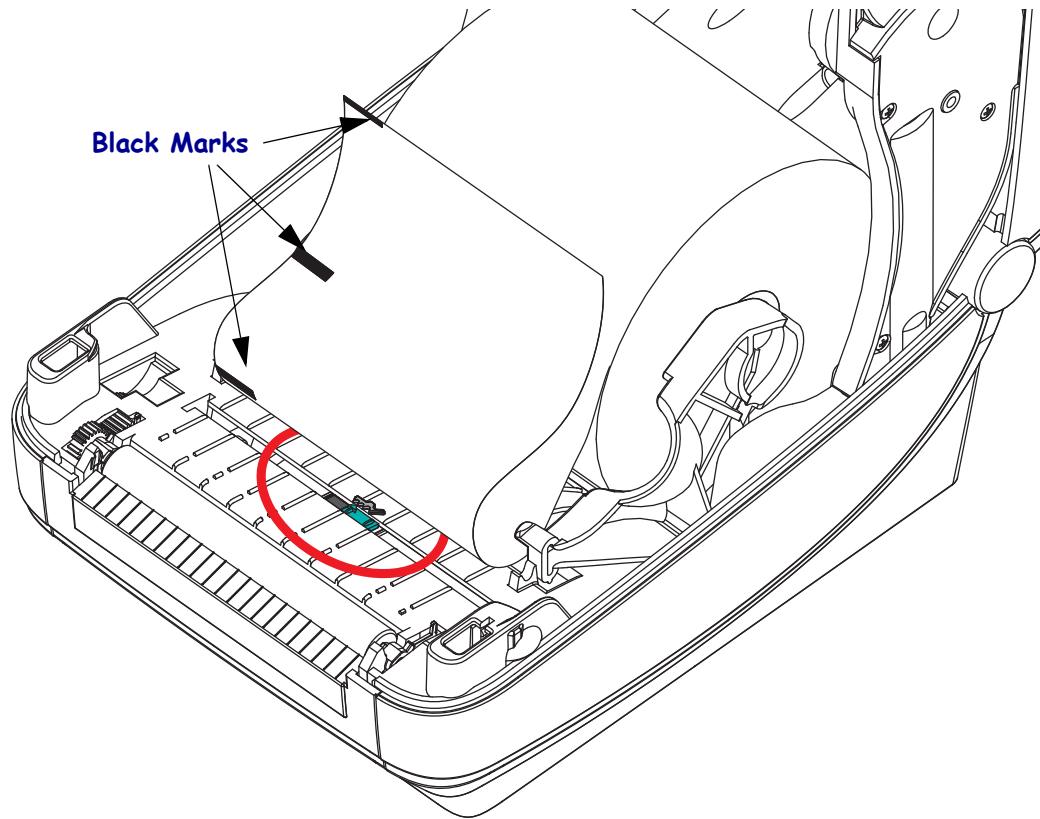
The sensor can be adjusted by two methods:

- 1) Preferred - Aligning the movable sensor's alignment arrow to the right side of a black mark or notch located in the center or left side of the printer, or
- 2) Align the center of the sensor cover under the oval notch or a right hand black mark (or notch).

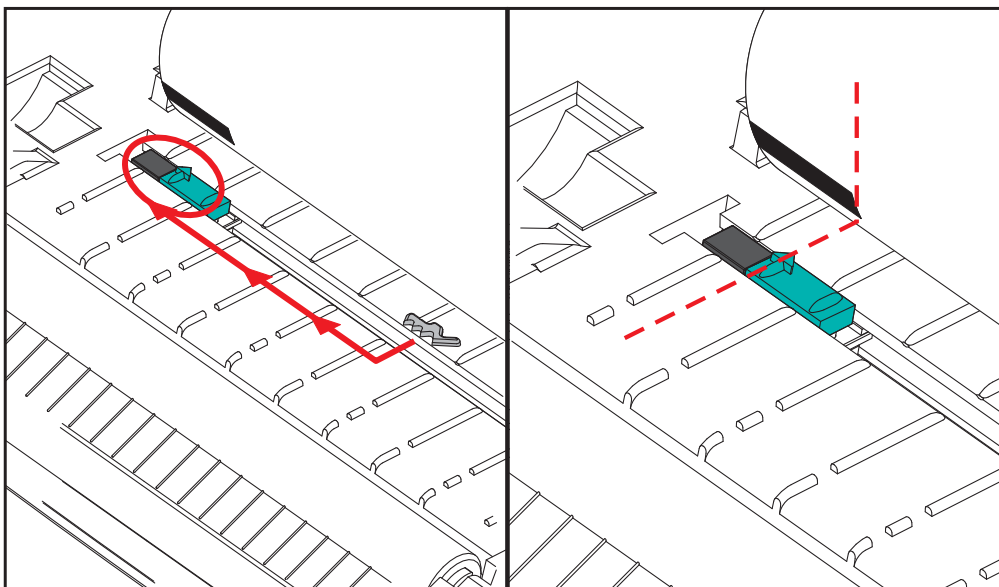
The sensor should be set as far as possible from the edge of media. The media can move from side to side and notches cut into the side of the media can become damaged.

Load the media. Before closing the printer, adjust the moveable blackline sensor as follows:

1. Flip the media back over the roll to expose the black marks or notches on the backside (non printing side) of the media.

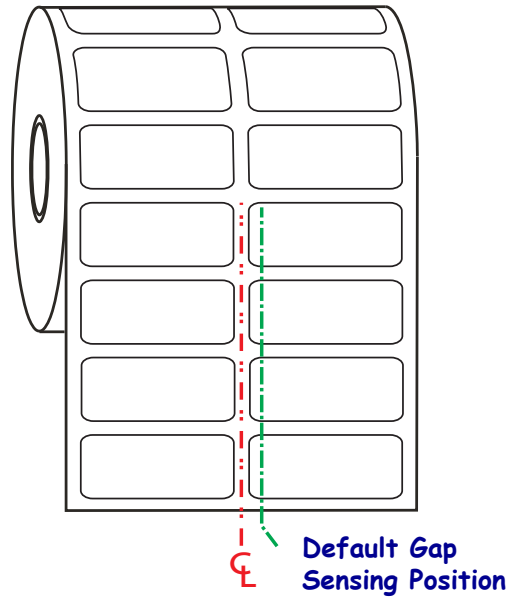


2. Slide the moveable sensor from the center default position to align with the black marks. The arrow on the moveable sensor should line up with the right edge of the black mark. This is true for marks or notches located on either side of the media (left side shown).

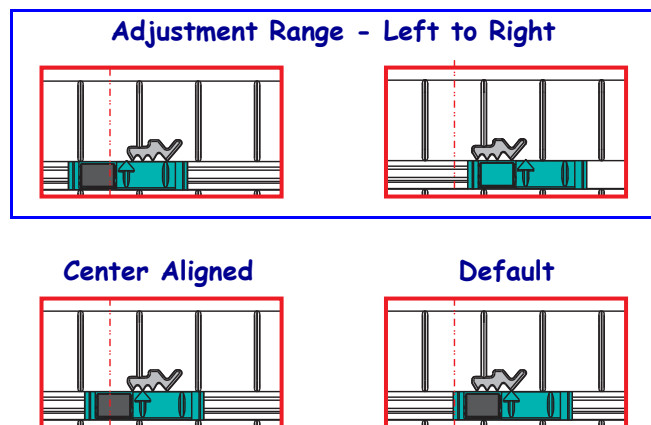


## Adjusting the Moveable Sensor for Web (Gap) Sensing

The moveable sensor supports the web/gap positions used by the legacy Zebra® desktop printer models and for some non standard media configurations. The G-Series™ printer's web (gap) sensing in the standard (fixed position) or the moveable sensor's default position are offset from the center of the printer to allow for gap sensing of labels that are side by side on a roll - see below.



Web (Gap) sensing with the movable sensor can only function when the sensor's alignment arrow points to any position on the alignment key. The sensor must be aligned with the labels (or other media) to sense the gap between the labels. The example above shows where the sensor is set if center aligned positioning is used. It will miss sensing labels with a 2-up label configuration and with the sensor in the 'default' position, it will be able to detect labels and the gaps between them.



- Default — Zebra Models: G-Series™ fixed position sensors, LP/TLP 2842™, LP/TLP 2844™, LP/TLP 2042™
- Center Aligned — Zebra Model: LP/TLP 2742™

## Cutter Option

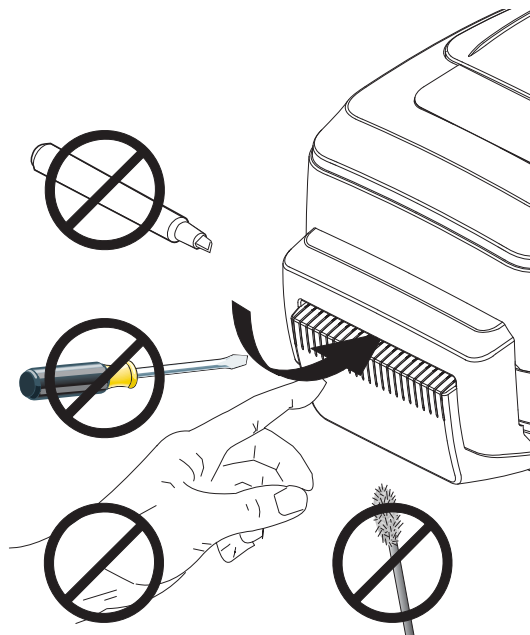
The printer supports a factory installed media cutter option for full width cutting of label liner, tag or receipt media. These cutter options are all supported in a single cutter housing design. The type of cutter option installed in your printer can be verified by printing a *printer configuration status* label. The cutter options are:

- **A medium duty cutter** for cutting label liner and light tag media (LINER/TAG)  
**Maximum Paper Weight (thickness):** up to 180 g/m2 (0.0077 in.)  
**Life\*:** 2 million cuts: 0.5 mil to 5 mil stock (10-120 g/m2)  
1 million cuts: 5 mil to 7.5 mil stock (120-180 g/m2)  
750,000 cuts: 7.5 mil to 10 mil (180-200 g/m2)  
  
\* - *Exceeding the maximum media weight (density/hardness) and thickness will shorten the cutters usable life or can cause the cutter to fail (jam or other error).*
- **Cut Width:** 4.25 inches (108 mm) maximum to 0.75 inches (19 mm) minimum
- **Minimum distance between cuts (label length):** 1 inch (25.4 mm). Cutting shorter media lengths between the cuts may cause the cutter to jam or error.
- By default, the cutter operations include a self cleaning preventive maintenance cleaning cut after every twenty-fifth cut. This feature can be disabled via ZPL or EPL SGD (Set/Get/DO) programming command (**cutter.clean\_cutter**) but is not recommended.



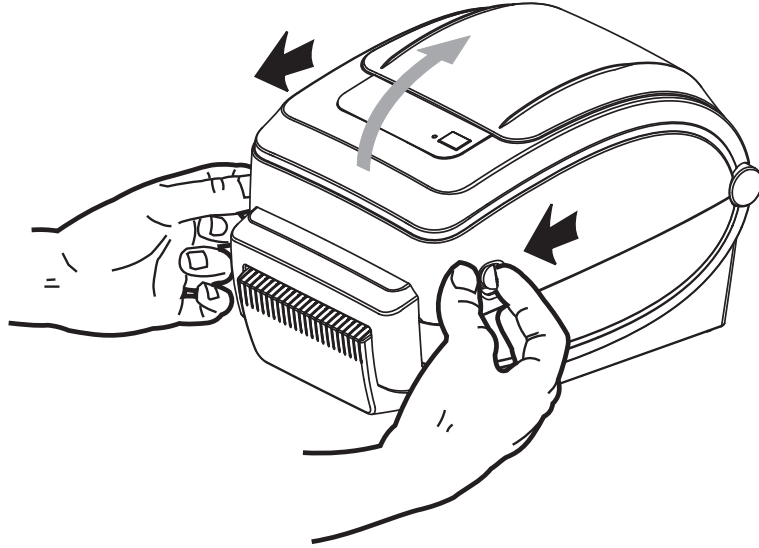
**Warning** • There are no operator serviceable parts in the cutter unit. Never remove the cutter cover (bezel). Never attempt to insert objects or fingers in to the cutter mechanism.

**Important** • Tools, cotton swabs, solvents (including alcohol), etc. all may damage or shorten the cutter's usable life or cause the cutter to jam.

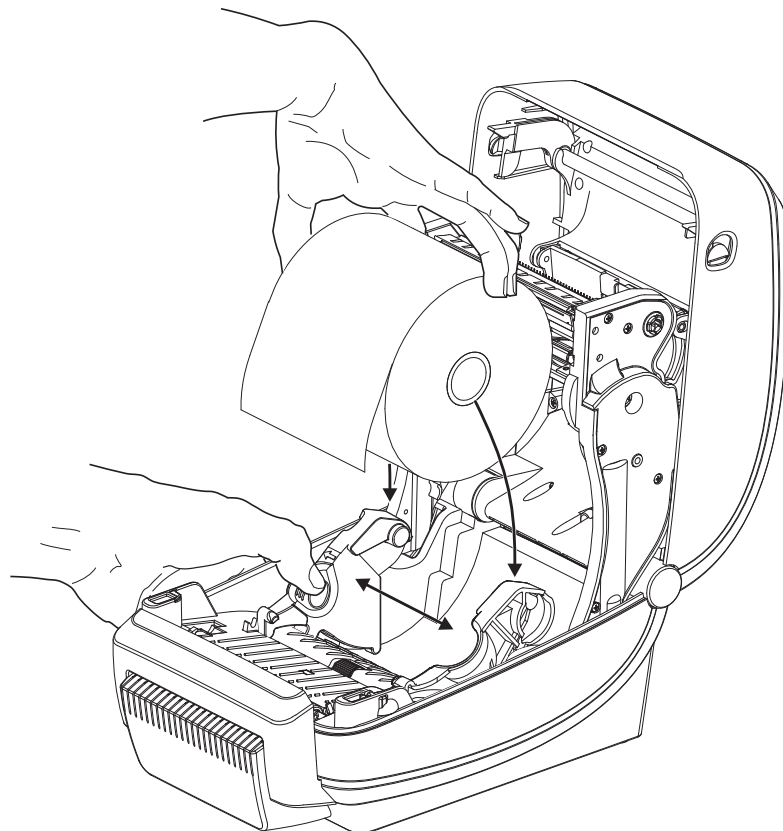


## Loading Media with the Cutter Option

1. Open the printer. Remember that you need to pull the release latch levers toward the front of the printer.

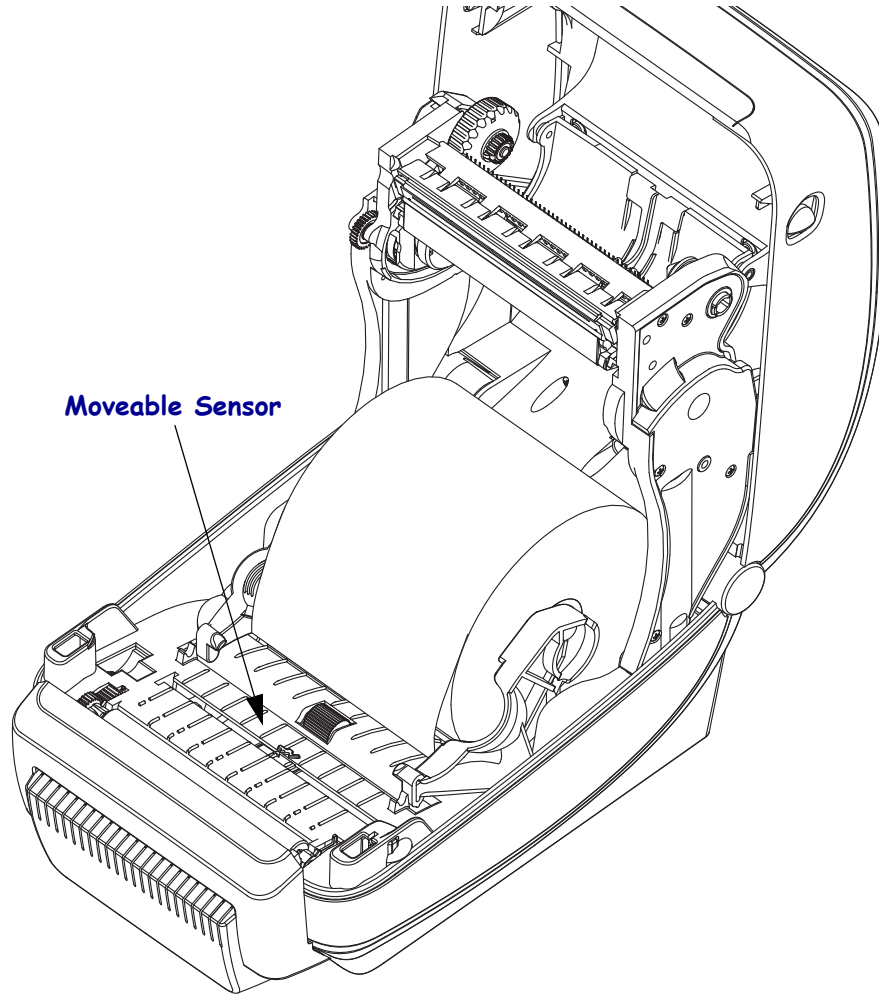


2. Open the media roll holders. Pull the media guides open with your free hand and place the media roll on the roll holders and release the guides. Orient the media roll so that its printing surface will be up as it passes over the platen (drive) roller.



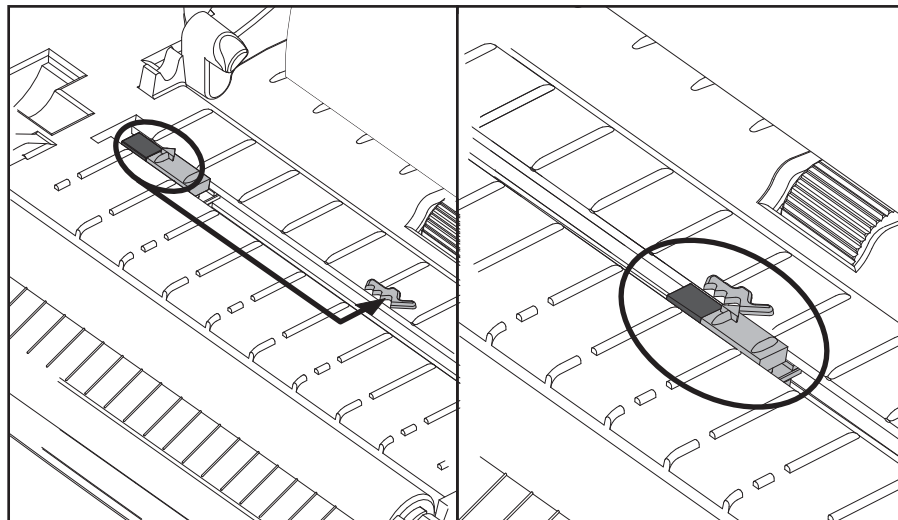


3. If your printer includes the moveable blackline sensor option, verify that the sensor is aligned in the slightly right of center default position. This is the standard operating position for media sensing. When the sensor is moved from the default position, the printer will only detect blackline, black mark and notched marked media.

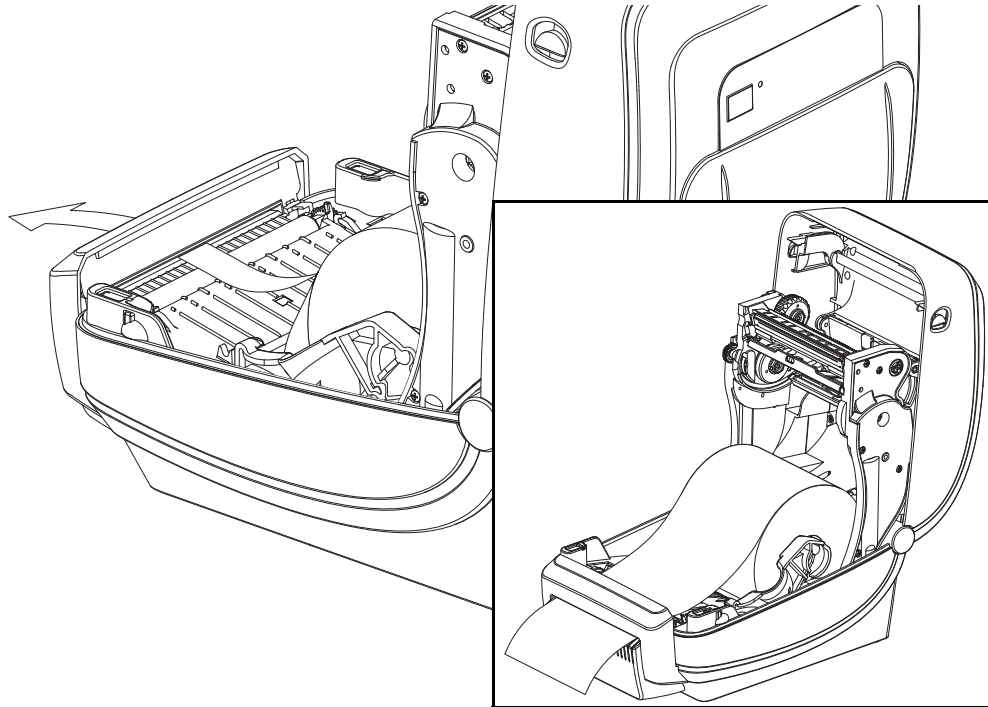


**Off Center  
Blackline Sensing Only**

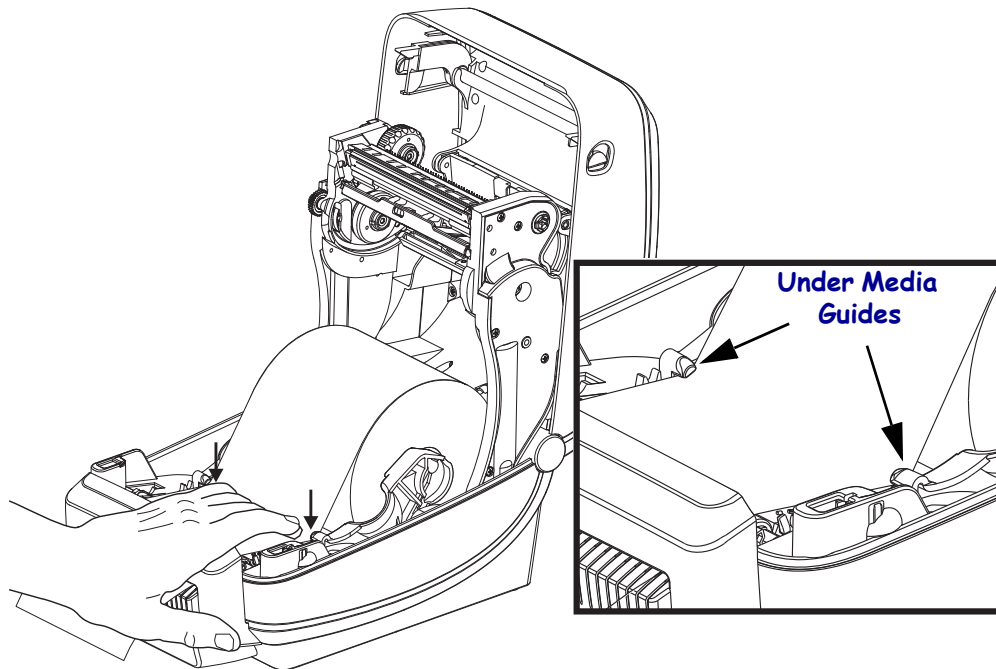
**Default - Web (Gap) Sensing  
Standard Operating Position**



4. Thread the media through the cutter's inside slot so that it extends out of the front of the printer. Verify the roll turns freely. The roll must not sit in the bottom of the media compartment. Verify that media's printing surface is facing up.



5. Push the media under both of the media guides.



6. Close the printer. Press down until the cover snaps closed.

## ZebraNet® 10/100 Internal (Wired) Print Server Option

The ZebraNet® 10/100 Internal Print Server (PS) is an optional factory installed device that connects the network and your ZebraLink™-enabled printer. The print server provides you with a browser interface for printer and print server settings. If you use the ZebraNet™ Bridge Zebra network printer management software, you can easily access the specialized features of a ZebraLink™ enabled printer.

Printers with the 10/100 Internal PS provide the following features:

- Print server and printer setup using a browser
- Remote monitoring and configuration capability of the 10/100 Internal PS using a browser
- Alerts
- The ability for you to send unsolicited printer status messages via E-mail-enabled devices

**ZebraNet™ Bridge** — ZebraNet™ Bridge is a software program that works with the 10/100 Internal PS and enhances the features of ZebraLink resident within ZPL-based printers. The features include the following:

- ZebraNet™ Bridge allows you to locate printers automatically. ZebraNet™ Bridge searches on parameters such as IP address, subnet, printer model, printer status, and many other user-defined characteristics.
- Remote Configuration — Manage all your Zebra label printers throughout the enterprise without having to travel to remote sites or physically handle any printers. Any Zebra printer connected to the enterprise network can be accessed from the ZebraNet™ Bridge interface and configured remotely through an easy-to-use graphical user interface.
- Printer Alerts, Status, Heartbeat Monitoring and event Notification — ZebraNet™ Bridge lets you configure multiple event alerts per device with different alerts directed to different people. Receive alerts and notifications by e-mail, cell phone/pager, or through the ZebraNet™ Bridge Events tab. View alerts by printer or group, and filter by date/time, severity, or trigger.
- Configure and Copy Printer Profiles — Copy and paste settings from one printer to another or broadcast them to an entire group. ZebraNet™ Bridge allows you to copy printer settings, printer-resident files (formats, fonts, and graphics), and alerts with a click of the mouse. Create printer profiles — virtual “golden printers” — with desired settings, objects, and alerts, and clone or broadcast them as if they were real printers, providing extensive savings in setup time. Printer profiles are also an excellent way to back up a printer’s configuration for crisis recovery.

## Printer Network Configuration Status Label

G-Series™ printers with the ZebraNet® 10/100 Internal Print Server option on-board print an additional printer configuration status label for information needed to establish and trouble shoot network printing. The following printout is printed with the ZPL ~WL command.

| Network Configuration                          |                  |
|------------------------------------------------|------------------|
| Zebra Technologies<br>ZTC GX420t<br>ZBR2834649 |                  |
| Internal Wired*                                |                  |
| ALL.....                                       | IP PROTOCOL      |
| 172.029.001.047....                            | IP ADDRESS       |
| 255.255.255.000....                            | SUBNET MASK      |
| 172.029.001.001....                            | DEFLT. GATEWAY   |
| 172.029.001.003....                            | WINS SERVER IP   |
| YES.....                                       | TIMEOUT CHECKING |
| 300.....                                       | TIMEOUT VALUE    |
| 000.....                                       | ARP INTERVAL     |
| 9100.....                                      | BASE RAW PORT    |
| 00:07:4d:2b:40:d9..                            | MAC ADDRESS      |
| FIRMWARE IN THIS PRINTER IS COPYRIGHTED        |                  |

The regular printout of the configuration status label has some of the printer's network settings such as IP Address, on the lower half of the printout.

The printer's IP address is needed to identify and configure the printer for operations on your network. See the ZebraNet® 10/100 Internal Print Server manual for more information.

## ZebraNet® Internal Wireless Print Server Option

The ZebraNet® Internal Wireless Print Server (PS) is an optional factory installed device that connects to Wi-Fi network access point on your network and your ZebraLink™-enabled printer. The print server provides you with a browser interface for printer and print server settings. If you use the ZebraNet™ Bridge, Zebra® network printer management software, you can easily access the specialized features of a ZebraLink™ enabled printer.

Printers with an Internal Wireless Print Server provide the following features:

- A printer status display for alerts, configuration settings and network settings.
- Print server and printer setup using a browser
- Remote monitoring and configuration capability of the Internal Wireless PS using a browser
- Alerts
- The ability for you to send unsolicited printer status messages via E-mail-enabled devices

**ZebraNet™ Bridge** — ZebraNet™ Bridge is a software program that works with the 10/100 Internal PS and enhances the features of ZebraLink resident within ZPL-based printers. The features include the following:

- ZebraNet™ Bridge allows you to locate printers automatically. ZebraNet™ Bridge searches on parameters such as IP address, subnet, printer model, printer status, and many other user-defined characteristics.
- Remote Configuration — Manage all your Zebra label printers throughout the enterprise without having to travel to remote sites or physically handle any printers. Any Zebra printer connected to the enterprise network can be accessed from the ZebraNet™ Bridge interface and configured remotely through an easy-to-use graphical user interface.
- Printer Alerts, Status, Heartbeat Monitoring and event Notification — ZebraNet™ Bridge lets you configure multiple event alerts per device with different alerts directed to different people. Receive alerts and notifications by e-mail, cell phone/pager, or through the ZebraNet™ Bridge Events tab. View alerts by printer or group, and filter by date/time, severity, or trigger.
- Configure and Copy Printer Profiles — Copy and paste settings from one printer to another or broadcast them to an entire group. ZebraNet Bridge allows you to copy printer settings, printer-resident files (formats, fonts, and graphics), and alerts with a click of the mouse. Create printer profiles — virtual “golden printers” — with desired settings, objects, and alerts, and clone or broadcast them as if they were real printers, providing extensive savings in setup time. Printer profiles are also an excellent way to back up a printer’s configuration for crisis recovery.

## Printer Network Configuration Status Label

G-Series™ printers with the ZebraNet® Internal Wireless Print Server option on-board print an additional printer configuration status label for information needed to establish and trouble shoot network printing. The following printout is printed with the ZPL **~WL** command.

| Network Configuration                           |                  |
|-------------------------------------------------|------------------|
| Zebra Technologies<br>ZTC GX420t<br>ZBR12861842 |                  |
| Wireless*                                       |                  |
| ALL.....                                        | IP PROTOCOL      |
| 172.029.016.027....                             | IP ADDRESS       |
| 255.255.255.000....                             | SUBNET MASK      |
| 172.029.016.001....                             | DEFAULT GATEWAY  |
| 172.029.001.003....                             | WINS SERVER IP   |
| YES.....                                        | TIMEOUT CHECKING |
| 300.....                                        | TIMEOUT VALUE    |
| 000.....                                        | ARP INTERVAL     |
| 9100.....                                       | BASE RAW PORT    |
| YES.....                                        | CARD INSERTED    |
| 02DFH.....                                      | CARD MFG ID      |
| 8103H.....                                      | CARD PRODUCT ID  |
| 000278c44192.....                               | MAC ADDRESS      |
| YES.....                                        | DRIVER INSTALLED |
| INFRASTRUCTURE.....                             | OPERATING MODE   |
| 125.....                                        | ESSID            |
| 100.....                                        | TX POWER         |
| 54 Mb/s.....                                    | CURRENT TX RATE  |
| DIVERSITY.....                                  | RECEIVE ANTENNA  |
| DIVERSITY.....                                  | XMIT ANTENNA     |
| OPEN.....                                       | WEP TYPE         |
| NONE.....                                       | WLAN SECURITY    |
| 1.....                                          | WEP INDEX        |
| 020.....                                        | POOR SIGNAL      |
| LONG.....                                       | PREAMBLE         |
| YES.....                                        | ASSOCIATED       |
| ON.....                                         | PULSE ENABLED    |
| 15.....                                         | PULSE RATE       |
| OFF.....                                        | INTL MODE        |
| 07FFH.....                                      | CHANNEL MASK     |
| FIRMWARE IN THIS PRINTER IS COPYRIGHTED         |                  |

The regular printout of the configuration status label has some of the printer's network settings such as IP Address, on the lower half of the printout.

The printer's IP address is needed to identify and configure the printer for operations on your network. See the ZebraNet® Internal Wireless Print Server manual for more information.



## Bluetooth Option

Bluetooth is an optional factory installed (internal wireless) device that can connect to other Bluetooth enabled devices and computers. The Bluetooth radio inside G-series printers complies with Bluetooth Specification 2.0 and supports the Serial Port Profile (SPP), a Windows virtual serial port. All print jobs sent to the printer on Windows PC's will be done through the SPP, which is basically a Bluetooth emulation of an RS-232 serial communication. This printer option includes a LCD display to provide connection status and configuration information to assist with installation and connection to a Bluetooth enabled host system.

Microsoft has supported most Bluetooth devices since the release of Windows XP Service Pack 2 (SP2). Prior to that, Bluetooth devices for earlier versions of Windows required drivers to be supplied with the device. Windows XP SP2 and above also include 'Add a Printer' wizard support for printers. The G-Series printer is accessed via a Bluetooth virtual serial port connection (SPP). The printer only needs Bluetooth 2.0 driver that supports SPP to be able to communicate with the Windows PC's Bluetooth devices.

In the Windows environment, the printer requires the ZebraDesigner™ driver to be installed on the computer to enable printing after Bluetooth communication has been established. Use the Zebra Setup Utility's printer install wizards to help you configure your Bluetooth connection.

## Printer Network Configuration Status Label

G-Series printers with the ZebraNet® Internal Bluetooth option on-board prints an additional printer configuration status label for information needed to establish and trouble shoot network printing. The following printout is printed with the ZPL **~WL** command.

| Network Configuration                   |                 |
|-----------------------------------------|-----------------|
| Zebra Technologies<br>ZTC GX430t        |                 |
| Bluetooth                               |                 |
| 2.1.8.....                              | FW VERSION      |
| 04/25/08.....                           | DATE            |
| Test1234.....                           | LOCAL NAME      |
| OFF.....                                | AUTHENTICATION  |
| ON.....                                 | DISCOVERABLE    |
| 2.0.....                                | RADIO VERSION   |
| ON.....                                 | ENABLED         |
| OFF.....                                | AFH MODE        |
| 00:03:7A:18:F4:47..                     | ADDRESS         |
| 7FFFFFFFFFFFFFFFFF                      | CURRENT AFH MAP |
| FIRMWARE IN THIS PRINTER IS COPYRIGHTED |                 |

The printer's Bluetooth settings help identify the printer for installation and print operations.

### Preparing the Windows PC for Bluetooth communication.

- Install a Bluetooth dongle or verify that the existing Bluetooth is enabled. *Note that some systems with on-board Bluetooth share the Bluetooth radio with Wi-Fi. Wi-Fi maybe disabled, which may disable Bluetooth also.*

### Preparing the Printer for Bluetooth Printing

- The printer must have the factory option Bluetooth installed. The printer will include a LCD display. The printer's main menu screen will display the word Bluetooth.



**Shows Disconnected or Connected Status**

- **The printer should have media loaded** to print configuration status labels (and Windows test print labels later).
- **Install the printer using the Zebra Setup Utility (ZSU) using the USB interface cable** (or a standard serial port cable). Verify printing using the printer properties or the ZSU to print a printer configuration status label.
- With the ZSU installed and open, select your printer. The buttons below the installed Zebra® printers will activate. Click on the 'Configure Printer Connectivity' button to start the 'Connectivity Setup Wizard'.
  - **Select Bluetooth** and click 'Next'.
  - **Enter an unique name for the printer** to the 'Friendly name' text input box.
  - **Leave 'Enabled' checked, Discoverable' On, and 'Authentication' to Off.** Click 'Next' to continue.
  - The window will display the SGD (Set/Get/Do) commands to be sent to the printer. Click 'Next'.
  - Leave the printer selected and click 'Finished'. The printer will reset to activate all the new Bluetooth commands.

### Bluetooth - Installing the Printer with a Windows XP® SP2

This process covers the printer installation for Windows XP SP2 (or above)

1. From the Windows 'Start' menu, open the 'Printers and Faxes' window. Click on 'Add Printer' and a wizard will start. Click 'Next' when the "Welcome" window displays to continue.
2. Leave the 'Local' radio button with the 'Automatic detect ...' checkbox (checked) selected and click 'Next'.
3. No printer should be found. If there is one or more new printers detected, then do not select any of those printers and click 'Next'.
4. Select the 'Create a new port' radio button and select 'Bluetooth Printer Port' from the associated pull down menu.



5. From the new 'Bluetooth Printer Port' window, select your printer from the Bluetooth devices in the list. This may take several minutes. Click on the 'Connect' button.
6. The 'Install Printer Software' window will open. Select 'ZDesigner' as 'Manufacture' to select supported Zebra printer models, and then select your printer model in the 'Printers' selection. Click 'Next' to continue.
7. Select 'Keep the existing drivers' and click 'Next'.
8. Name your printer and choose your default printer setting as desired. Click 'Next'.
9. Answer 'Yes', you want to print a test page. Click 'Next' to continue.
10. Click 'Finish' on the 'Completing the Add Printer Wizard' window and the Bluetooth printer will complete installation and print a Windows test print page (a Windows logo at minimum).

### Bluetooth - Installing the Printer with a Windows Vista® SP2 and Windows 7®

The Windows Vista (SP2 or higher) and Windows 7 bluetooth install differs from the XP install.

- **Windows Vista:** Open the 'Add Printer' wizard by clicking the 'Start' button, clicking 'Control Panel', clicking 'Hardware and Sound', clicking 'Printers', and then clicking 'Add a printer'.
- **Windows 7:** Open Devices and Printers by clicking the Start button Picture of the Start button, and then, on the Start menu, clicking Devices and Printers
- Some non-Microsoft Bluetooth dongles and built-in Bluetooth devices in the host PC's have marginal driver support for SPP printing and may not complete the 'Add printer' wizard normally. You may need to go to the 'Bluetooth Devices' in the 'Control Panel' or from the system tray on the Windows 'Start' bar and activate SPP for the 'device', your Bluetooth printer you are installing. Install the printer to a local printer (USB or Serial Port for your G-Series printer) and then change the 'Port' after the install has completed to the SPP (virtual serial port) COM port.

## RTC and Full Flash Memory Option

Printers with the factory installed RTC (Real Time Clock) and Full Flash Memory option allow you to create highly customizable printer applications, improve printer productivity and flexibility, and provide extra memory to store more forms, graphics (logos), fonts, and the large Asian font sets.

The RTC allows the label form developer to include:

- Time Date Stamping
- Pre or Post Dating
- Allows Date/Time date to be included on printed form without altering the form
- Provide dates and times without needing to generate them with the host system, application software or programmatically by another process (e.g. a database report).

The RTC enhances the printer's ability to operate in a stand-alone configuration (e.g. not connected to a host or PC). A common use would be to provide a date stamp to a tag, ticket, label or receipt printed on demand. See [Stand Alone Printing on page 49](#) for more details.

The RTC feature supports both ZPL and EPL printer programming. The two programming behaviors that display the time and date are independent of each other. The commands that set time and date change them for both printer languages.



**Caution** • The optional RTC assembly has a three volt lithium battery. You can discern a low or flat battery if the printer provides a consistently delayed date stamp. Battery replacement must be performed by a qualified service technician. Only use a Zebra approved replacement battery.



**Important** • Recycle batteries according to local your guidelines and regulations. Wrap the battery when disposing (or storing) to avoid a short circuit.

**DO NOT** short circuit the battery. Short circuiting the battery may result in heat generation, fire or bursting.

**DO NOT** heat, disassemble or dispose of battery in fire.

The Full Flash Memory provides an additional 64 Megabytes to the printer's standard 4 Megabytes of printer memory for a total of 68 Megabytes printer memory. This option originally supported only 12MB of Full Flash Memory for printer built prior to July 24, 2009.

## Asian Fonts - Printer Accessory

Asian font sets are downloaded into the printer by the user or integrator. Your G-Series printer must have the RTC and Full Flash Printer Memory option installed to support the large Asian font files. The fonts are obtained separately from the printer.

- Simplified and Traditional Chinese (ZPL and EPL)
- Japanese — JIS and Shift-JIS mappings (ZPL and EPL)
- Korean (ZPL including Johab and EPL)
- Thai (ZPL only)

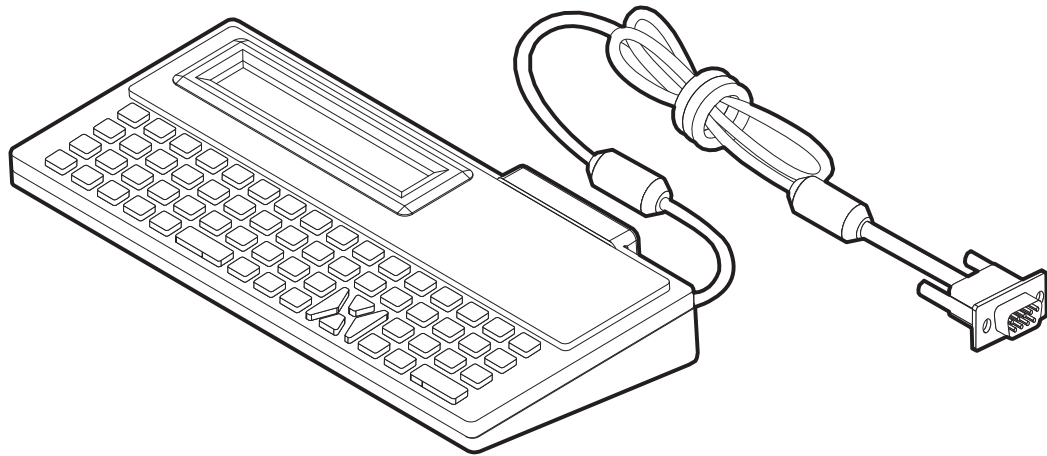
See [Fonts and Your Printer on page 47](#) for more details on Asian fonts.

## Zebra® KDU — Printer Accessory

The KDU (keyboard display unit) is a small terminal unit that interfaces with the printer for accessing EPL label forms stored in the printer.

The KDU is strictly a terminal and does not have the capability to store data or set up parameters. The KDU is used for the following functions:

- List label forms stored in printer
- Retrieve label forms stored in printer
- Input variable data
- Printing labels



## KDU Plus™ — Printer Accessory

The KDU Plus is a terminal device with memory for storing files and maintaining one or more remote printers. The KDU Plus has a larger laptop style keyboard than the Zebra's original KDU design.

The KDU Plus is designed for the following functions:

- List label forms stored in printer
- Retrieve label forms stored in printer
- Input variable data
- Printing labels
- File storage and transfer

The KDU Plus is designed to work with ZPL and EPL printers by dynamically detecting and switching the printer's communications configuration (DTE or DCE) to match your Zebra printer. By default, the KDU Plus will start up forms mode. This is compatible with the operation of Zebra's original KDU. The KDU Plus is configurable to work in ZPL or terminal modes too.

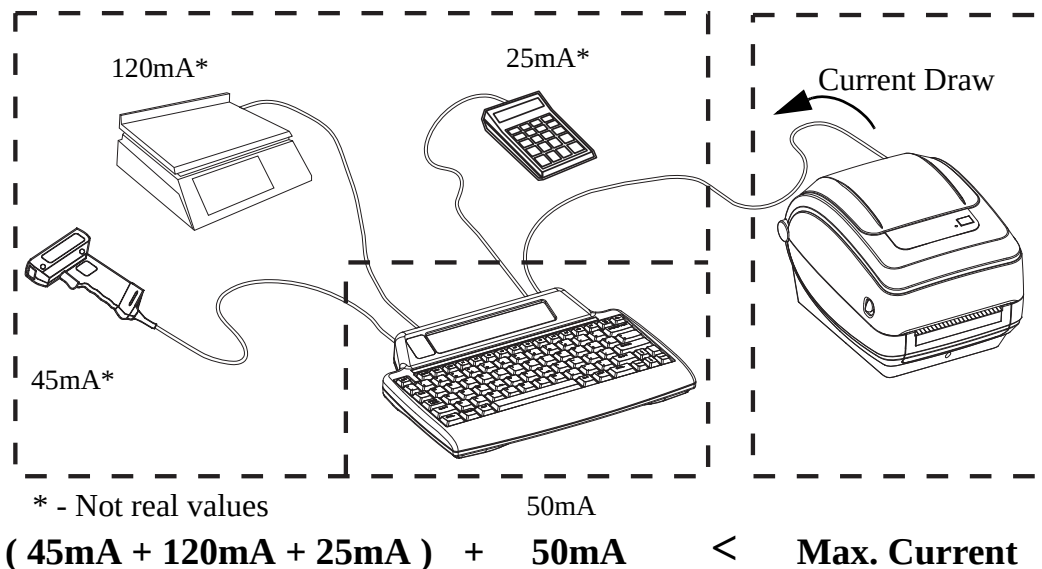
The KDU Plus has file transfer and storage capabilities. This is great for maintaining remote and stand alone printers with label forms, logos and smaller font sets.

The KDU Plus has two configurable serial port and a PS/2 port to connect to other data input devices such as scanners, weigh scales, etc..

The +5 volts from the printer's serial (pin 9) provides power for the KDU Plus and any additional input devices attached to the KDU Plus.

Maximum total current draw for the G-Series printer's serial port: **750mA**

The KDU Plus's current without external input devices attached is: **50mA**



**Caution** • The KDU Plus's input ports (AUX1, AUX2, and PS/2 style) are not fused. Exceeding the total current draw available could damage an input device, KDU Plus, or the printer.

## **ZBI 2.0™ — Zebra Basic Interpreter**

Customize and enhance your printer with ZBI 2.0™ optional programming language, ZBI 2.0 allows Zebra printers to run applications, prompt users through LCD menu commands and take input from scales, scanners and other peripherals—all without a PC or network connection. ZBI 2.0 works with the ZPL printer command language so that printers can understand non-ZPL data streams and convert them into labels. That means the Zebra printer can create bar codes and text from input received, non-ZPL label formats, sensors, keyboards and peripherals. Printers can also be programmed to interact with PC based database applications to retrieve information to be used on printed labels.

ZBI 2.0 can be activated by ordering a ZBI 2.0 Key Kit, or activated by purchasing a key from the ZBI 2.0 store at [www.zebrasoftware.com](http://www.zebrasoftware.com).

Use the ZDownloader utility to apply the key. Zdownloader is available on the users CD or at the Zebra Web site: [www.zebra.com](http://www.zebra.com).

An intuitive ZBI-Developer™ programming utility is used to create, test and distribute ZBI 2.0 applications is available on the users CD or at the Zebra Web site: [www.zebra.com](http://www.zebra.com).



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# Control Panel Operation

This section describes the operation, navigation and parameters displayed by printers equipped with the LCD control panel option.

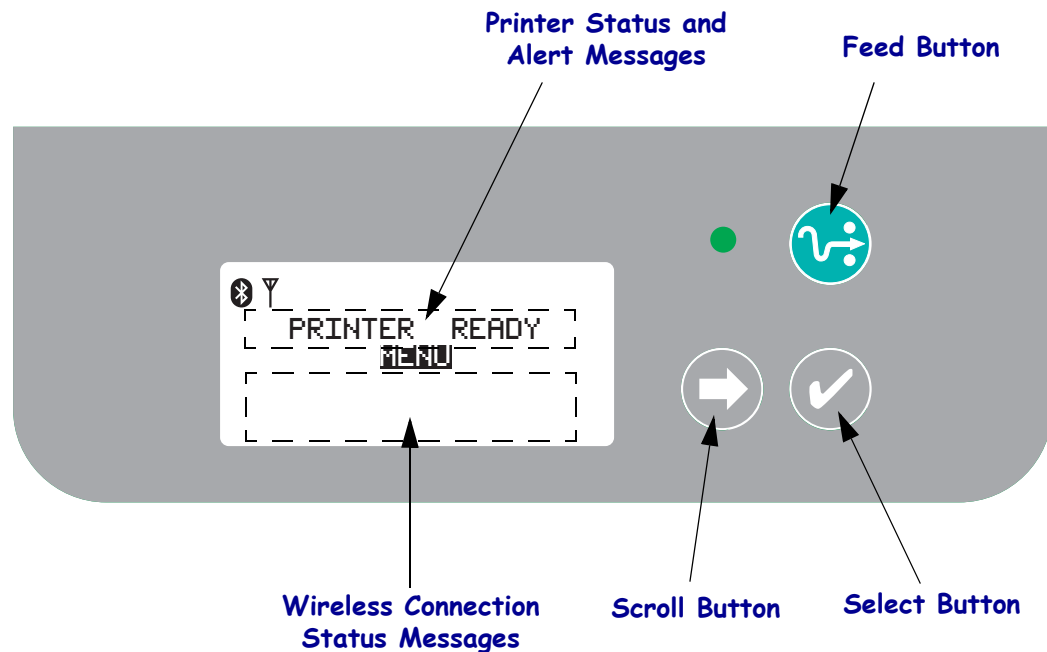
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## LCD Display Overview

The LCD display for G-Series printers with the Wi-Fi (Wireless) or Bluetooth options provide user feedback for printer operations, some basic controls for printer functionality, printer display controls, and wireless connectivity information.

The control panel features three operator controls: the Feed button, the Scroll button and the Select button. The Feed button is used to advance one label or a pre-set length of continuous media. You can control display operation by using the Scroll and Select buttons. The Scroll button allows scrolling through the various options and settings. The Select button allows selection of the option or function displayed on the screen.



**Note** • While the control panel can be used to change some printer settings, many settings can only be changed using Set / Get / Do (SGD) commands. Refer to the ZPL II Programming Guide for information on the use of SGD commands.

When the number of selections are too numerous to be displayed on one screen, MORE will be listed at the bottom of the display. To access the additional selections, use the Scroll button to move down through the current screen and the display will list the next set of choices.

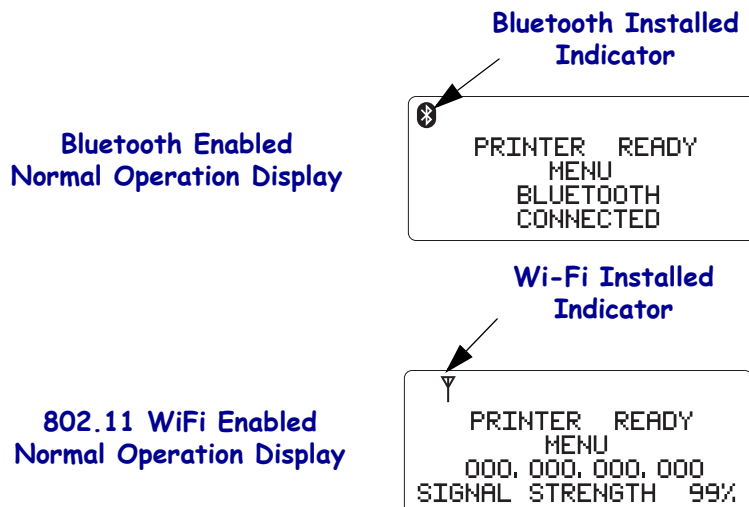
## Menu Customization and WML

The G-Series™ printers allows for the use of Zebra's WML menu customization language. The WML language features both display and command abilities. See the WML section in the ZPL Programmer's guide (for this printer) for details.



## Normal Operation Display

When the printer is On and Ready, the LCD display will resemble one of the examples shown in the figure below.



## Display Settings

The display has a back light feature which allows viewing the display in dark environments. The back light will illuminate the display when the Select button is pressed, entering the printers option menus. If there is no activity, the back light will extinguish after a selectable time-out. You can also adjust the display contrast as well as select an appropriate language for the display. The user selectable Display settings and are shown in the table below.

| Display Settings |               |                                                                                                                                    |
|------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| Setting          | Default Value | Optional Values                                                                                                                    |
| BACK LIGHT       | ON            | ON / OFF                                                                                                                           |
| TIMEOUT          | 10            | 10 / 20 / 30                                                                                                                       |
| LCD CONTRAST     | 4             | 1 - 7                                                                                                                              |
| LANGUAGE         | ENGLISH       | ENGLISH<br>ESPAÑOL<br>FRANÇAIS<br>DEUTSCH<br>ITALIANO<br>NORSK<br>PORTUGUES<br>SVENSKA<br>DANSK<br>EASANOL2<br>NEDERLANDS<br>SUOMI |

## Printer Settings

The user selectable and programmable Printer settings are shown in the table below.

| Printer Settings                          |                  |                                                                                   |
|-------------------------------------------|------------------|-----------------------------------------------------------------------------------|
| COMM. (user selectable COM port settings) |                  |                                                                                   |
| Setting                                   | Default Value    | Optional Values                                                                   |
| BAUD                                      | 9600             | 300<br>600<br>1200<br>9600<br>14400<br>19200<br>28800<br>32400<br>57600<br>115200 |
| DATA BITS                                 | 8                | 7<br>8                                                                            |
| PARITY                                    | NONE             | NONE<br>EVEN<br>ODD                                                               |
| FLOW                                      | DTR & XON / XOFF | XON / XOFF<br>DTR & XON / XOFF<br>DSR / DTR<br>RTS / CTS                          |
| MEDIA (display only)                      |                  |                                                                                   |
| Setting                                   | Default Value    | Optional Values                                                                   |
| MEDIA                                     | GAP              | GAP<br>MARK<br>RECEIPT                                                            |
| RIBBON                                    | YES              | YES<br>NO                                                                         |
| PRINT (display only)                      |                  |                                                                                   |
| Setting                                   | Default Value    | Optional Values                                                                   |
| SPEED                                     | 6 IPS            | 2 IPS<br>3 IPS<br>4 IPS<br>5 IPS<br>6 IPS                                         |
| ZPL Darkness                              | 10.0             | 0.0 - 30.0                                                                        |

## 802.11 WLAN Settings

The programmable 802.11 WLAN settings are shown in the table below.

| 802.11 WLAN Settings (display only) |                 |                 |
|-------------------------------------|-----------------|-----------------|
| Setting                             | Default Value   | Optional Values |
| IP ADDRESS                          | 000.000.000.000 |                 |
| DEFAULT GATEWAY                     | 000.000.000.000 |                 |
| SUBNET MASK                         | 000.000.000.000 |                 |
| WLAN SECURITY                       | OPEN            |                 |
| ESSID                               | 125             |                 |

## Bluetooth Settings

The programmable Bluetooth settings are shown in the table below.

| Bluetooth Settings (display only) |               |                 |
|-----------------------------------|---------------|-----------------|
| Setting                           | Default Value | Optional Values |
| ADDRESS                           |               |                 |
| DISCOVERABLE                      | ON            | ON / OFF        |
| AUTHENTICATION                    | OFF           | ON / OFF        |



Notes • \_\_\_\_\_

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# Maintenance

This section provides routine cleaning and maintenance procedures.

## Contents

|                                       |    |
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| Cleaning the Printhead .....          | 83 |
| Media Path Considerations .....       | 84 |
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## Cleaning

When you clean the printer, use one of the following supplies that best suits your needs:

| Cleaning Supplies           | Order Quantity | Intended Purpose                     |
|-----------------------------|----------------|--------------------------------------|
| Cleaning pens (105950-035)  | Set of 12      | Clean printhead                      |
| Cleaning swabs (105909-057) | Set of 25      | Clean media path, guides and sensors |

You can obtain cleaning supplies at [www.zipzebra.com](http://www.zipzebra.com).

The cleaning process takes just a couple of minutes using the steps outlined below.

| Printer Part  | Method                                                                                                                                                                                                                     | Interval                                                                                             |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Printhead     | Let the printhead to cool for a minute, then use a new cleaning pen to swab the dark line on the printhead cleaning from the center to the outside edges of the printhead.<br>See <i>Cleaning the Printhead on page 83</i> | When using ribbon: after every roll of ribbon; when using direct thermal: after every roll of media. |
| Platen roller | Remove the platen roller to clean. Clean the roller thoroughly with 90% medical-grade alcohol and a cleaning swab or lint-free cloth.<br>See <i>Platen Cleaning and Replacement on page 86</i>                             | As needed.                                                                                           |
| Peel bar      | Clean it thoroughly with 90% medical-grade alcohol and a fiber-free cleaning swab.<br>Let alcohol dissipate and the printer dry completely.                                                                                |                                                                                                      |
| Media path    |                                                                                                                                                                                                                            |                                                                                                      |
| Exterior      | Water-dampened cloth.                                                                                                                                                                                                      |                                                                                                      |
| Interior      | Gently brush out printer.                                                                                                                                                                                                  |                                                                                                      |



**Caution** • Adhesives and media material can over time transfer onto the printer components along the media path including the platen and printhead. This build-up can accumulate dust and debris. Failure to clean the printhead, media path and platen roller could result in inadvertent loss of labels, label jams and possible damage to the printer.



**Important** • Using too much alcohol can result in contamination of the electronic components requiring a much longer drying time before the printer will function properly.

## Cleaning the Printhead

Always use a new cleaning pen on the printhead (an old pen carries contaminants from its previous uses that may damage the printhead).



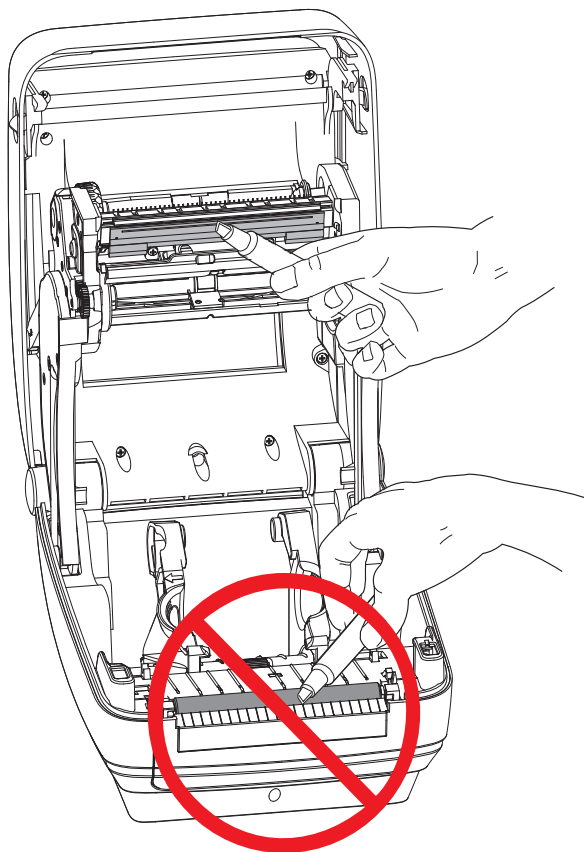
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**Caution** • The printhead becomes hot while printing. To protect from damaging the printhead and risk of personal injury, avoid touching the printhead. Use only the cleaning pen to perform maintenance.

---

When you load new media, you can also clean the printhead.

1. Rub the cleaning pen across the dark area of the printhead. Clean from the middle to the outside. This will move adhesive transferred from the edges of media to the printhead outside of media path.
2. Wait one minute before closing the printer.

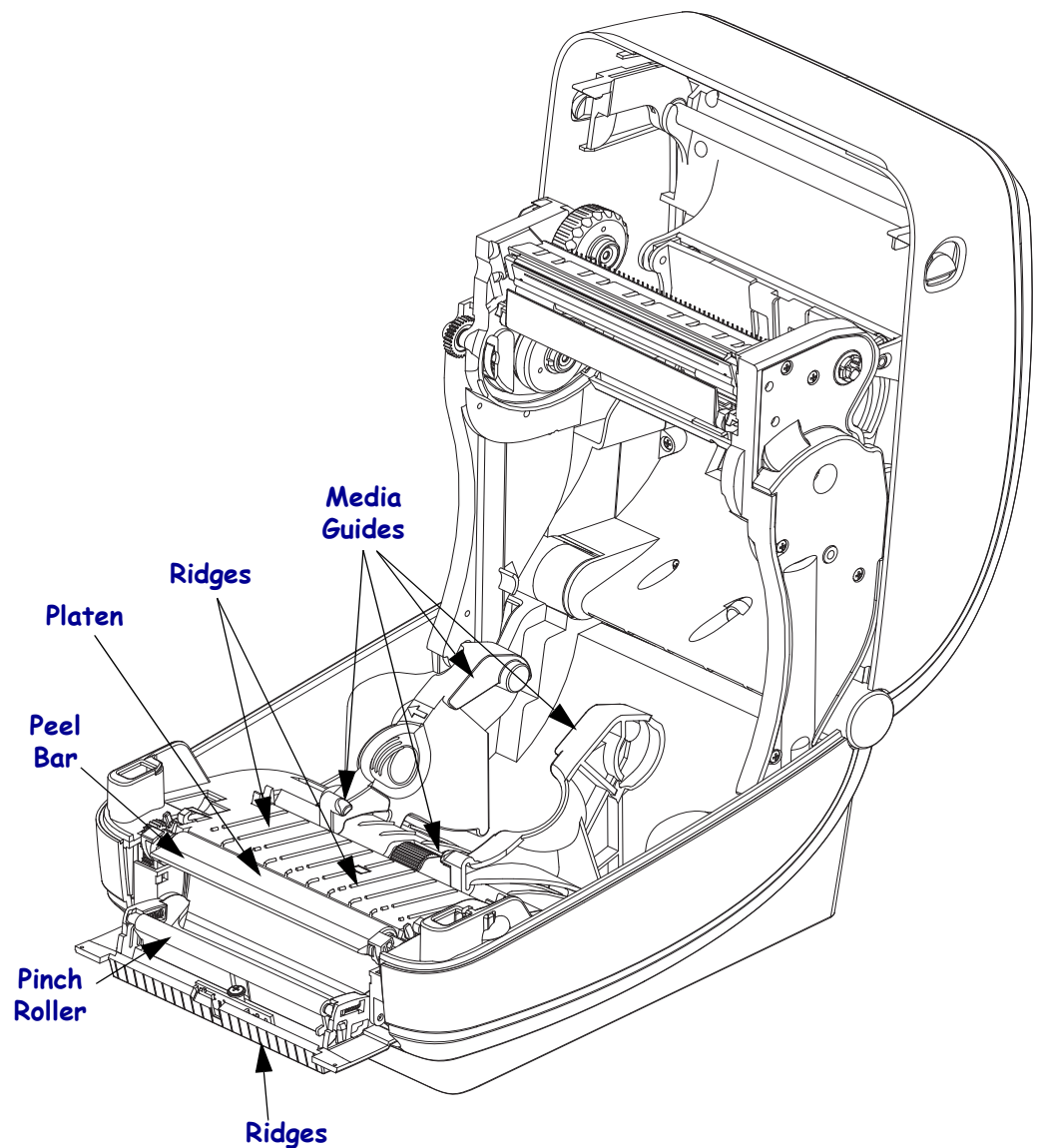


## Media Path Considerations

Use a cleaning swab to remove debris, dust or crust that has built-up on the holders, guides and media path surfaces.

1. Use the alcohol in the cleaning swab to soak the debris to break up the adhesive.
2. Wipe the ridges to remove accumulated debris.
3. Wipe the inside edges of both edge guides to remove any built-up residue.
4. Wait one minute before closing the printer.

Discard the cleaning swab after use.

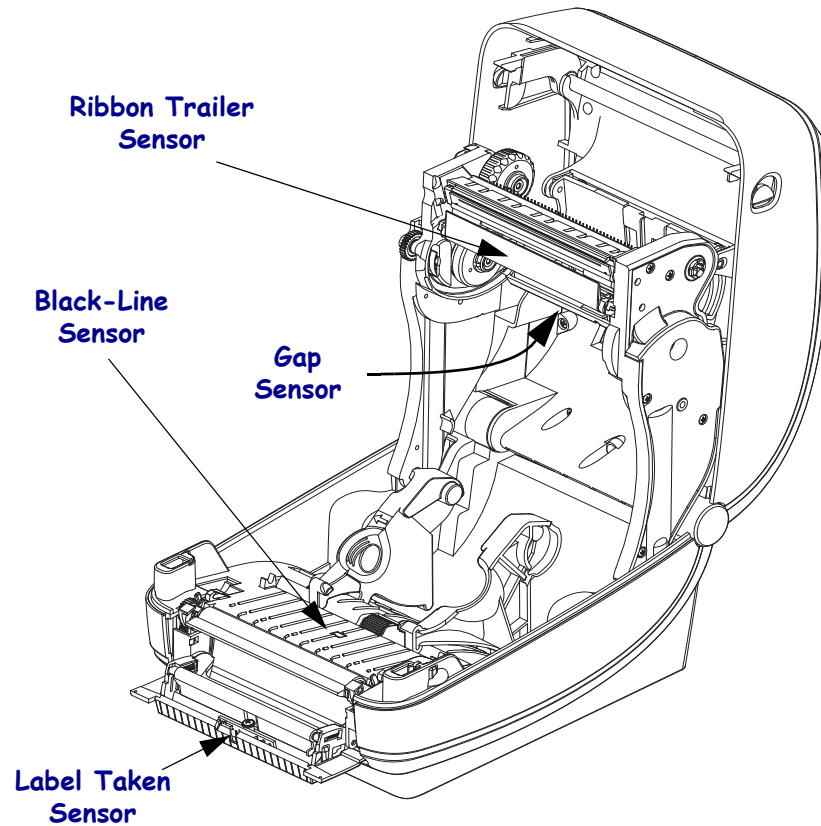




## Sensor Cleaning

Dust can accumulate on the media sensors.

1. Gently brush away dust; if necessary, use a dry swab to brush away dust. If adhesives or other contaminants remain, use an alcohol moistened swab to break it up.
2. Use a dry swab to remove any residue that may be left from the first cleaning.
3. Repeat steps 1 and 2 as required until all residue and streaks are removed from the sensor.



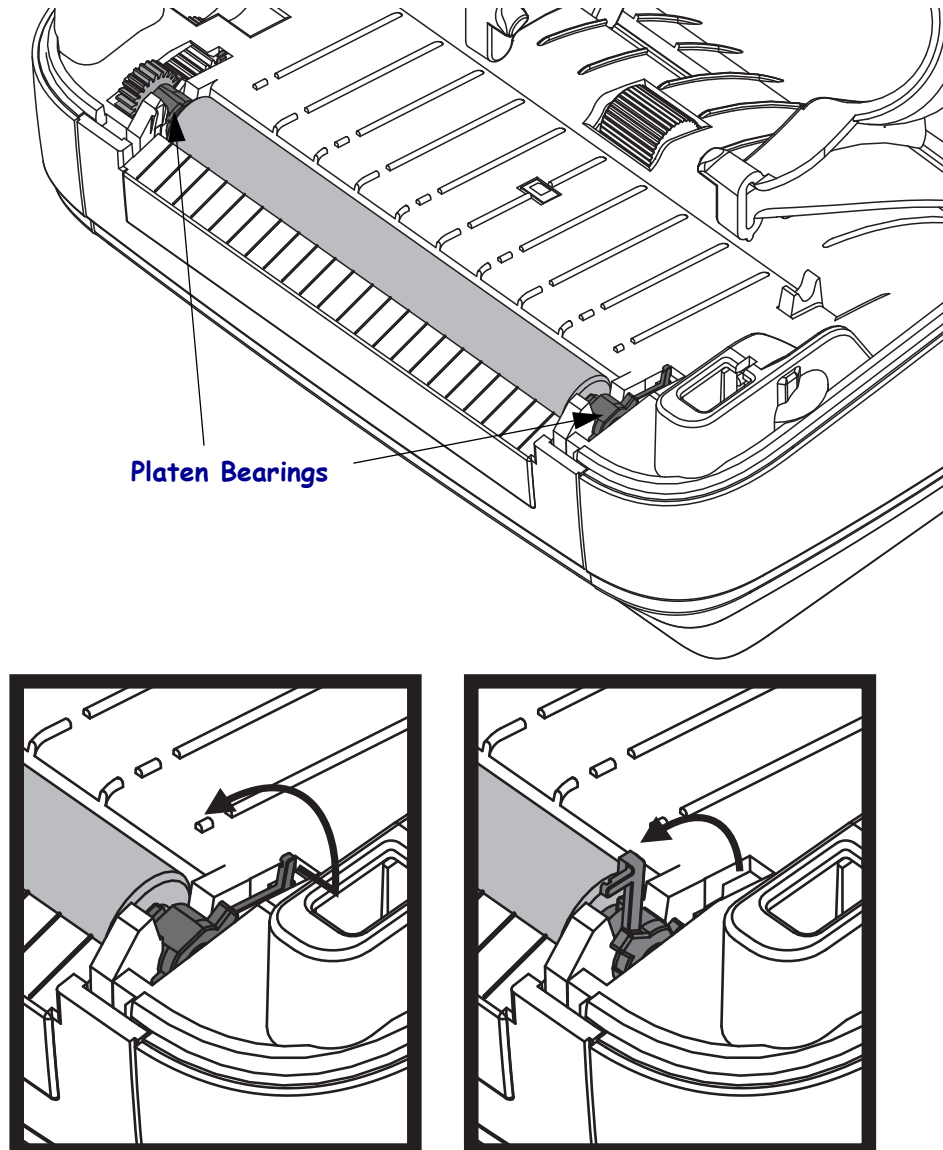
## Platen Cleaning and Replacement

The standard platen (drive roller) normally does not require cleaning. Paper and liner dust can accumulate without effecting print operations. Contaminates on the platen roller can damage the printhead or cause the media to slip or stick when printing. Adhesive, dirt, general dust, oils and other contaminants should be cleaned immediately off the platen.

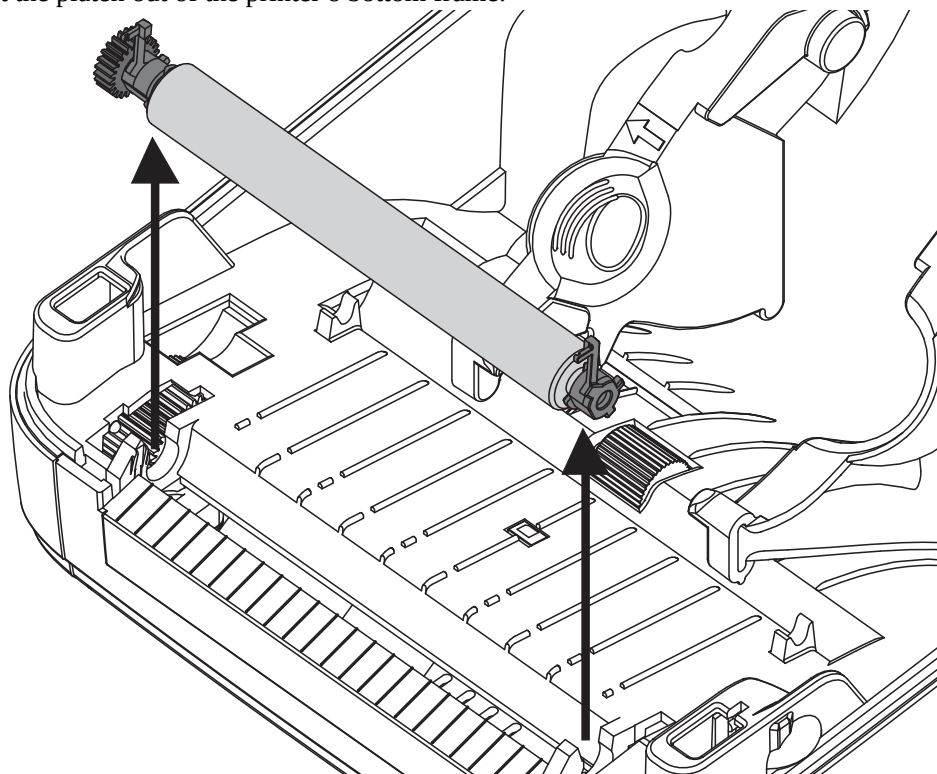
Clean the platen (and media path) whenever the printer has significantly poorer performance, print quality or media handling. The platen is the print surface and drive roller for your media. If sticking or jamming continues even after cleaning, you must replace the platen.

The platen can be cleaned with a fiber-free swab (such as a Texpad swab) or a lint free, clean, damp cloth very lightly moistened with medical grade alcohol (90% pure or better).

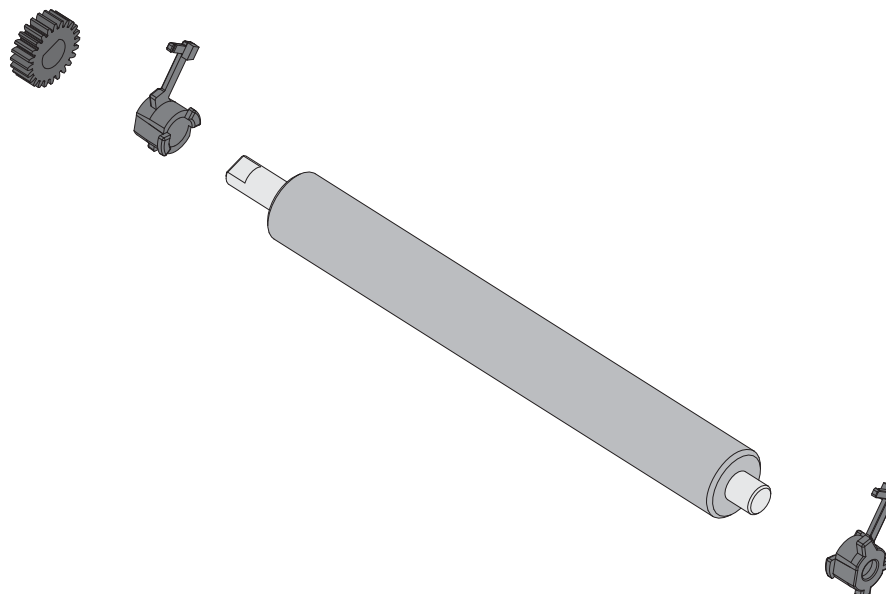
1. Open the cover (and dispenser door). Remove media from platen area.
2. Pull the platen bearing latch release tabs on the right and left sides towards the front of the printer and rotate them up.



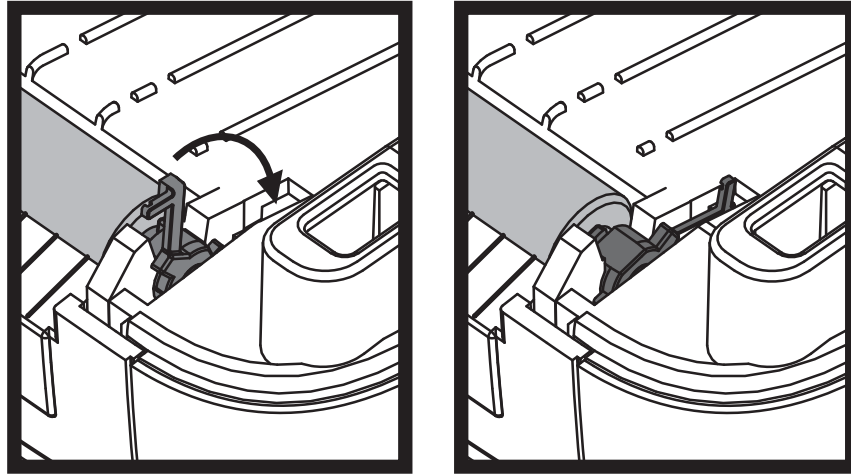
3. Lift the platen out of the printer's bottom frame.



4. Clean the platen with the alcohol moistened swab. Clean from the center out. Repeat this process until the all of the roller surface has been cleaned. If there has been heavy adhesive build-up or label jam, repeat with a new swab to remove residual contaminates. Adhesives and oils, for example, may be thinned by the initial cleaning but not completely removed.
5. Install the platen in the printer. Discard the cleaning swabs after use - do not reuse.
6. Make sure the bearings and drive gear are on the shaft of the platen.



7. Align the platen with the gear to the left and lower it into the printer's bottom frame.
8. Rotate the platen bearing latch release tabs down on the right and left sides towards the rear of the printer and snap them into place.



Allow the printer to dry for one minute before closing the dispenser door, media cover or loading labels.

## Other Printer Maintenance

There are no user level maintenance procedures beyond those detailed in this section. See the [Troubleshooting on page 95](#) for more information on diagnosing printer and print problems.

## Replacing the Printhead

If you need to replace the printhead, read the procedure and review the removal and installation steps before actually replacing the printhead.



**Caution** • Prepare your work area by protecting against static discharge. Your work area must be static-safe and include a properly grounded conductive cushioned mat to hold the printer and a conductive wrist strap for yourself.

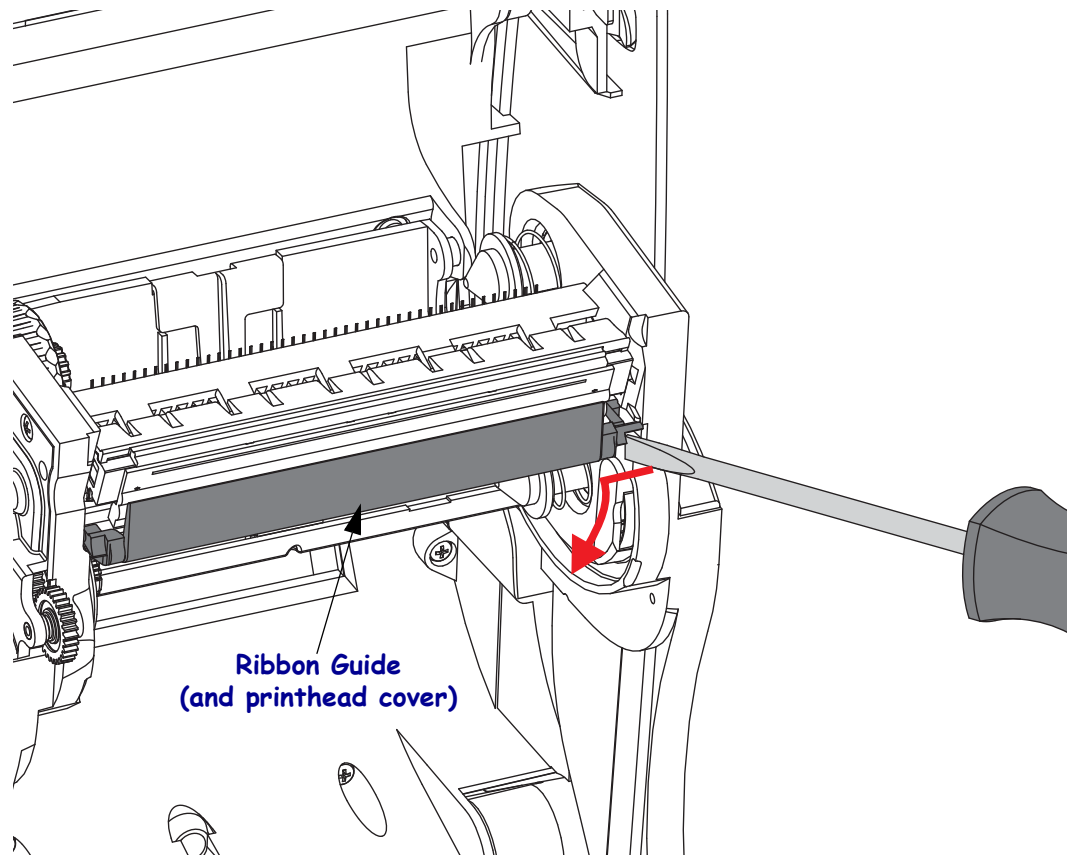


**Caution** • Turn the printer power off and unplug the power cord before replacing the printhead.

Before following the steps in this procedure, open the printer by pulling the release latches forward then lifting the top cover.

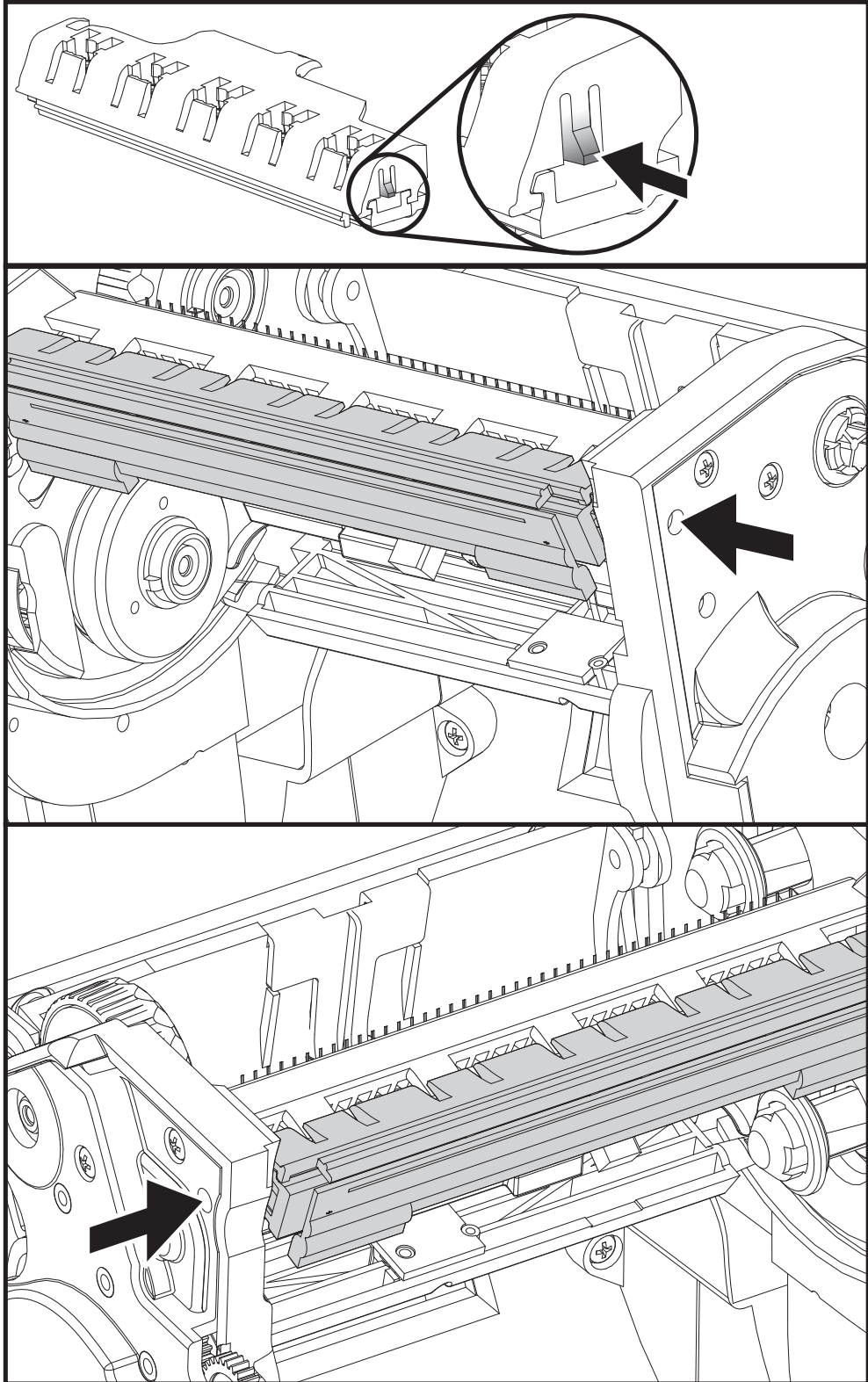
### Printhead Removal

1. Open the printer. Remove any transfer ribbon from the printer.
2. Use a small flat bladed screwdriver to remove the ribbon guide. Gently pry the guide right-side tab from the ribbon frame. Pull the left side of the ribbon guide out.

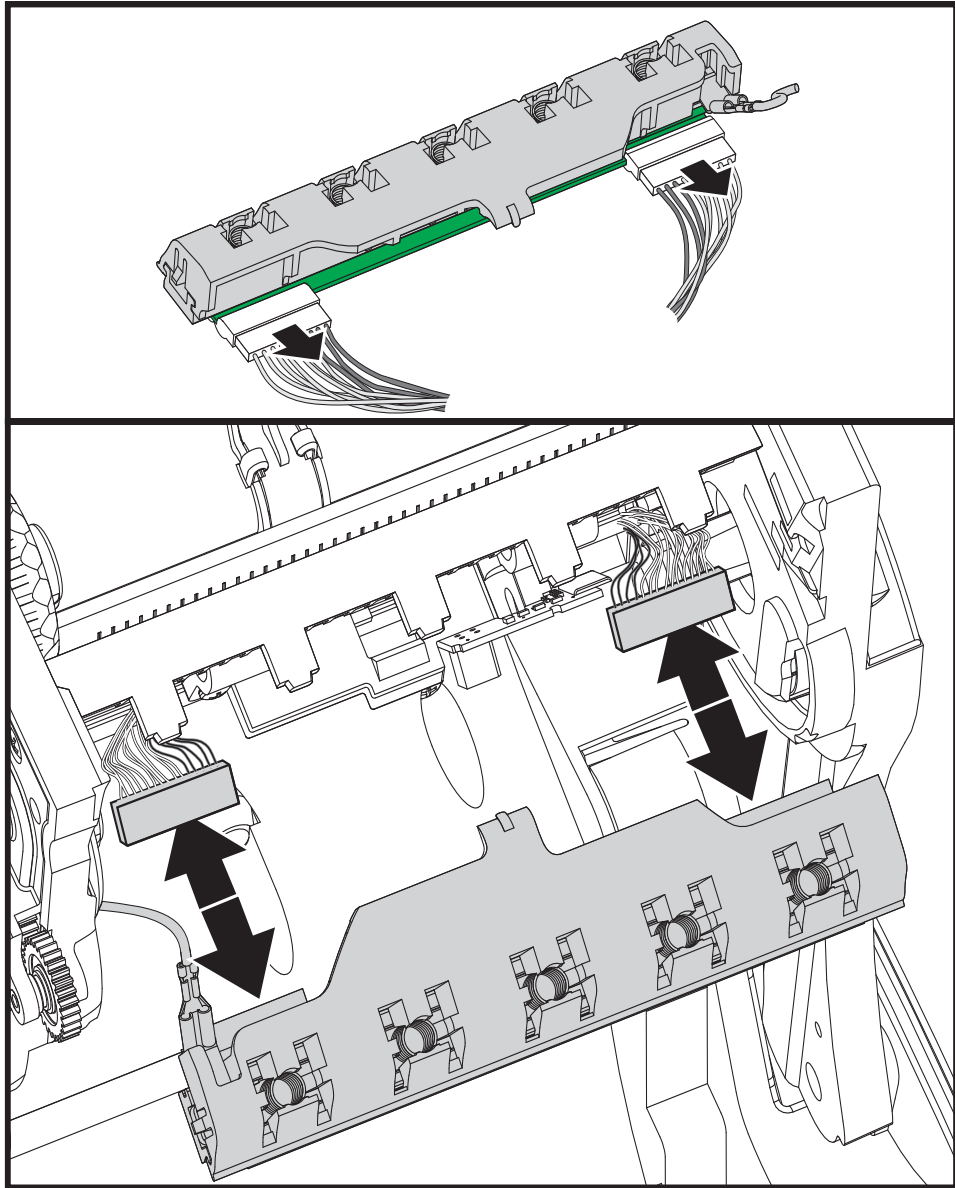


3. Use a blunt tipped tool to press the release tab on the right side and then the left side of the printhead. The tool diameter can be from 0.10 to 0.15 inches (2.5 to 3.8 mm).

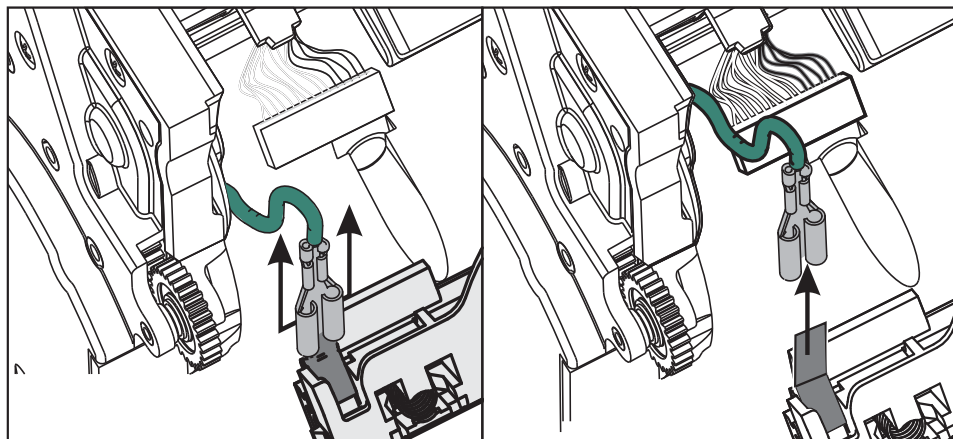
Insert the tool into the ribbon frame's printhead release access (the round) opening. Push the release tab and gently pull the printhead bracket down.



4. Gently but firmly pull the two printhead cable bundles' connectors off of the printhead.



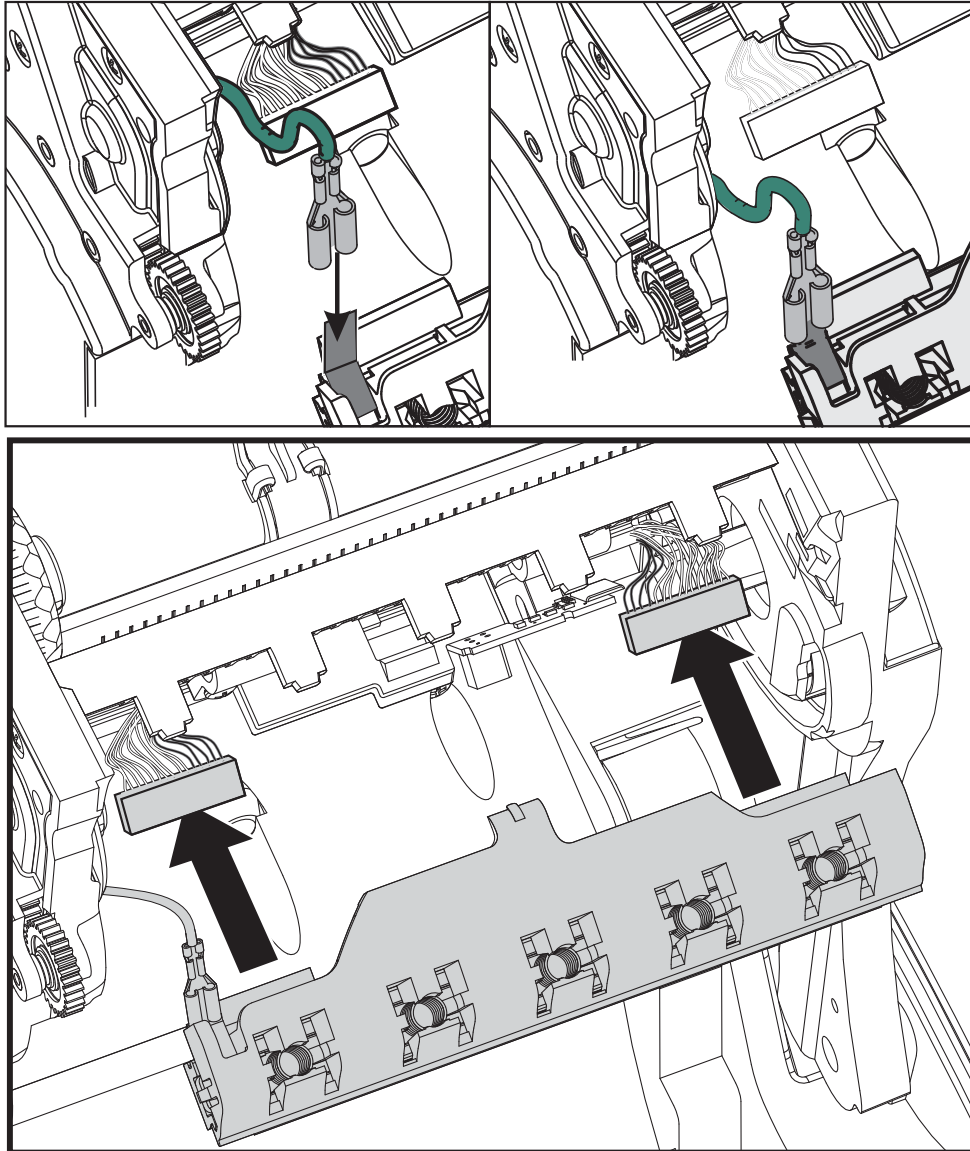
5. Gently but firmly pull the green ground wire off the printhead.





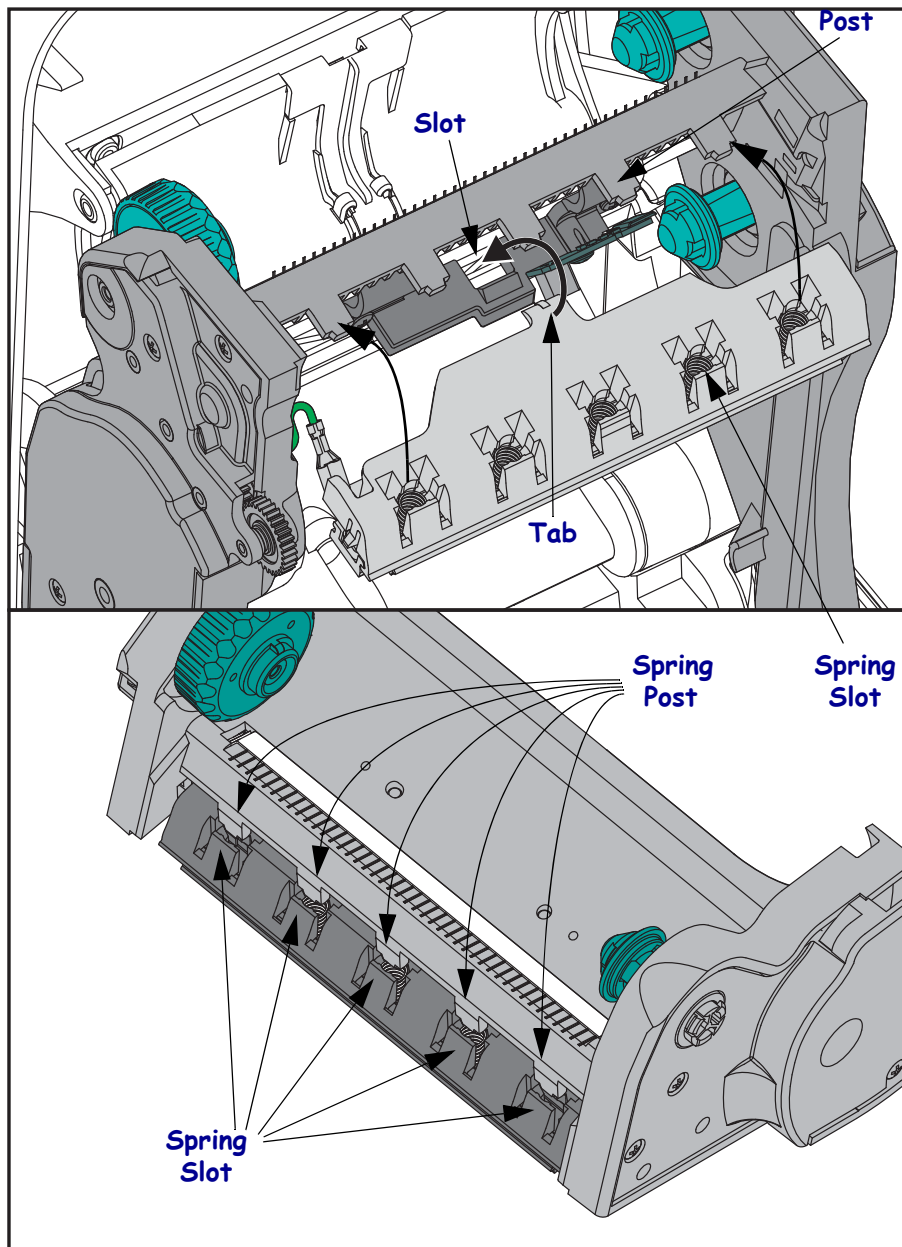
### Replacing the Printhead

1. Push the left side printhead connector into the printhead. The connector is keyed to only insert one way.
2. Connect the green ground wire to the printhead.
3. Push the right side printhead wire connector onto the printhead.
4. Check that the ground wire and wire bundles are still connected to the printhead.

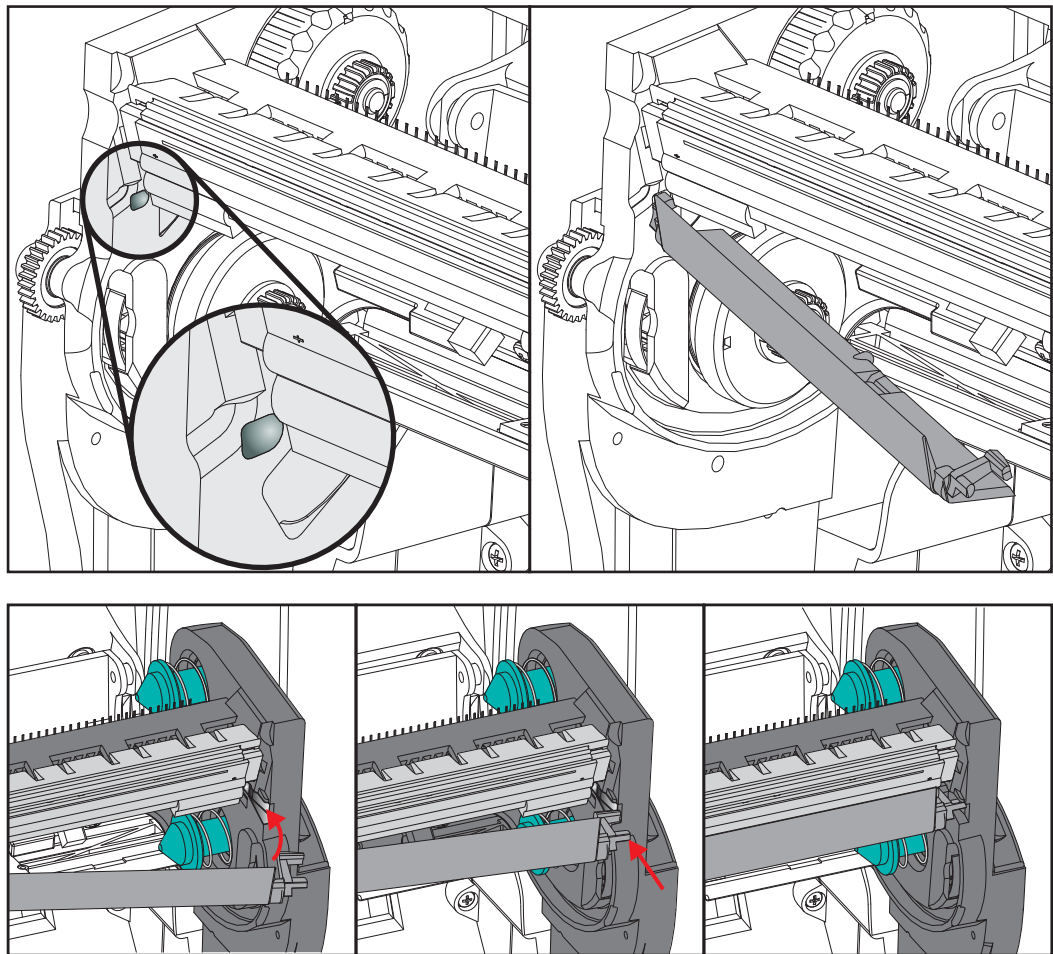




5. Insert the printhead bracket's tab into slot in the ribbon frame assembly. Align the printhead spring slots to the five posts and snap the printhead into the ribbon frame.



6. Place the left hand side of the ribbon guide into the ribbon frame. Swing the right side of the ribbon guide into the slot and snap it into position.



7. Verify that the printhead moves up and down freely when pressure is applied and remains locked when released.
8. Clean the printhead. Use a new pen to wipe body oils (finger prints) and debris of the printhead. Clean from the center of the printhead to the outside. See [Cleaning the Printhead on page 83](#).
9. Reload media. Plug in the power cord, turn on the printer and print a status report to ensure proper function. See [Printing a Test \(Printer Configuration\) Label on page 23](#).



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# Troubleshooting

This section provides information about printer error reporting that you might need for printer troubleshooting. Assorted diagnostic tests are included.

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## Status Light Descriptions

| What the Status Light is Telling You                               |                        |                                    |
|--------------------------------------------------------------------|------------------------|------------------------------------|
| LED Status and Color                                               | Printer Status         | For a Resolution, Refer to Number: |
| Off                                                                | Off                    | 1                                  |
| Solid Green                                                        | On                     | 2                                  |
| Solid Amber                                                        | Stopped                | 3                                  |
| Flashing Green                                                     | Normal Operation       | 4                                  |
| Flashing Red                                                       | Stopped                | 5                                  |
| Double Flashing Green                                              | Paused                 | 6                                  |
| Flashing Amber                                                     | Paused                 | 7                                  |
| Alternately Flashing Green and Red                                 | Needs Service          | 8                                  |
| Flashing Red, Red and Green                                        | Needs Service          | 9                                  |
| Flashing Red, Amber Green<br>(* - Do NOT reset or turn off power!) | Memory Defragmentation | 10*                                |

## Status Light Error Resolutions

The following Status Error Light Resolution numbers correspond to the Status Light Description table on the previous page. Each error number may have one or more resolution to correct that error listed.

### 1. The printer is not receiving power.

- Have you turned on the printer power?
- Check power connections from the wall outlet to the power supply, and from the power supply to the printer. See [Attaching Power on page 14](#).
- Disconnect the printer from the wall outlet for 30 seconds and then reconnect the printer to the wall outlet.

### 2. The printer is on and in an idle state.

No action necessary.

### 3. The printer has failed its power on self test (POST).

- If this error occurs right after you turn on the printer, contact an authorized reseller for assistance. When the printer is operating normally, the printer status light will be amber for about 10 seconds before turning green (solid or blinking).

### There is a memory error.

- If this error occurs after you have been printing, turn the printer power off and on, and then resume printing.

### The printhead needs to cool down.

- If this error continues, turn the printer power off for five minutes or more, and then turn on. If the amber light persists, then the printer requires service.

### 4. The printer is receiving data.

- As soon as all of the data has been received, the status LED will turn green and the printer will automatically resume operation.

## 5. The media is out.

- Follow the instructions for [Loading Roll Media on page 15](#) in the Getting Started section, and then press the Feed button to resume printing.

## The ribbon is out.

- The printer has sensed the end of the ribbon roll. Replace the ribbon.

## The printhead is open.

- Close the top cover and then press the Feed button to resume printing.

## 6. The printer is paused.

- Press the Feed button to resume printing.

## 7. The printhead is over temperature.

- Printing will stop until the printhead cools to an acceptable printing temperature. When it does, the printer will automatically resume operation.

## 8. FLASH memory is not programmed.

- Return the printer to an authorized reseller.

## 9. The printhead or motor has had a critical failure.

- Return the printer to an authorized reseller.

## 10. The printer is defragmenting memory.

**Caution** • Do NOT turn off the printer power during defragmentation. Doing so can damage the printer.

- Defragmentation is a normal operation of the printer – necessary to manage the memory space for optimal use. The printer will defragment its memory both after a factory default and when the printer detects that a defragment is required.

When the printer is in this condition, allow it to finish defragmenting. If this warning occurs frequently, check the label formats. Formats that frequently and repeatedly write to and erase from memory may cause the printer to defragment often. Using formats that avoid frequent and repeated memory write/erase event will minimize the need for defragmentation.

- If this warning condition does not go away, contact Technical Support. The printer requires service.

## Print Quality Problems

### No print on the label.

- The media may not be direct thermal media when printing without ribbon (i.e. thermal transfer). The media may not be direct thermal media. See the test procedure [Determining Thermal Media Types on page 39](#).
- For thermal transfer printers, the media may not be outside wound or approved for use in the G-Series printer. See the following ribbon test procedures: [Ribbon Test with Adhesive on page 42](#) and [Ribbon Scratch Test on page 43](#).
- Is the media loaded correctly? Follow the instructions for [Loading Roll Media on page 15](#) in the Getting Started section. For printing using transfer ribbon, see [Loading Transfer Ribbon on page 19](#).

### The printed image does not look right.

- The printhead is dirty. Clean the printhead.
- The printhead is under temperature.
- Adjust the print darkness and/or print speed.
  - Use the **^PR** (speed) and **~SD** (darkness) commands referenced in the ZPL Programming Guide.
  - Use the **D** (darkness/density) and **S** (speed) commands in the *EPL Programmer's Guide*.
  - Manually adjust print darkness with the six-flash sequence of [Feed Button Modes on page 106](#).
  - The Windows printer driver or application software may change these settings and may require a change to optimize print quality.
- The media being used is incompatible with the printer. Be sure to use the recommended media for your application, and always use Zebra-approved labels and tags.
- Verify that the printer power supply in use is rated at 100 Watts of DC output.
- The printhead has worn out. The printhead is a consumable item and will wear out due to friction between the media and printhead. Using unapproved media may shorten life or damage your printhead. Replace the printhead.
- The platen may need cleaning or replacement. The platen (driver) roller maybe losing traction due to:
  - Foreign objects attached to its surface,
  - The rubbery smooth surface has become polished and slippery, or
  - There is damage to the normally smooth and flat print surface such as box knife cuts.

## There are long tracks of missing print (blank vertical lines) on several labels.

- The printhead is dirty. Clean the printhead.
- The printhead elements are damaged.

## The printing does not start at the top of the label or misprinting of one to three labels.

- The media may not be threaded correctly. Follow the instructions for [Loading Roll Media on page 15](#) in the Getting Started section.
- The printer needs to be calibrated. Refer to the two-flash sequence of [Feed Button Modes on page 106](#) in this section.
- ZPL Label Formats - The correct media sensor may not be activated. Manual calibration selects the media sensing method for the labels being used (refer to the **^MM** command in the *ZPL Programming Guide*).
- ZPL Label Formats - Verify that the Label Top (**^LT**) command is correctly set for your application (consult the *ZPL Programming Guide*).
- EPL Label Formats - The correct media sensor may not be activated for label dispensing, blackline or notch sensing, or for Gap/Web sensing. Manual calibration selects the media sensing method for the labels being used (refer to the **O** and **Q** commands in the *EPL Programmer's Guide*).
- EPL Label Formats - Verify that the Set Label Length (**Q**) command is correctly set for your application (consult the *EPL Programmer's Guide*).

## A ZPL label format was sent to, but not recognized by, the printer.

- Is the printer in pause mode? If so, press the Feed button.
- If the status LED is on or flashing, refer to [What the Status Light is Telling You on page 96](#).
- Make sure the data cable is correctly installed.
- A communications problem has occurred. First, make sure that the correct communications port on the computer is selected. Refer to [Connecting your Printer to a Computer on page 26](#) in the Getting Started section.
- Verify the correct Format and Control Prefix on the printer match what you are using in your ZPL programmed label format. The default Format (COMMAND CHAR) is Caret (^) character and the Control (CONTROL CHAR) is a Tilde (~) character. Verify the characters with the Configuration Status label printout. Refer to the one-flash sequence of [Feed Button Modes on page 106](#) to print this label.



## A EPL label format was sent to, but not recognized by, the printer.

- Is the printer in pause mode? If so, press the Feed button.
- If the printer has label dispensing enabled, the printer may be waiting for the label to be removed. The liner/web must be properly threaded through the label dispenser mechanism (peeler) to correctly operate in label dispense mode, refer to [Label Dispenser Option on page 52](#).
- If the status LED is on or flashing, refer to [What the Status Light is Telling You on page 96](#).
- Make sure the data cable is correctly installed.
- A communications problem has occurred. First, make sure that the correct communications port (USB) on the computer is selected. Refer to [Connecting your Printer to a Computer on page 26](#) in the Getting Started section.

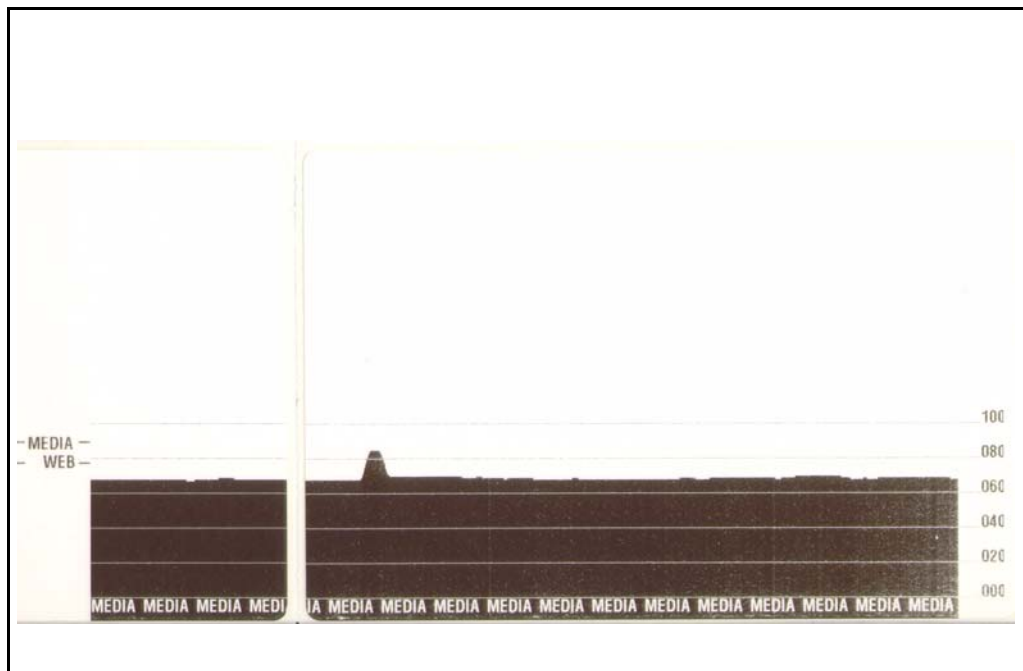
## Manual Calibration

Manual calibration is recommended whenever you are using pre-printed media or if the printer will not correctly auto calibrate.

1. Make sure media is loaded.
2. Turn on the printer power.
3. Press and hold the Feed button until the green status light flashes once, then twice and then continuing until the flash groups reach the group of seven flashes. Release the Feed button.
4. The printer will set the media sensor for the label backing being used. After it is done making this adjustment, the roll will automatically feed until a label is positioned at the printhead. A profile of the media sensor settings (similar to the example below) will print. Upon completion, the printer will save the new settings in memory and the printer is ready for normal operation.
5. Press the Feed button. One entire blank label will feed. If this does not happen, try defaulting (refer to the four-flash sequence in “Feed Button Modes” later in this chapter) and recalibrating the printer.



**Note** • Performing a manual calibration disables the auto calibration function. To return to auto calibration, default the printer (refer to the four-flash sequence in [Feed Button Modes](#) on page 106 in this section).



## Troubleshooting Tests

### Printing a Configuration Label

To print out a listing of the printer's current configuration, refer to the one-flash sequence in [Feed Button Modes on page 106](#) in this section. See [Appendix: ZPL Configuration on page 119](#) to interpret the label.

| PRINTER CONFIGURATION                   |                 |
|-----------------------------------------|-----------------|
| Zebra Technologies<br>ZTC GX420t        |                 |
| 10.0.....                               | DARKNESS        |
| 6 IPS.....                              | PRINT SPEED     |
| +000.....                               | TEAR OFF        |
| TEAR OFF.....                           | PRINT MODE      |
| GAP/NOTCH.....                          | MEDIA TYPE      |
| WEB.....                                | SENSOR TYPE     |
| AUTO.....                               | SENSOR SELECT   |
| THERMAL-TRANS.....                      | PRINT METHOD    |
| 832.....                                | PRINT WIDTH     |
| 1236.....                               | LABEL LENGTH    |
| 39.0IN 989MM.....                       | MAXIMUM LENGTH  |
| NOT CONNECTED.....                      | USB COMM.       |
| BIDIRECTIONAL.....                      | PARALLEL COMM.  |
| 9600.....                               | BAUD            |
| 8 BITS.....                             | DATA BITS       |
| NONE.....                               | PARITY          |
| DTR & XON/XOFF.....                     | HOST HANDSHAKE  |
| NONE.....                               | PROTOCOL        |
| AUTO.....                               | SER COMM. MODE  |
| LINER/TAG FULL.....                     | CUTTER TYPE     |
| <~> 7EH.....                            | CONTROL CHAR    |
| <^> 5EH.....                            | COMMAND CHAR    |
| <, > 2CH.....                           | DELIM. CHAR     |
| ZPL II.....                             | ZPL MODE        |
| NO MOTION.....                          | MEDIA POWER UP  |
| FEED.....                               | HEAD CLOSE      |
| DEFAULT.....                            | BACKFEED        |
| +000.....                               | LABEL TOP       |
| +0000.....                              | LEFT POSITION   |
| NO.....                                 | HEXDUMP         |
| 043.....                                | WEB S.          |
| 096.....                                | MEDIA S.        |
| 009.....                                | WEB GAIN        |
| 050.....                                | MARK S.         |
| 015.....                                | MARK GAIN       |
| 095.....                                | MARK MED S.     |
| 032.....                                | MARK MEDIA GAIN |
| 095.....                                | CONT MEDIA S.   |
| 007.....                                | CONT MEDIA GAIN |
| 075.....                                | RIBBON OUT      |
| 040.....                                | RIBBON GAIN     |
| 066.....                                | TAKE LABEL      |
| CWF.....                                | MODES ENABLED   |
| .....                                   | MODES DISABLED  |
| 832 8/MM FULL.....                      | RESOLUTION      |
| V56.17.7Z <-.....                       | FIRMWARE        |
| 1.3.....                                | XML SCHEMA      |
| V25.00.0207.....                        | HARDWARE ID     |
| CUSTOMIZED.....                         | CONFIGURATION   |
| 2104k.....R:                            | RAM             |
| 65536k.....B:                           | OPTION MEMORY   |
| 1536k.....E:                            | ONBOARD FLASH   |
| NONE.....                               | FORMAT CONVERT  |
| DISABLED.....                           | ZBI             |
| 2.1.....                                | ZBI VERSION     |
| 4.986 IN.....                           | LAST CLEANED    |
| 4.986 IN.....                           | HEAD USAGE      |
| 4.986 IN.....                           | TOTAL USAGE     |
| 4.986 IN.....                           | RESET CNTR1     |
| 4.986 IN.....                           | RESET CNTR2     |
| TOP-04.....                             | SERIAL NUMBER   |
| MAINT. OFF.....                         | EARLY WARNING   |
| FIRMWARE IN THIS PRINTER IS COPYRIGHTED |                 |

### Recalibration

Recalibrate the printer if it starts to display unusual symptoms, such as skipping labels. Refer to the two-flash sequence in [Feed Button Modes on page 106](#) in this section.

## Resetting the Factory Default Values

Sometimes, resetting the printer to the factory defaults may solve some problems. Refer to the four-flash sequence in [Feed Button Modes on page 106](#) in this section.

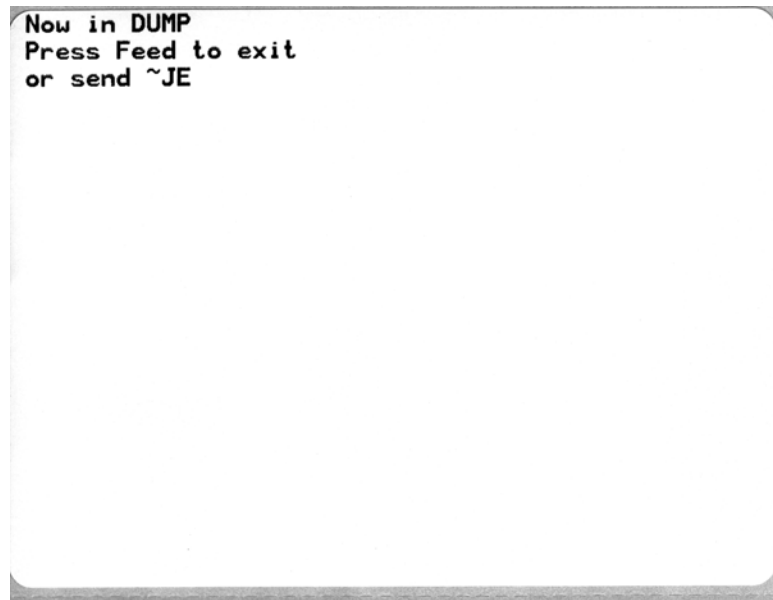
## Communications Diagnostics

If there is a problem transferring data between the computer and printer, try putting the printer in the communications diagnostics mode. The printer will print the ASCII characters and their respective hexadecimal values (a sample is shown below) for any data received from the host computer. To find out how,

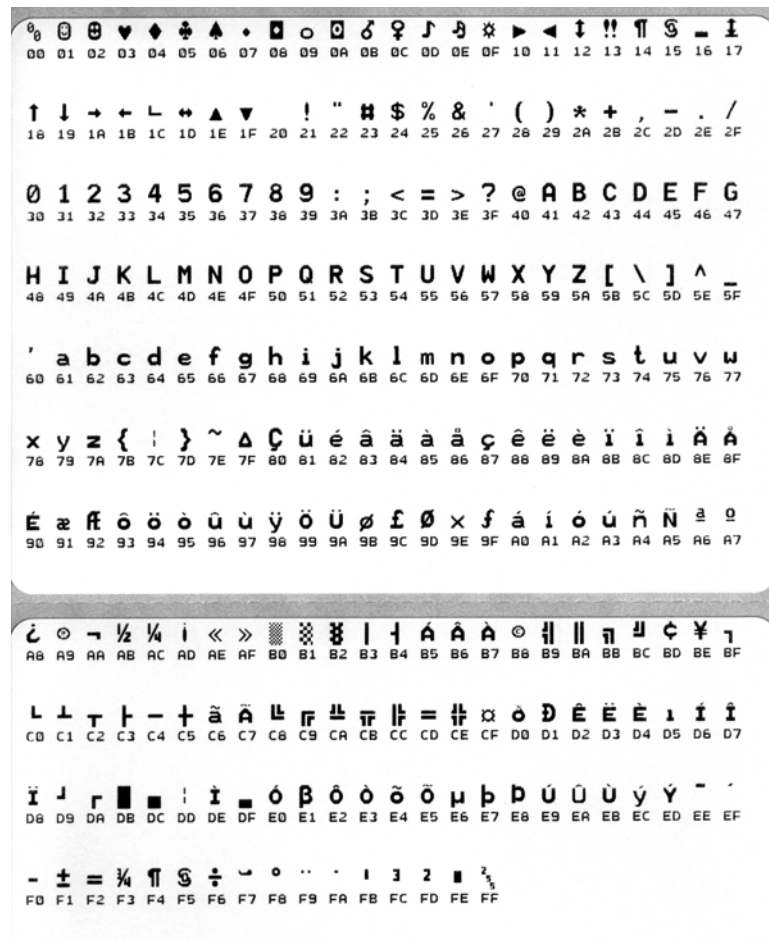
There are multiple ways to enter hex data dump mode:

- The **~JD** ZPL command
- The **dump** EPL command
- On power up with the Feed button pressed. Refer to the power off mode procedure in [Feed Button Modes on page 106](#) in this section.

The printer will print 'Now in DUMP' (see below) and advance to the top of the next label.



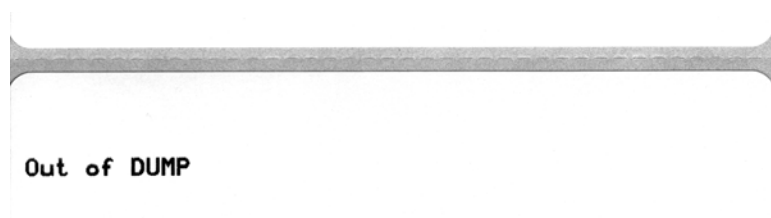
The following example of a communications DUMP mode printed out. The printout displays hexadecimal data 00h-FFh (0-255 decimal) with a unique character for each hexadecimal value displayed above the hexadecimal data.



The blank lines between the lines of data are where serial port and Bluetooth data handling errors get logged. The errors are:

- F = Frame Error
- P = Parity Error
- N = Noise Error
- O = Data Overrun Error

To exit the diagnostic mode and resume printing, turn off and then turn on the printer. An alternate method for exiting the diagnostic mode is to press the Feed button as many times as it takes to clear the printer's command buffer and print 'Out of DUMP' on the label.



## Feed Button Modes

| Power Off Modes                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| With the printer power off, press and hold the Feed button while you turn on the power. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Flash Sequence                                                                          | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Amber - Red Flashing</b>                                                             | <p><b>Firmware Download Mode</b> - The printer starts rapidly flashing red to denote entry into the Firmware Download mode. Releasing the Feed button here will start initializing the printer for download. The printer is ready to start downloading firmware when the status light begins to slowly flash between red and green.</p> <p>See <a href="#">Sending Files to the Printer on page 50</a> for more information on using the Firmware (and File) Download utility available for use with this printer. Firmware updates for your printer, if available, are posted on the Zebra on our web site at: <a href="http://www.zebra.com">www.zebra.com</a></p>                                                     |
| <b>Amber</b>                                                                            | <p><b>Normal Operations Mode</b> - The printer continues into a normal printer initialization. Releasing the Feed button here will allow the printer to start normally without firmware download or operating in communications diagnostics mode.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Green</b>                                                                            | <p><b>Communications Diagnostic (Dump) Mode</b> - Release the Feed button immediately after the printer status light turns green. The printer will print 'Now in DUMP' at the top of the label and then advance to the next label. After printing the first label, the printer will automatically enter into diagnostic mode in which the printer prints out a literal representation of all data subsequently received.</p> <p><i>To exit the diagnostic mode and resume printing, turn off and then turn on the printer. An alternate method for exiting the diagnostic mode is to press the Feed button as many times as it takes to clear the printer's command buffer and print 'Out of DUMP' on the label.</i></p> |

| Power On Modes                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| With the printer power on and top cover closed, press and hold the Feed button for several seconds. The green status LED will flash a number of times in sequence. The explanation at the right (Action) shows what happens when you release the key after the start specific number of flashes and before the next flash sequence starts. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Flash Sequence                                                                                                                                                                                                                                                                                                                             | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| *                                                                                                                                                                                                                                                                                                                                          | <b>Configuration Status</b> - Prints a detailed printer configuration status label. The label can be used to verify printing, assist printer to computer communication configuration, maintenance, troubleshooting, and help us with customer care communications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| *<br>**                                                                                                                                                                                                                                                                                                                                    | <b>Standard Auto Media Calibration</b> - The printer detects and sets media type and media length, and it adjusts the media sensors for optimal performance with the installed media (equivalent ZPL command <b>~JC</b> ). The printer will feed one to four labels while calibrating.<br><i>Note: Users familiar with the Zebra EPL desktop printer use this Feed mode to replace power-up AutoSensing calibration (equivalent EPL command <b>XA</b>).</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| *<br>**<br>***                                                                                                                                                                                                                                                                                                                             | <b>Serial Port Configuration</b> - Applies only to printers with serial interface ports.<br>To reset the serial port communication parameters, except for Flow Control. Press and release the Feed button while the LED rapidly flashes amber and green.<br>For autobaud synchronization: Send the <b>^XA^XZ</b> command sequence to the printer while the LED rapidly flashes amber and green. When the printer and host are synchronized, the LED changes to solid green. NOTE: No labels will print during autobaud synchronization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| *<br>**<br>***<br>****                                                                                                                                                                                                                                                                                                                     | <b>Factory Defaults</b> - Resets the printer to the factory default settings (equivalent to ZPL command <b>^JUN</b> ). See <a href="#">Appendix: ZPL Configuration on page 119</a> for a description of the primary configuration settings and their associated ZPL commands. Some configuration settings are not returned to their initial 'out of box' settings. Other settings are exclusively set, viewed and controlled by programming are also reset. The performs a standard media calibration and then a memory defragmentation routine.<br><br><b>Once the printer has entered the Factory Default mode</b> , the status light will turn amber for three (3) seconds. During that time you may do two things: Do nothing and the printer will reset the factory defaults automatically as described above OR press and hold the feed button to enter a factory default reset modes for printers with a network (Ethernet, Wi-Fi or Bluetooth) printer option (equivalent ZPL command <b>^JUF</b> ). Releasing the button after the first flash resets the network factory options only (equivalent ZPL command <b>^JUN</b> ). Releasing the button after the second flash sequence (two flashes) will reset the printer defaults only. Releasing the button after the third flash sequence (three flashes) will reset both printer and network settings (equivalent ZPL commands <b>^JUN</b> and <b>^JUF</b> ) |
| *<br>**<br>***<br>****<br>*****                                                                                                                                                                                                                                                                                                            | <b>Print Width Adjustment</b> - Prints a succession of boxes starting at the minimum print width and ending in the printer's maximum print width in 4mm increments. Press the Feed button once when the printer has reached the desired maximum print width. Note that the printer driver and applications can override this setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| *<br>**<br>***<br>****<br>*****<br>*****                                                                                                                                                                                                                                                                                                   | <b>Print Darkness (Density) Adjustment</b> - Prints a succession of bar code simulation patterns starting at the minimum darkness (print density/heat) and ending in the printer's maximum darkness in increments of four (4) using the ZPL darkness setting range values. Press the Feed button once the pattern is clear and legible. Do not continue to increase the darkness setting or bar code line widths may become distorted reducing readability. Note that the printer driver and applications can override this setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| *<br>**<br>***<br>****<br>*****<br>*****<br>*****                                                                                                                                                                                                                                                                                          | <b>Manual Media Calibration</b> - The printer runs extensive tests to detect and set media type and media length, and then it adjusts the media sensors for optimal performance with the installed media (equivalent ZPL command <b>~JG</b> ). Manual calibration is recommended whenever you are using pre-printed media, print on the liner or if the printer will not correctly auto calibrate. A graphical profile of the media sensing will print. See <a href="#">Manual Calibration on page 102</a> for more details and considerations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| If the Feed button remains pressed after 7th flash sequence cycle finishes, the printer exits the configuration mode when the Feed button is released.                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



Notes • \_\_\_\_\_

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# Appendix: Interface Wiring

This section provides printer interface connection information.

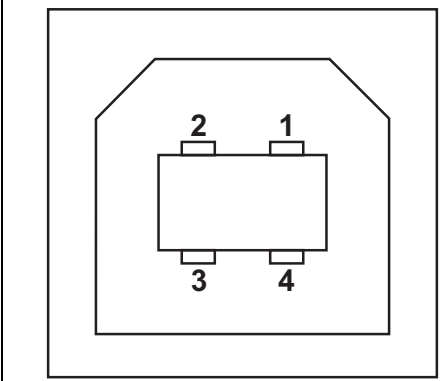
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| Parallel Interface . . . . .                   | 111 |
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| Serial Port Interface . . . . .                | 113 |

## Universal Serial Bus (USB) Interface

The figure below displays the cable wiring required to use the printer's USB interface.

The printer requires cable or cable packaging that bears the "Certified USB™" mark to guarantee USB 2.0 compliance.

|  | Pin   | Signal              |
|-----------------------------------------------------------------------------------|-------|---------------------|
|                                                                                   | 1     | Vbus - N/C          |
|                                                                                   | 2     | D-                  |
|                                                                                   | 3     | D+                  |
|                                                                                   | 4     | Ground              |
|                                                                                   | Shell | Shield / Drain Wire |

For printer supported operating systems and drivers, see the software and documentation CD or visit the Zebra Web site at:

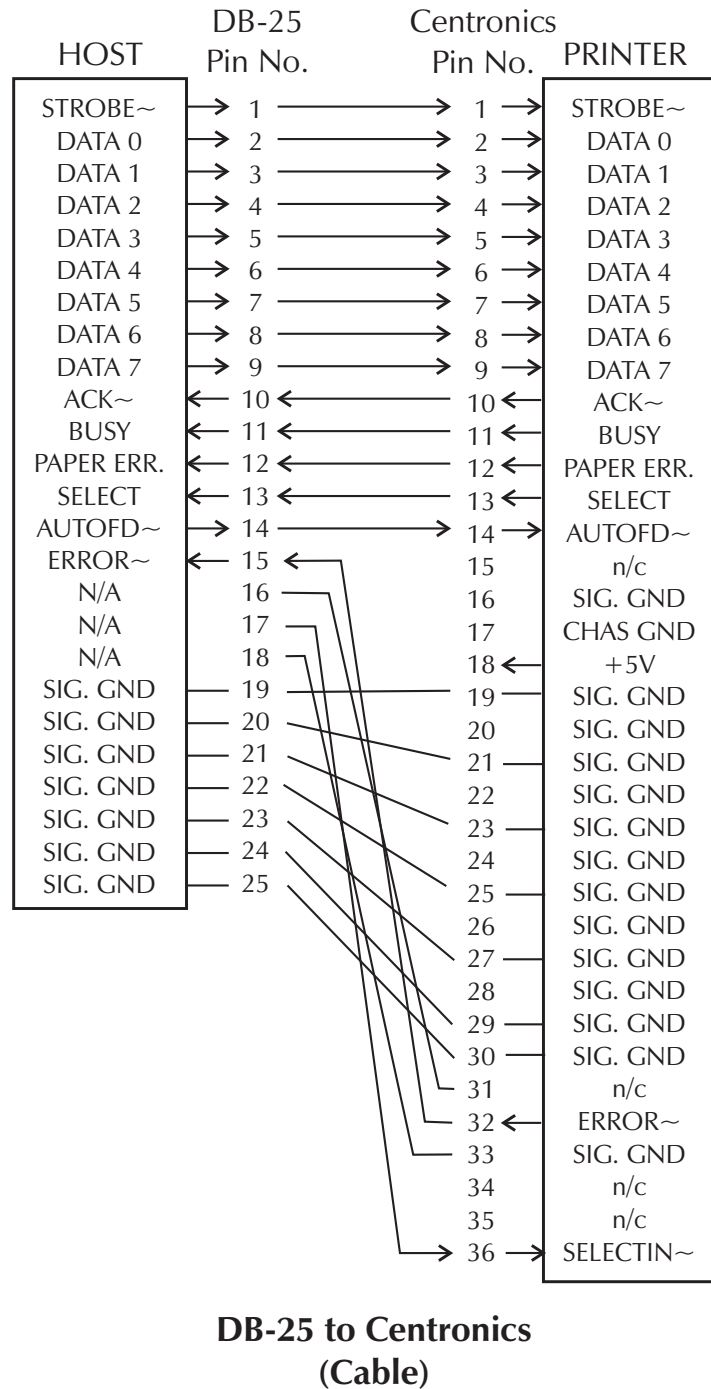
<http://www.zebra.com>

For information on the USB interface, go to the USB web site at:

<http://www.usb.org>

# Parallel Interface

The printer’s parallel interface uses the IEEE 1284-A to 1284-B parallel interface cables, the host connector has a DB-25 pin male connector. The printer connector is a Centronics style connector. Early model G-Series printers had DB-25 pin connector on the printer side, as well as, the host connection side (IEEE 1284-A to A parallel cable).

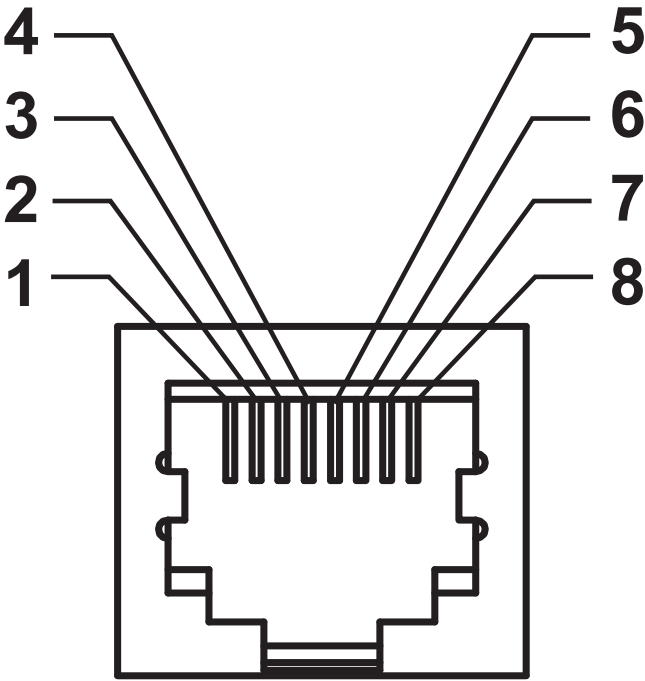


## Ethernet Interface

This interface requires a UTP RJ45 Ethernet cable rated CAT-5 or better.

The table below provides the cable’s pinout assignments.

| Signal | Pin | Pin | Signal |
|--------|-----|-----|--------|
| Tx+    | 1   | 1   | Tx+    |
| Tx-    | 2   | 2   | Tx-    |
| Rx+    | 3   | 3   | Rx+    |
| —      | 4   | 4   | —      |
| —      | 5   | 5   | —      |
| Rx-    | 6   | 6   | Rx-    |
| —      | 7   | 7   | —      |
| —      | 8   | 8   | —      |



## Serial Port Interface

| Pin | Description                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------|
| 1   | Not used                                                                                                      |
| 2   | RXD (receive data) input to the printer                                                                       |
| 3   | TXD (transmit data) output from the printer                                                                   |
| 4   | DTR (data terminal ready) output from the printer -- controls when the host may send data                     |
| 5   | Chassis ground                                                                                                |
| 6   | DSR (data set ready) input to the printer                                                                     |
| 7   | RTS (request to send) output from the printer -- always in the ACTIVE condition when the printer is turned on |
| 8   | CTS (clear to send) - Not used by the printer                                                                 |
| 9   | +5 V @ 0.75 A fused                                                                                           |

The maximum current available through the serial and/or parallel port is not to exceed a total of 0.75 Amps.

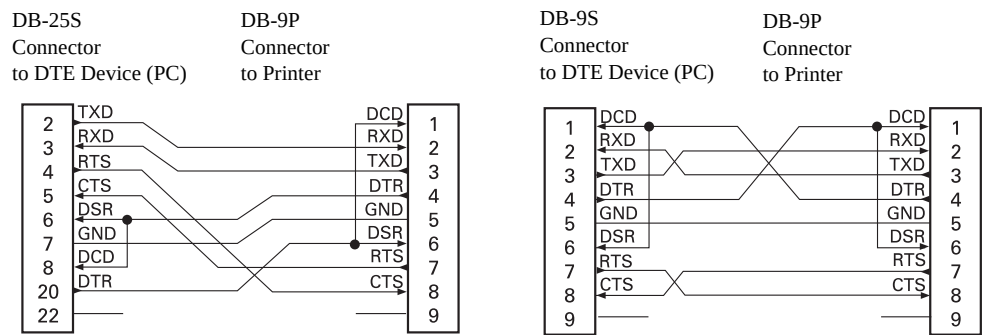
When XON/XOFF handshaking is selected, data flow is controlled by the ASCII control codes DC1 (XON) and DC3 (XOFF). The DTR control lead will have no effect.

Interconnecting to DTE Devices — The printer is configured as data terminal equipment (DTE). To connect the printer to other DTE devices (such as the serial port of a personal computer), use an RS-232 null modem (crossover) cable. Figure 31 shows the required cable connections.

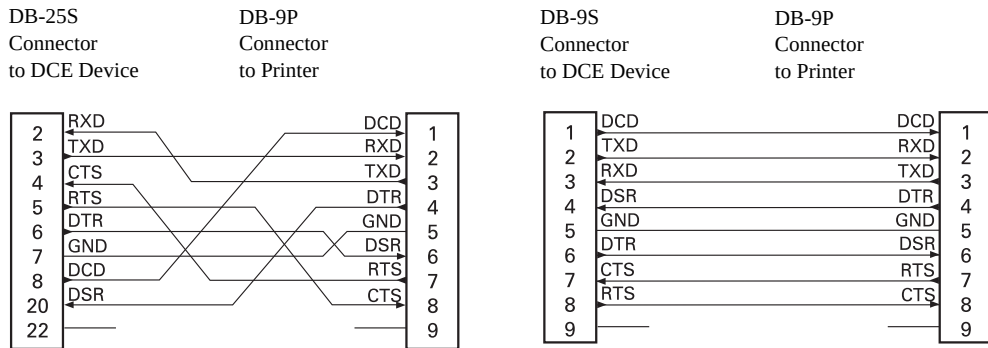
Interconnecting to DCE Devices — When the printer is connected via its RS-232 interface to data communication equipment (DCE) such as a modem, a STANDARD RS-232 (straight-through) interface cable must be used. Figure 32 shows the connections required for this cable.

Connecting to the KDU (Keyboard Display Unit) — The KDU was designed for DCE printer connections and requires a Zebra custom serial port gender changing adapter. The KDU now includes the KDU adapter and the Zebra kit part number for the KDU Adapter is 105934-088.

Connecting the Printer to a DTE Device



Connecting the Printer to a DCE Device





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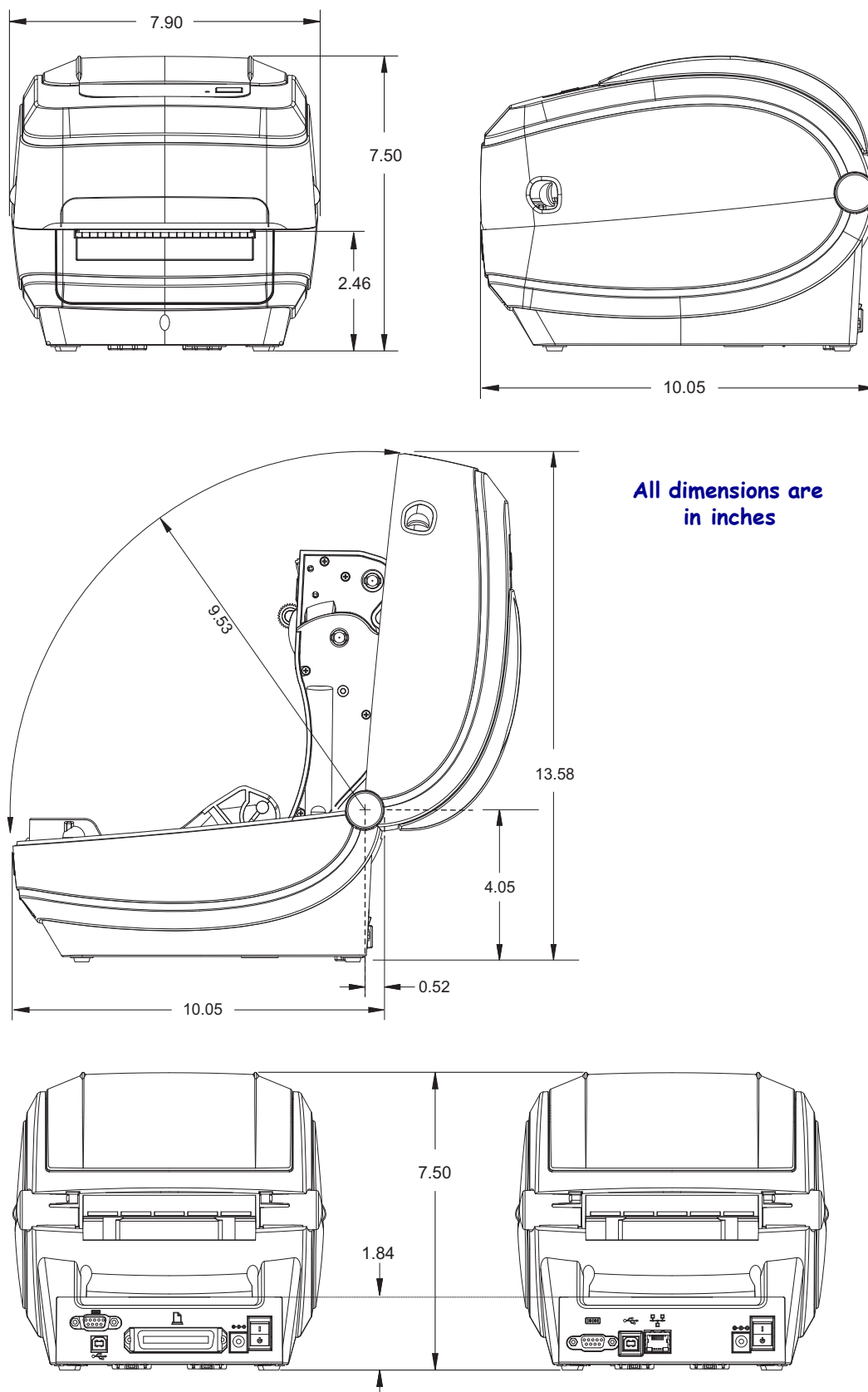
# Appendix: Dimensions

This section provides external printer dimensions.

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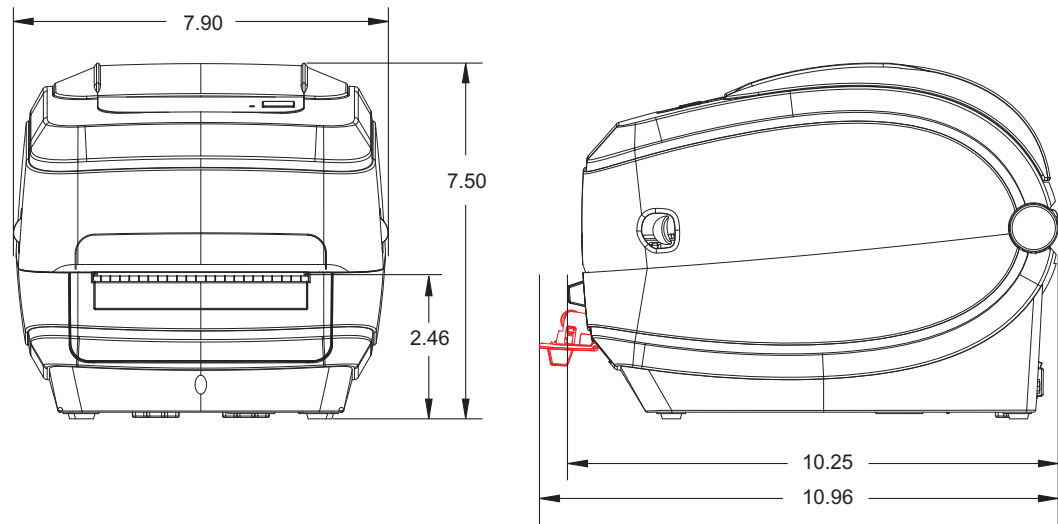
|                                                 |     |
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## External GX Printer Dimensions

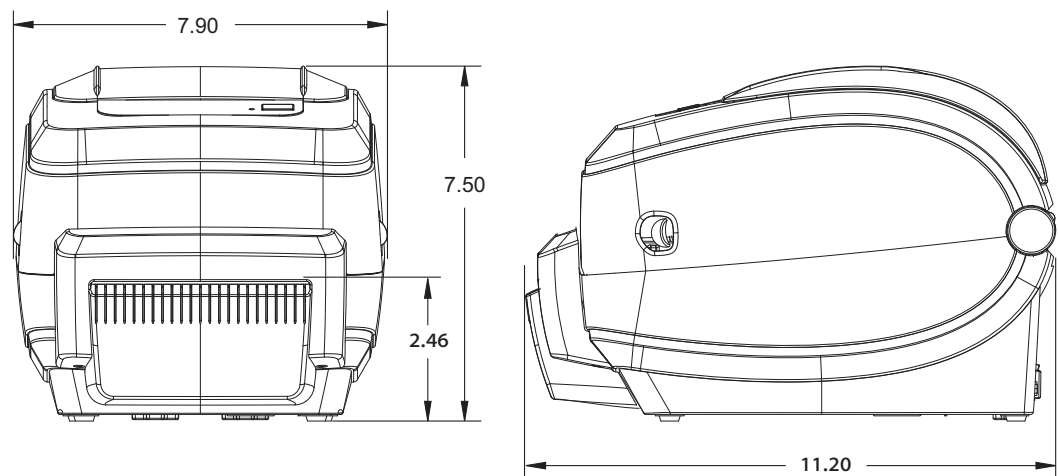


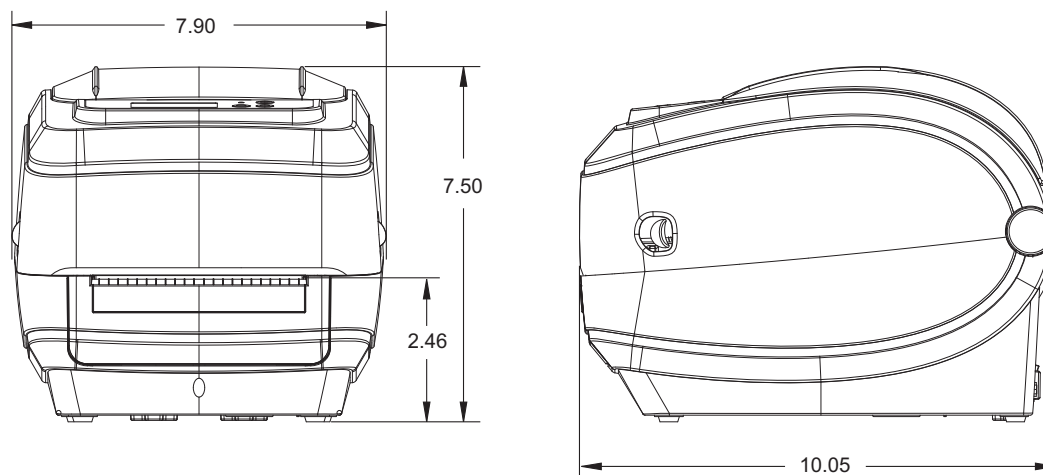
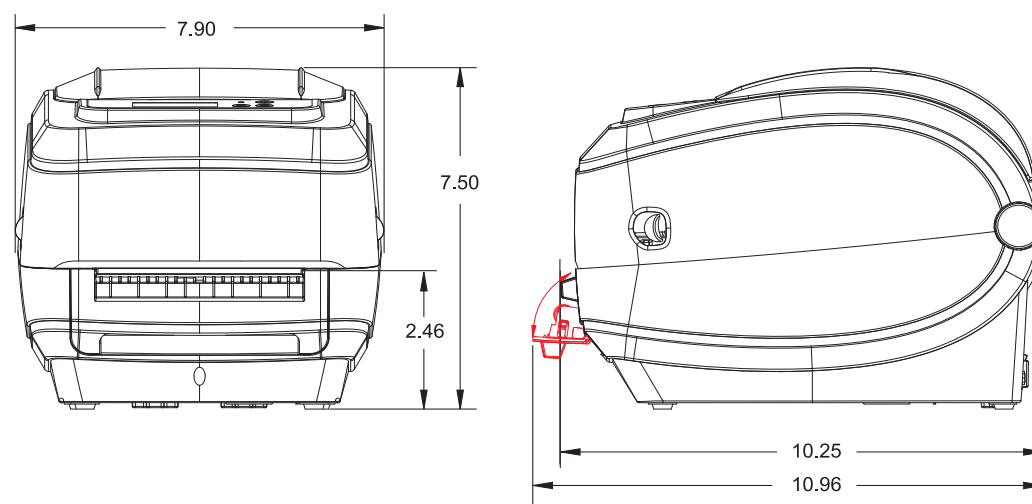
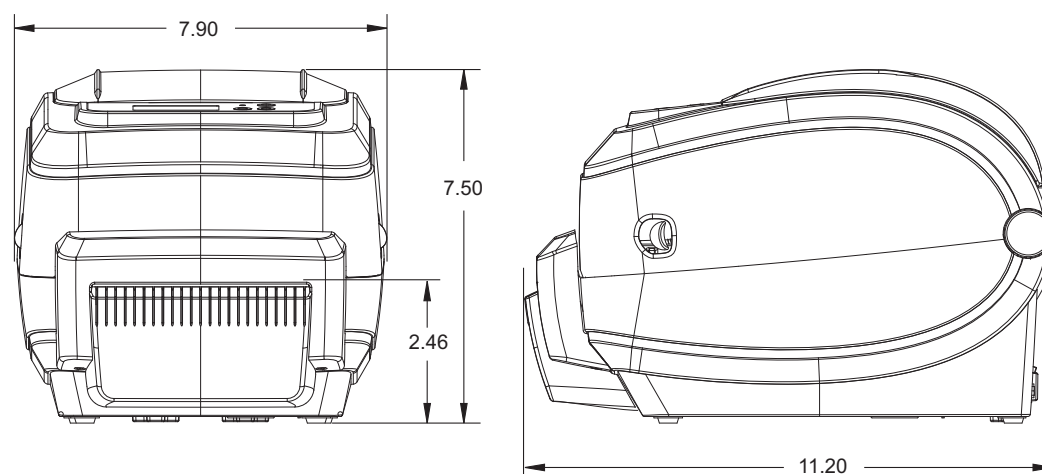


## Label Dispenser



## Cutter



**Wireless Printer****Wireless Printer with Label Dispenser****Wireless Printer with Cutter**



---

# Appendix: ZPL Configuration

This section provides a basic overview of managing printer configuration, the Configuring Status Printout, and Printer Memory Printouts.

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## Managing the ZPL Printer Configuration

The ZPL printer is designed to allow you to change printer settings dynamically for fast first label out printing. Printer parameters that are persistent, will be retained for following formats to use. Those settings will remain in effect until they are changed by subsequent commands, the printer is reset, power is cycled, or when you restore a parameter that has a factory default by using the Feed Button Mode four (4) flash routine. The ZPL Configuration Update command (**^JU**) saves and restores printer configurations to initialize (or re-initialize) the printer with pre-configured settings.

- To keep settings after a power cycle or printer reset, a **^JUS** can be sent to the printer to save all current persistent settings.
- The values are recalled with a **^JUR** command to restore the last saved values to you printer.

ZPL stores all parameters at once with a single command discussed above. The legacy EPL programming language (supported by this printer) changes and saves individual commands immediately. Most configuration and settings are shared between ZPL and EPL. For example, changing the speed setting with EPL will also change the speed set for ZPL operations. The changed EPL setting will persist even after a power cycle or reset issued by either printer language.

To assist the developer, the printer has a listing of the operating parameters, a Printer Configuration Label. It can be accessed by using the printer's Feed button and the [Printing a Test \(Printer Configuration\) Label on page 23](#). The Zebra Setup Utility and the ZebraDesigner™ Windows driver also print this label and other printer status labels to help you manage your printer.

### ZPL Printer Configuration Format

Managing more than one printer can be done creating a printer configuration programming file to send to one or more printers or ZebraNet™ Bridge can be used to clone a printer's setup. Figure 1 below shows the basic structure of a ZPL programming configuration file.

See the ZPL programmer's guide and the [ZPL Configuration Status to Command Cross-reference on page 121](#) to create a programming file. The Zebra Setup Utility (ZSU) can be used to send programming files to the printer. Windows Notepad (text editor) can be used to create programing files.

**Figure 1 • Configuration Parameter Format Structure**

**^XA** — Start Format Command

Format Commands are order sensitive

- a) General Print and command settings
- b) Media handling and behaviors
- c) Media print size

**^ JUS** command to save

**^XZ** — End Format Command

## ZPL Configuration Status to Command Cross-reference

The Printer Configuration Label, shown below, provides a listing of a majority of the configurations settings that can be set by ZPL command.

Figure 2 • Configuration Label Printout

| PRINTER CONFIGURATION                   |                 |
|-----------------------------------------|-----------------|
| Zebra Technologies<br>ZTC GX420t        |                 |
| 10.0.....                               | DARKNESS        |
| 6 IPS.....                              | PRINT SPEED     |
| +000.....                               | TEAR OFF        |
| TEAR OFF.....                           | PRINT MODE      |
| GAP/NOTCH.....                          | MEDIA TYPE      |
| WEB.....                                | SENSOR TYPE     |
| AUTO.....                               | SENSOR SELECT   |
| THERMAL-TRANS.....                      | PRINT METHOD    |
| 832.....                                | PRINT WIDTH     |
| 1236.....                               | LABEL LENGTH    |
| 39.0IN 989MM.....                       | MAXIMUM LENGTH  |
| NOT CONNECTED.....                      | USB COMM.       |
| BIDIRECTIONAL.....                      | PARALLEL COMM.  |
| 9600.....                               | BAUD            |
| 8 BITS.....                             | DATA BITS       |
| NONE.....                               | PARITY          |
| DTR & XON/XOFF.....                     | HOST HANDSHAKE  |
| NONE.....                               | PROTOCOL        |
| AUTO.....                               | SER COMM. MODE  |
| LINER/TAG FULL.....                     | CUTTER TYPE     |
| <~> 7EH.....                            | CONTROL CHAR    |
| <^> 5EH.....                            | COMMAND CHAR    |
| <, > 2CH.....                           | DELIM. CHAR     |
| ZPL II.....                             | ZPL MODE        |
| NO MOTION.....                          | MEDIA POWER UP  |
| FEED.....                               | HEAD CLOSE      |
| DEFAULT.....                            | BACKFEED        |
| +000.....                               | LABEL TOP       |
| +0000.....                              | LEFT POSITION   |
| NO.....                                 | HEXDUMP         |
| 043.....                                | WEB S.          |
| 096.....                                | MEDIA S.        |
| 009.....                                | WEB GAIN        |
| 050.....                                | MARK S.         |
| 015.....                                | MARK GAIN       |
| 095.....                                | MARK MED S.     |
| 032.....                                | MARK MEDIA GAIN |
| 095.....                                | CONT MEDIA S.   |
| 007.....                                | CONT MEDIA GAIN |
| 075.....                                | RIBBON OUT      |
| 040.....                                | RIBBON GAIN     |
| 066.....                                | TAKE LABEL      |
| CWF.....                                | MODES ENABLED   |
| .....                                   | MODES DISABLED  |
| 832 8/MM FULL.....                      | RESOLUTION      |
| V56.17.7Z <-.....                       | FIRMWARE        |
| 1.3.....                                | XML SCHEMA      |
| V25.00.0207.....                        | HARDWARE ID     |
| CUSTOMIZED.....                         | CONFIGURATION   |
| 2104k.....R:                            | RAM             |
| 65536k.....B:                           | OPTION MEMORY   |
| 1536k.....E:                            | ONBOARD FLASH   |
| NONE.....                               | FORMAT CONVERT  |
| DISABLED.....                           | ZBI             |
| 2.1.....                                | ZBI VERSION     |
| 4,986 IN.....                           | LAST CLEANED    |
| 4,986 IN.....                           | HEAD USAGE      |
| 4,986 IN.....                           | TOTAL USAGE     |
| 4,986 IN.....                           | RESET CNTR1     |
| 4,986 IN.....                           | RESET CNTR2     |
| TOP-04.....                             | SERIAL NUMBER   |
| MAINT. OFF.....                         | EARLY WARNING   |
| FIRMWARE IN THIS PRINTER IS COPYRIGHTED |                 |

Sensor Settings  
used for Service  
Purposes

Table 2 • ZPL Commands and Configuration Receipt Callout Cross-Reference

| Command               | Listing Name          | Description                                                                                                       |
|-----------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>~SD</b>            | <b>DARKNESS</b>       | <i>Default: 10.0</i>                                                                                              |
| <b>^PR</b>            | <b>PRINT SPEED</b>    | <i>Default: 6 IPS / 152.4 mm/s (max.)</i>                                                                         |
| <b>~TA</b>            | <b>TEAR OFF</b>       | <i>Default: +000</i>                                                                                              |
| <b>^MN</b>            | <b>MEDIA TYPE</b>     | <i>Default: GAP/NOTCH</i>                                                                                         |
|                       | <b>SENSOR TYPE</b>    | <i>Default: WEB</i>                                                                                               |
|                       | <b>SENSOR SELECT</b>  | <i>Default: AUTO (^MNA - Auto-Detect)</i>                                                                         |
| <b>^MT</b>            | <b>PRINT METHOD</b>   | <i>Default: NONE</i>                                                                                              |
| <b>^MT</b>            | <b>PRINT METHOD</b>   | <b>THERMAL-TRANS</b> or <b>DIRECT-THERMAL</b>                                                                     |
| <b>^PW</b>            | <b>PRINT WIDTH</b>    | <i>Default: 832 (dots)</i>                                                                                        |
| <b>^LL</b>            | <b>LABEL LENGTH</b>   | <i>Default: 1225 (dots) (the value constantly re-calibrates with automatic Web-Gap detection for label media)</i> |
| <b>^ML</b>            | <b>MAXIMUM LENGTH</b> | <i>Default: 39.0IN 989MM</i>                                                                                      |
| <b>—</b>              | <b>USB COMM.</b>      | <i>Connection Status: Connected / Not Connected</i>                                                               |
| <b>—</b>              | <b>PARALLEL COMM.</b> | <i>Connection Available: BIDIRECTIONAL</i>                                                                        |
| <b>^SCa</b>           | <b>BAUD</b>           | <i>Default: 9600</i>                                                                                              |
| <b>^SC, b</b>         | <b>DATA BITS</b>      | <i>Default: 8 BITS</i>                                                                                            |
| <b>^SC, , c</b>       | <b>PARITY</b>         | <i>Default: NONE</i>                                                                                              |
| <b>^SC, , , , e</b>   | <b>HOST HANDSHAKE</b> | <i>Default: DTR &amp; XON/XOFF</i>                                                                                |
| <b>^SC, , , , , f</b> | <b>PROTOCOL</b>       | <i>Default: NONE</i>                                                                                              |
|                       | <b>SER COMM. MODE</b> | <i>Default: AUTO</i>                                                                                              |
|                       | <b>CUTTER TYPE</b>    | <i>Option Present: LINER/TAG FULL (cut)</i>                                                                       |
| <b>^CT / ~CT</b>      | <b>CONTROL CHAR</b>   | <i>Default: &lt;~&gt; 7EH</i>                                                                                     |
| <b>^CC / ~CC</b>      | <b>COMMAND CHAR</b>   | <i>Default: &lt;^&gt; 5EH</i>                                                                                     |
| <b>^CD / ~CD</b>      | <b>DELIM./CHAR</b>    | <i>Default: &lt;, &gt; 2CH</i>                                                                                    |
| <b>^SZ</b>            | <b>ZPL MODE</b>       | <i>Default: ZPL II</i>                                                                                            |
| <b>^MFa</b>           | <b>MEDIA POWER UP</b> | <i>Default: NO MOTION</i>                                                                                         |
| <b>^MF, b</b>         | <b>HEAD CLOSE</b>     | <i>Default: FEED</i>                                                                                              |
| <b>~JS</b>            | <b>BACKFEED</b>       | <i>Default: DEFAULT</i>                                                                                           |
| <b>^LT</b>            | <b>LABEL TOP</b>      | <i>Default: +000</i>                                                                                              |
| <b>^LS</b>            | <b>LEFT POSITION</b>  | <i>Default: +0000</i>                                                                                             |
| <b>~JD / ~JE</b>      | <b>HEXDUMP</b>        | <i>Default: NO (~JE)</i>                                                                                          |

From this point in the Configuration Receipt listing, the printout has sensor settings and values are displayed to troubleshoot sensor and media operations. These are typically used by Zebra Tech Support to diagnose printer problems.

The configuration settings listed here resume after the **TAKE LABEL** sensor value. These listings contain printer features seldom changed from default or provide status information (e.g. Firmware Version).

**Table 3 • ZPL Commands and Configuration Receipt Callout Cross-Reference**

| Command                                | Listing Name          | Description                                                                                       |
|----------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------|
| <b>^MP</b>                             | <b>MODES ENABLED</b>  | <i>Default: CWF</i> (See <b>^MP</b> Command)<br>( <b>CWFM</b> for wireless printers with display) |
|                                        | <b>MODES DISABLED</b> | <i>Default: (nothing set)</i>                                                                     |
| <b>^JM</b>                             | <b>RESOLUTION</b>     | Default: <b>832 8/mm FULL</b> (203 dpi)<br><b>1280 8/mm FULL</b> (300 dpi)                        |
| —                                      | <b>FIRMWARE</b>       | Lists ZPL Firmware Version                                                                        |
| —                                      | <b>XML SCHEMA</b>     | <b>1.3</b>                                                                                        |
| —                                      | <b>HARDWARE ID</b>    | Lists Firmware Boot-block Version                                                                 |
| —                                      | <b>CONFIGURATION</b>  | <b>CUSTOMIZED</b> (after first use)                                                               |
| —                                      | <b>RAM</b>            | <b>2104k..... R:</b>                                                                              |
| —                                      | <b>OPTION MEMORY</b>  | <b>65536k.....B:</b> ( <i>only displays this if installed</i> )                                   |
| —                                      | <b>ONBOARD FLASH</b>  | <b>1536k.....E:</b>                                                                               |
| <b>^MU</b>                             | <b>FORMAT CONVERT</b> | <b>NONE</b>                                                                                       |
| <b>^JI / ~JI</b>                       | <b>ZBI</b>            | <b>DISABLED</b> ( <i>Requires key to enable</i> )                                                 |
| —                                      | <b>ZBI VERSION</b>    | <b>2.1</b>                                                                                        |
| <b>^JH</b><br><b>^MA</b><br><b>~R0</b> | <b>LAST CLEANED</b>   | <b>X,XXX IN</b>                                                                                   |
|                                        | <b>HEAD USAGE</b>     | <b>X,XXX IN</b>                                                                                   |
|                                        | <b>TOTAL USAGE</b>    | <b>X,XXX IN</b>                                                                                   |
|                                        | <b>RESET CNTR1</b>    | <b>X,XXX IN</b>                                                                                   |
|                                        | <b>RESET CNTR1</b>    | <b>X,XXX IN</b>                                                                                   |
| —                                      | <b>SERIAL NUMBER</b>  | <b>XXXXXXXXXXXX</b>                                                                               |
| <b>^JH</b>                             | <b>EARLY WARNING</b>  | <b>MAINT. OFF</b>                                                                                 |

The printer has the ability to set a command or a group of commands once for all receipts (or labels) that follow. Those settings will remain in effect until they are changed by subsequent commands, the printer is reset, or you restore factory defaults.

## Printer Memory Management and Related Status Reports

To help you manage printer resources, the printer supports a variety of format commands to manage memory, transfer objects (between memory areas, import and export), object naming, and provide various printer operating status reports. They are very similar to the old DOS commands like DIR (directory listing) and DEL (delete file). The most common reports are also part of the Zebra Setup Utility and ZebraDesigner™ Windows driver.

**Figure 3 • Memory Management Format Structure**

**^XA — Start Format Command**

A Single Format Command is recommended for purposes of reuse

**^XZ — End Format Command**

It is recommended that a single command be processed within a this type of format (form). A single command is easily reused as a maintenance and development tool.

Many of the commands that transfer objects, manage and report on memory are control (~) commands. They do not need to be within a format (form). They will be processed immediately upon receipt by the printer whether in a format (form) or not.



**Note •** To maximize available printer memory, the printer includes an automatic memory defragmentation (defrag). Several factors can trigger a defrag operation. Changing memory by deleting or adding objects to memory can cause a defrag. The printer status light will begin flashing Red, Amber and Green when a memory defrag is in process. Do not turn off the printer's power while the status light is flashing. This operation may take several minutes if memory use and file fragmentation are high.

## ZPL Programming for Memory Management

ZPL has various printer memory locations that are used to run the printer, assemble the print image, store formats (forms), graphics, fonts and configuration settings.

- ZPL treats Formats (Forms), Fonts, and Graphics like files; and memory locations like disk drives in the DOS operating system environment:
  - Memory Object Naming: Up to sixteen (16) alphanumeric characters followed by a three (3) alphanumeric character file extension:  
i.e. 123456789ABCDEF.TTF  
Legacy ZPL printers with V60.13 firmware and earlier can only use the 8.3 file name format versus today's 16.3 file name format.
- Allows moving objects between memory locations and deleting objects.
- Supports DOS directory style file list reports as printouts or status to the host.
- Allows use of 'wild cards' (\*) in file access



Table 4 • Object Management and Status Report Commands

| Command    | Name                             | Description                                                                                                                                                                                                                                        |
|------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>^WD</b> | Print Directory Label            | Prints a list of objects and resident bar codes and fonts in all addressable memory locations                                                                                                                                                      |
| <b>~WC</b> | Print Configuration Label        | Prints a configuration Status Receipt (Label)<br>Same as Feed Button mode one flash routine                                                                                                                                                        |
| <b>^ID</b> | Object Delete                    | Deletes objects from printer memory                                                                                                                                                                                                                |
| <b>^TO</b> | Transfer Object                  | Used to copy an object or group of objects from one memory area to another.                                                                                                                                                                        |
| <b>^CM</b> | Change Memory Letter Designation | Reassigns a letter designation to a printer memory area.                                                                                                                                                                                           |
| <b>^JB</b> | Initialize Flash memory          | Similar to formatting a disk - erases all objects from the specified memory locations B: or E: .                                                                                                                                                   |
| <b>~JB</b> | Reset Optional Memory            | Similar to formatting a disk - erases all objects from the B: memory (factory option).                                                                                                                                                             |
| <b>~DY</b> | Download Objects                 | Downloads and installs wide variety of printer usable programming objects: fonts (OpenType and TrueType), graphics, and other object data types.<br><i>Recommendation: Use ZebraNet™ Bridge for downloading graphics and fonts in the printer.</i> |
| <b>~DG</b> | Download Graphic                 | Downloads an ASCII Hex representation of a graphic image. <i>This is used by ZebraDesigner™ (label creation application) for graphics.</i>                                                                                                         |
| <b>^FL</b> | Font Linking                     | Appends secondary TrueType font or fonts to the primary TrueType font to add glyphs (characters).                                                                                                                                                  |
| <b>^LF</b> | List Font Links                  | Prints a list of the linked fonts                                                                                                                                                                                                                  |
| <b>^CW</b> | Font Identifier                  | Assigns a single alphanumeric character as an alias to a font stored in memory.                                                                                                                                                                    |



**Notes •** \_\_\_\_\_

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